THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Bioenergetics and Mitochondropathy

Mechanisms – Mediators – Manifestations

DAVID HAASE, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

David Haase, MD, is a speaker for Fresenius and a consultant for FBB Biomed.

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Explain the distinguishing features of primary mitochondrial disease and secondary mitochondropathy.

2. Explain physiological mechanisms by which environmental agents and insults contribute to mitochondrial dysfunction.

3. Explain physiological mechanisms by which mitochondrial dysfunction contributes to systemic physiological dysfunction, clinical imbalances, and medical conditions.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify and document clinical findings, antecedents, triggers, and mediators of mitochondrial dysfunction on the IFM Timeline and Matrix.

2. Explain to a patient how their diet, lifestyle behaviors, chronic stress, and other factors in their life (antecedents, triggers, and mediators) represent root causes of mitochondrial dysfunction that contribute to their clinical findings and current concerns.

3. Co-prioritize and initiate dietary, lifestyle, and gut health interventions as foundational therapies to address the patient's root cause contributors (ATMs) of mitochondrial dysfunction; initiate supplements as clinically indicated.

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ____________ CC: ____________

Clinical Focus

- Primary mitochondrial disease vs secondary mitochondropathy.
- Key features of secondary mitochondropathy.

Primary Mitochondrial Disease vs Secondary Mitochondropathy

Genetic Disorders
(Antecedents)

Primary Mitochondrial Disease

Environmental Agents & Insults, Genetic & Epigenetic Disorders
(Antecedents, Triggers, & Mediators)

Secondary Mitochondropathy

Primary Mitochondrial Diseases (PMD)

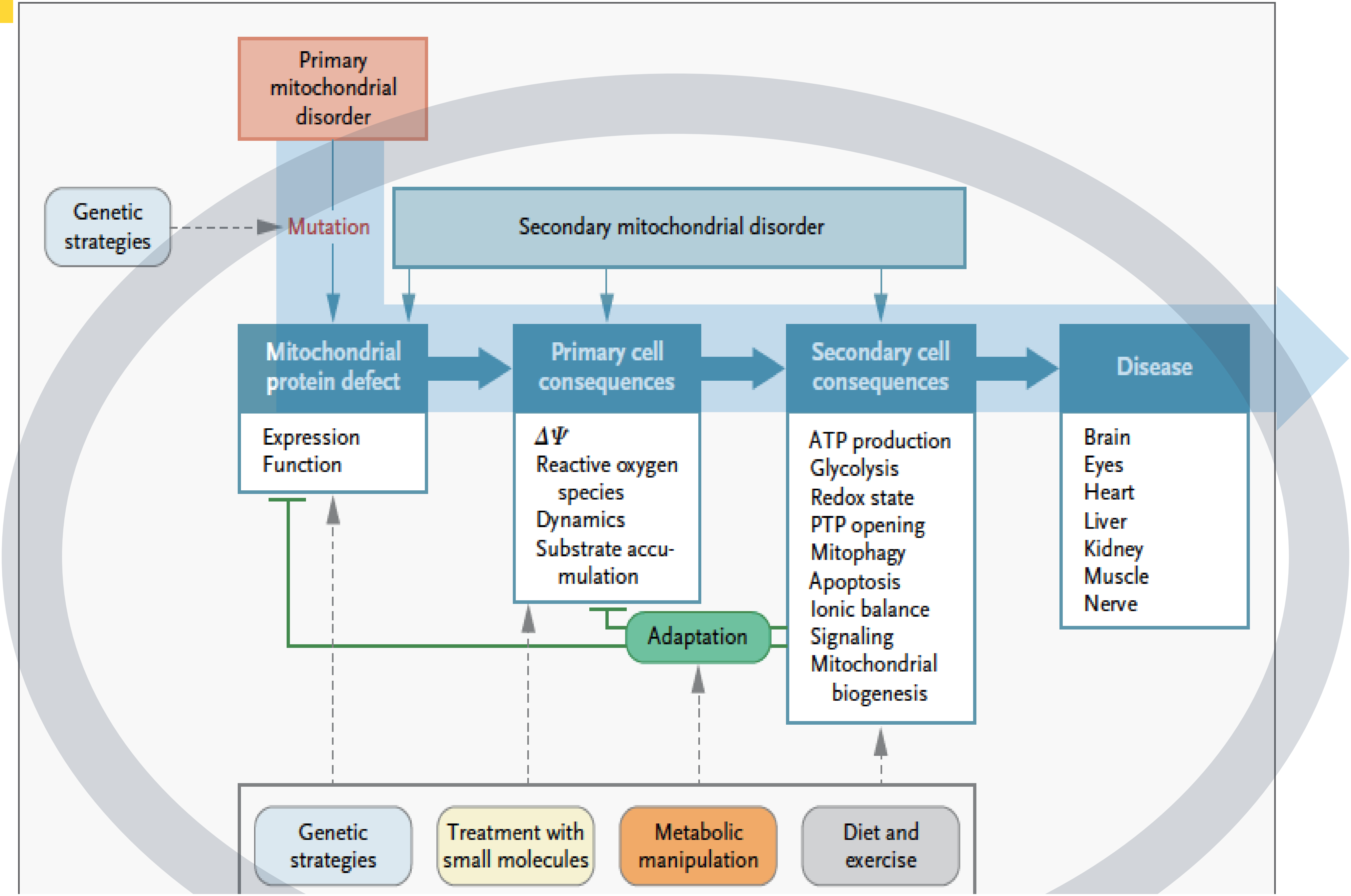
- Leber's hereditary optic neuropathy
- Leigh syndrome
- Maternally inherited diabetes and deafness
- Mitochondrial complex III deficiency
- Myoclonic epilepsy with ragged red fibers
- Neuropathy, ataxia, and retinitis pigmentosa
- Non-syndromic deafness
- Mitochondrial encephalomyopathy and lactic acidosis with stroke-like episodes (MELAS)
- Palmoplantar keratoderma with deafness
- Progressive external ophthalmoplegia

- Chinnery PF. Primary mitochondrial disorders overview. In: Adam MP, Feldman J, Mirzaa GM, et al, eds. *GeneReviews®*. University of Washington; 2000. Updated July 29, 2021. Accessed June 6, 2025. https://www.ncbi.nlm.nih.gov/books/NBK1224/
- Niyazov DM, Kahler SG, Frye RE. Primary mitochondrial disease and secondary mitochondrial dysfunction: importance of distinction for diagnosis and treatment. *Mol Syndromol*. 2016;7(3):122-137. doi:10.1159/000446586

Hallmarks of Mitochondrial Diseases

- **An atypical presentation** of a common disease
- Recurrent **setbacks or flare-ups** in a chronic disease
- **Worse with infection or stress or fasting**
- **Unpredictable** in time, severity, constellation, & presentation
- Otherwise **unexplained** disease
- **"Functional"** – (antiquated use of the term)
- Cluster into **"episodes"**
- **Affected relatives**

Munnich A, Rötig A, Chretien D, et al. Clinical presentation of mitochondrial disorders in childhood. *J Inherit Metab Dis*. 1996;19(4):521-527. doi:10.1007/BF01799112



From New England Journal of Medicine, Koopman, W. J. H., Willems, P. H. G. M., Smeitink, J. A. M., Monogenic mitochondrial disorders, Volume 366, Pages 1132-1141.

Long Description: Consequences of Mitochondrial Dysfunction

- This graphic illustrates the cascade of events from a primary mitochondrial disorder as well as a secondary mitochondrial disorder to disease outcomes. It also outlines how different therapeutic strategies may intervene at various stages of disease progression.
- **Starting Point:** A mutation causes a **primary mitochondrial disorder**, leading to a **mitochondrial protein defect**. This defect impairs protein expression and function within mitochondria.
- **Cascade of Consequences:**
  - The mitochondrial protein defect leads to **primary cell consequences**, which include:
    - Loss of mitochondrial membrane potential ($\Delta\Psi$)
    - Increased reactive oxygen species
    - Abnormal mitochondrial dynamics
    - Accumulation of substrates
  - These primary cell effects cause **secondary cell consequences**, including:
    - Impaired A T P production
    - Altered glycolysis
    - Changes in redox state
    - Opening of the mitochondrial permeability transition pore (P T P)
    - Increased mitophagy and apoptosis
    - Disruption of ionic balance and cell signaling
    - Impaired mitochondrial biogenesis
- **Outcome:** These cellular changes lead to **disease** affecting multiple organs: brain, eyes, heart, liver, kidney, muscle, and nerve.
- **Intervention Strategies:**
  - The process can be influenced or interrupted by different therapeutic strategies, which are indicated at relevant steps:
    - **Genetic strategies** can target the mutation or mitochondrial protein defect.
    - **Treatment with small molecules** can influence primary or secondary cell consequences.
    - **Metabolic manipulation** can affect adaptation and downstream consequences.
    - **Diet and exercise** can support mitochondrial biogenesis and overall function.
- **Adaptation:** A central "Adaptation" process attempts to regulate or compensate for cellular disturbances, providing feedback to earlier steps in the cascade.
- **Secondary Mitochondrial Disorder:**
  - This is depicted as an overlapping category that can arise independently or be amplified by primary disorders, feeding into the same pathways of cellular consequences and disease.

Secondary Mitochondropathy:
Egg and Chicken

Secondary mitochondropathy may result from a wide variety of environmental insults (exogenous or endogenous).

It may present with nearly any symptom, in any organ, at any age (and any disease manifestation or severity).

Secondary mitochondropathy is a notorious masquerader.

Baker MJ, Crameri JJ, Thorburn DR, Frazier AE, Stojanovski D. Mitochondrial biology and dysfunction in secondary mitochondrial disease. *Open Biol*. 2022;12(12):220274. doi:10.1098/rsob.220274

Secondary Mitochondropathy:
Chicken and Egg

Secondary mitochondropathy contributes to a wide variety of systemic physiological dysfunctions & clinical imbalances.

It may present with any symptom, in any organ, at any age (and any disease manifestation or degree of severity).

Secondary mitochondropathy is a notorious masquerader.

Baker MJ, Crameri JJ, Thorburn DR, Frazier AE, Stojanovski D. Mitochondrial biology and dysfunction in secondary mitochondrial disease. *Open Biol*. 2022;12(12):220274. doi:10.1098/rsob.220274

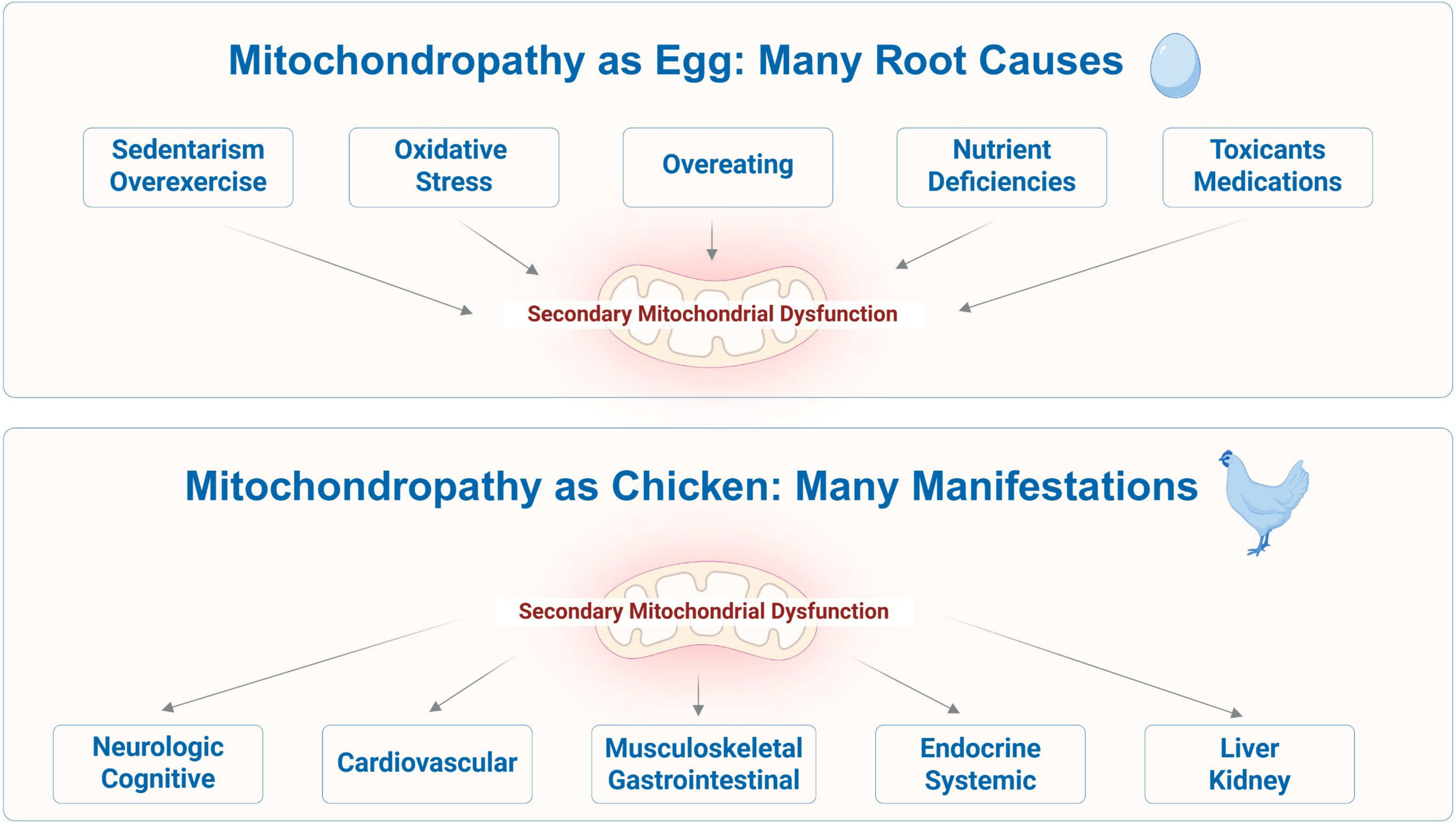
Epigenetic Mechanisms of Secondary Mitochondropathy

Secondary mitochondropathies reflect functional disruption of mitochondrial processes via extrinsic or “downstream” mechanisms.

- Environmental exposures leading to epigenetic modifications
- Diet-induced epigenetic changes
- Epigenetic repression of mitochondrial biogenesis genes

These mechanisms do not involve direct heritable or spontaneous pathogenic variants in mitochondrial or nuclear genes coding for mitochondrial proteins.

Niyazov DM, Kahler SG, Frye RE. Primary mitochondrial disease and secondary mitochondrial dysfunction: importance of distinction for diagnosis and treatment. *Mol Syndromol.* 2016;7(3):122-137. doi:10.1159/000446586



See: “References: Mitochondropathy: Many Root Causes and Manifestations”

Long Description:
Mitochondropathy: Many Root Causes and Manifestations

This graphic uses an "egg and chicken" metaphor to explain the two-way relationship between secondary mitochondrial dysfunction and chronic disease. It is divided into two panels:

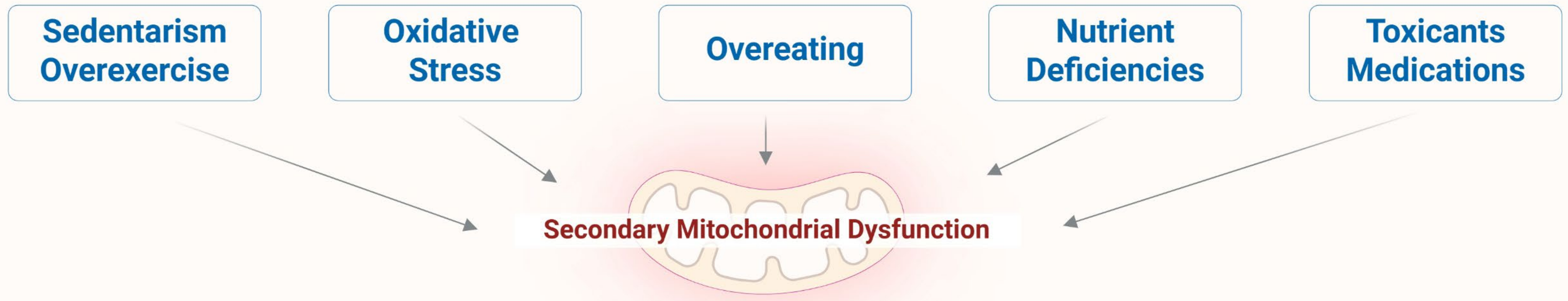
Top Panel: "Mitochondropathy as Egg: Many Root Causes“

- This panel explains that secondary mitochondrial dysfunction can be caused by various lifestyle and environmental factors. These root causes include:
- Sedentary lifestyle or overexercise
- Oxidative stress
- Overeating
- Nutrient deficiencies
- Exposure to toxicants or medications
- These factors lead to **secondary mitochondrial dysfunction**, depicted as the “egg” in the metaphor.

Bottom Panel: "Mitochondropathy as Chicken: Many Manifestations"

- This panel illustrates that secondary mitochondrial dysfunction can also be a contributor to a wide range of health issues, referred to as the “chicken” in the metaphor. The potential disease manifestations include:
- Neurologic and cognitive symptoms
- Cardiovascular disease
- Musculoskeletal and gastrointestinal issues
- Endocrine and systemic dysfunction
- Liver and kidney disorders

Mitochondropathy as Egg: Many Root Causes



See: "References: Mitochondropathy: Many Root Causes and Manifestations"

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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Sedentarism/Overexercise:

- Ostojic SM, Ratgeber L, Olah A, Betlehem J, Pongras A. What do over-trained athletes and patients with neurodegenerative diseases have in common? Mitochondrial dysfunction. *Exp Biol Med (Maywood)*. 2021;246(11):1241-1243. doi:10.1177/1535370221990619
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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

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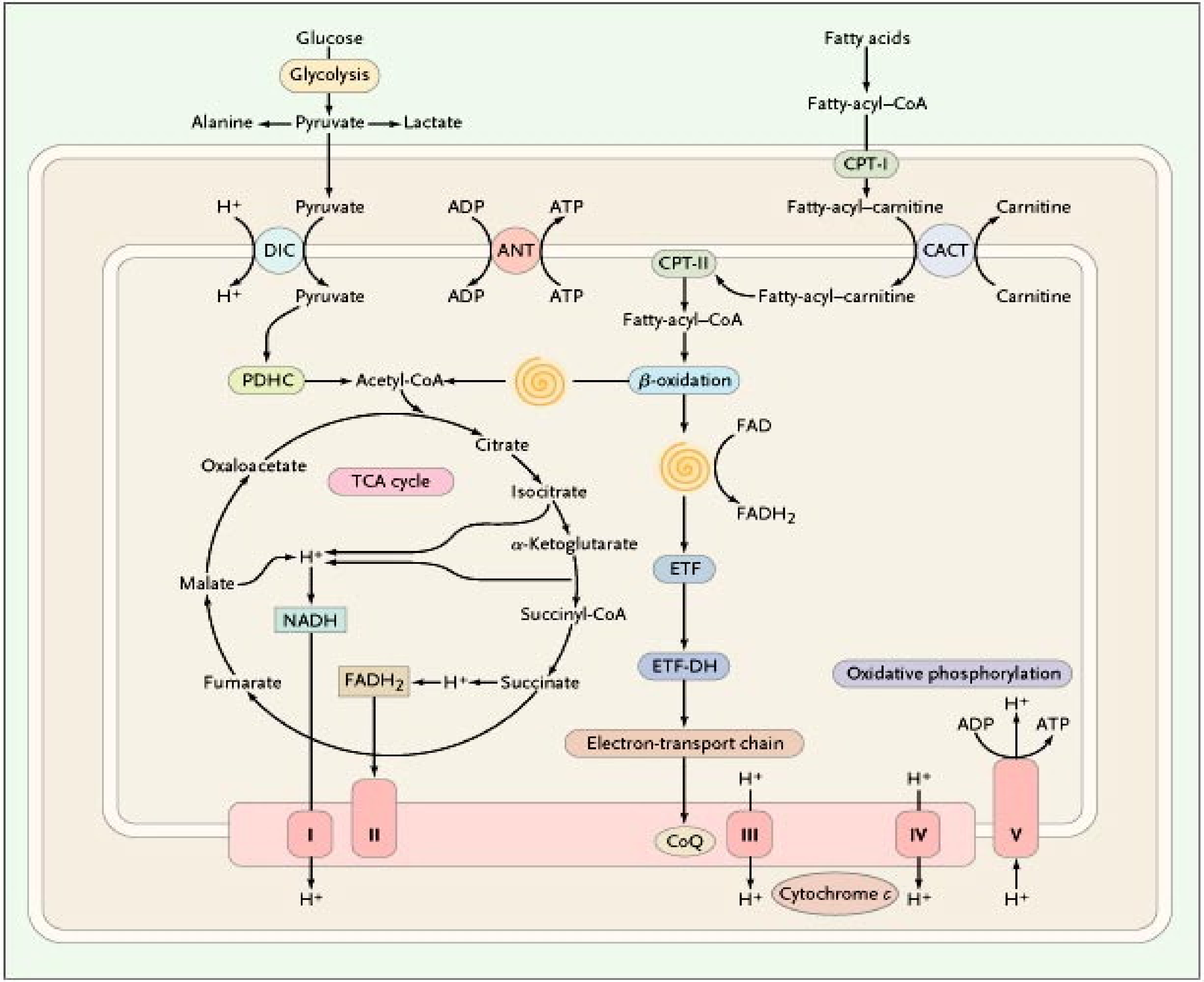
Clinical Focus

- Normal physiology and functions of mitochondria.

Nutrient Inputs of the TCA (Krebs) Cycle



From New England Journal of Medicine, DiMauro, S., Schon, E. A., Mitochondrial respiratory-chain diseases, Volume No. 348, Page 2656-68 Copyright © 2003 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

Long Description: Nutrient Inputs of the T C A (Krebs) Cycle

This graphic illustrates how glucose and fatty acids are metabolized within the mitochondria to produce A T P, highlighting the integration of glycolysis, the tricarboxylic acid (T C A) or Krebs cycle, fatty acid β-oxidation, and the electron transport chain.

- Glucose enters glycolysis, producing pyruvate, which is converted to acetyl-Co A and enters the T C A cycle.
- The T C A cycle generates N A D H and F A D H_2, which supply electrons to the electron transport chain.
- Fatty acids are converted to fatty-acyl-Co A, transported into mitochondria via the carnitine shuttle, and broken down through β-oxidation to produce acetyl-Co A, N A D H, and F A D H_2.
- Electrons from N A D H, F A D H_2, and electron transfer flavoprotein (ETF) flow through the electron transport chain to generate a proton gradient used by A T P synthase to produce ATP.

The diagram emphasizes how carbohydrate and fat metabolism converge to drive oxidative phosphorylation for cellular energy production.

Macronutrient Inputs of the TCA Cycle

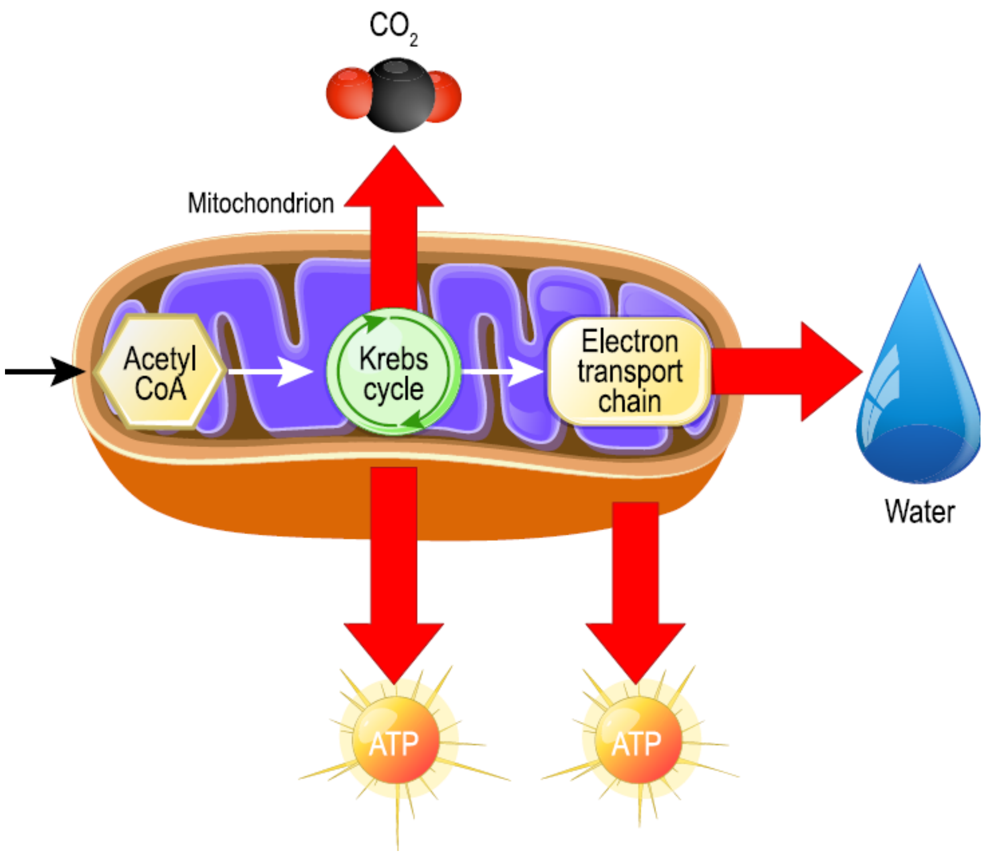
Protein

Glycolysis → Pyruvate

Carbohydrates

→ acyl CoA transport → Beta-oxidation

Fats

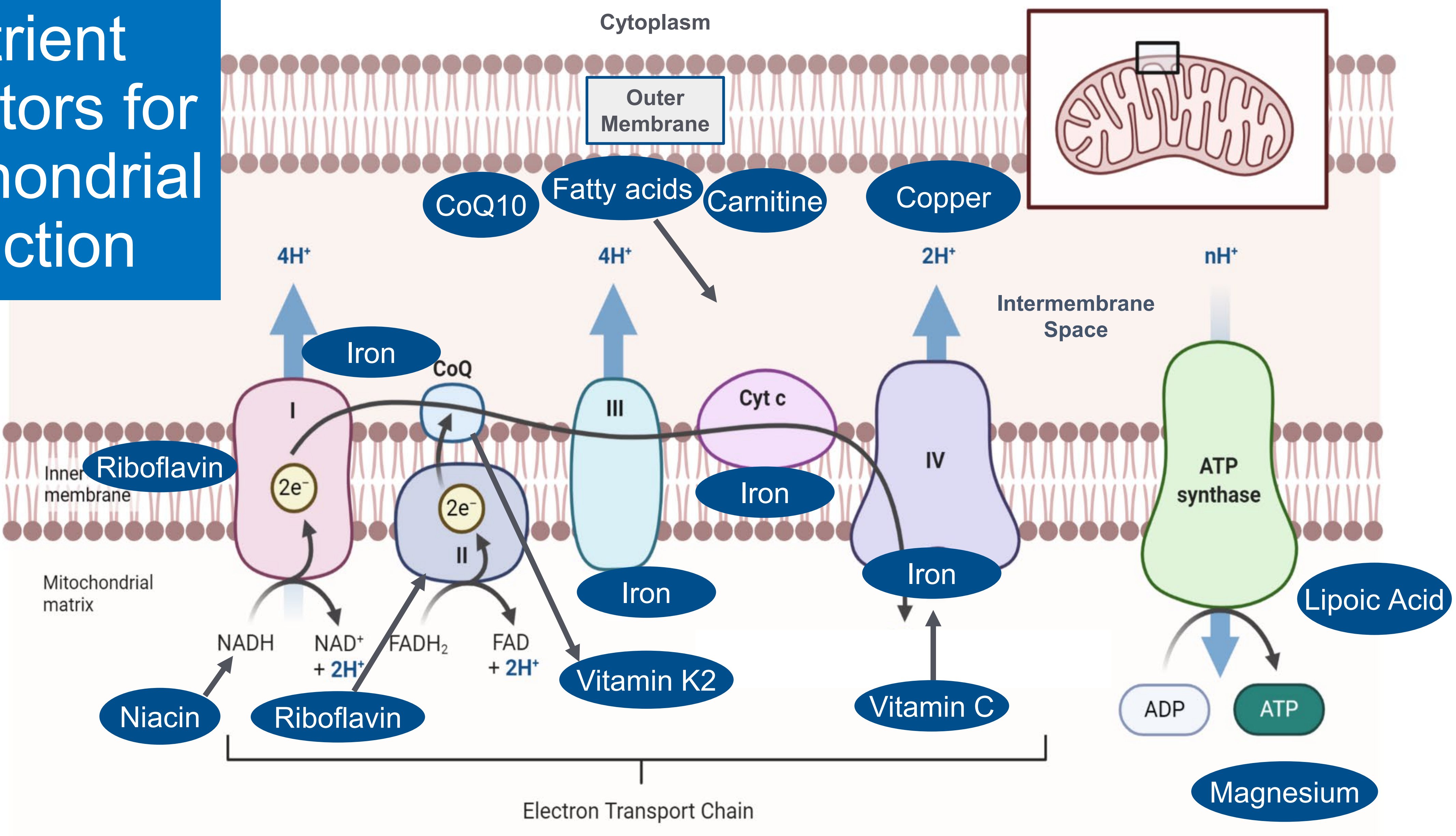


Mukherjee A, Ghosh KK, Chakrabortty S, Gulyás B, Padmanabhan P, Ball WB. Mitochondrial reactive oxygen species in infection and immunity. *Biomolecules*. 2024;14(6):670. doi:10.3390/biom14060670

Long Description: Macronutrient Inputs of the T C A Cycle

- Proteins, carbohydrates, and fats feed into the tricarboxylic acid (T C A) cycle to produce energy.
- Proteins and carbohydrates are broken down through glycolysis into pyruvate, which is converted into acetyl-CoA and enters the TCA cycle.
- Fats are converted into acyl-CoA, transported into mitochondria, and broken down through beta-oxidation to produce acetyl-CoA.
- Within the mitochondria, acetyl-CoA enters the TCA cycle, producing carbon dioxide, water, and energy carriers that drive the electron transport chain. This results in the production of ATP, the cell's main energy currency.

Nutrient Cofactors for Mitochondrial Function



Wesselink E, Koekkoek WAC, Grefte S, Witkamp RF, van Zanten ARH. Feeding mitochondria: potential role of nutritional components to improve critical illness convalescence. *Clin Nutr.* 2019;38(3):982-995. doi:10.1016/j.clnu.2018.08.032

Created with BioRender.com by The Institute for Functional Medicine

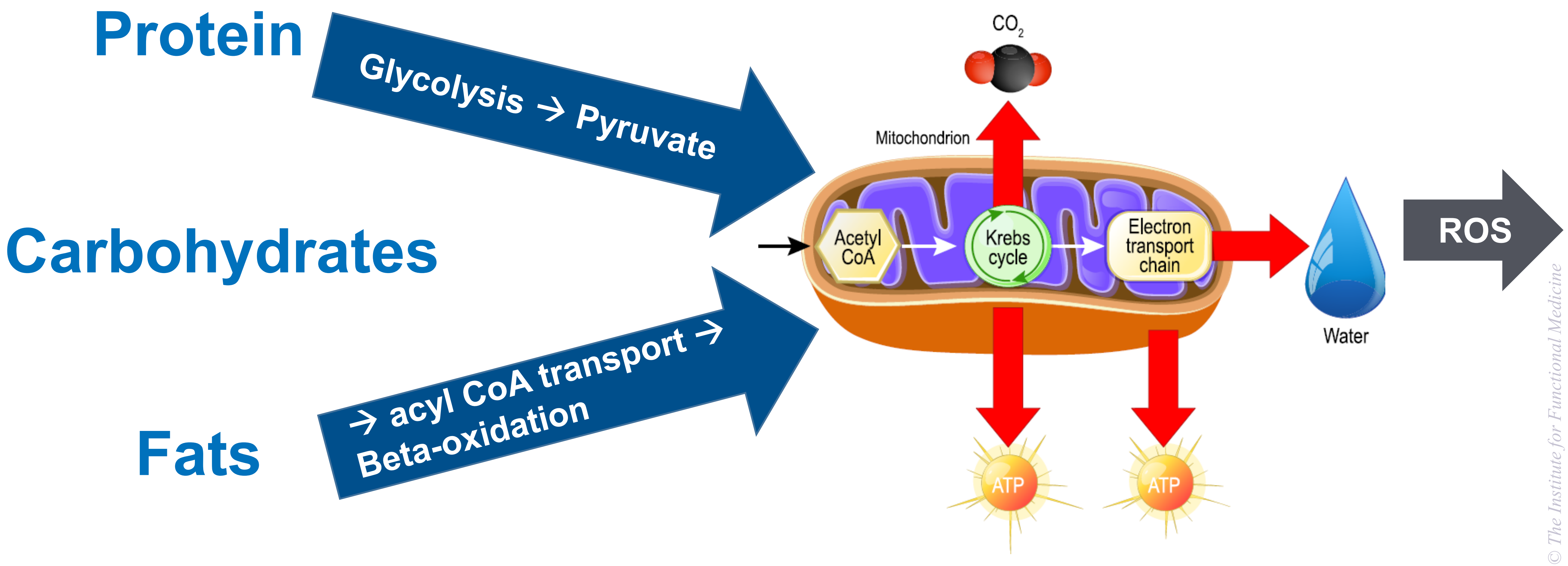
Long Description
Nutrient Cofactors for Mitochondrial Function

This diagram shows the key vitamins, minerals, and other nutrients required for proper mitochondrial energy production through the electron transport chain.

- It depicts the inner mitochondrial membrane with protein complexes 1 through 4 and A T P synthase.
- The nutrients and nutrient cofactors needed are as follows:
  - Niacin and riboflavin are needed for N A D H and F A D H_2 production.
  - Riboflavin also supports Complex 2.
  - Iron is required at multiple points in Complexes 1, 2, 3, and 4, while Coenzyme Q 10 carries electrons between Complexes 1 or 2 and 3.
  - Vitamin K 2, carnitine, and fatty acids are linked to Complex 3 function.
  - Copper and iron are needed for Complex 4.
  - Vitamin C supports iron function and electron transfer. Lipoic acid and magnesium are needed for A T P synthase, which converts A D P to A T P.

The graphic emphasizes that these micronutrients are essential cofactors for mitochondrial oxidative phosphorylation and energy generation.

Reactive Oxygen Species as Outputs of the TCA Cycle

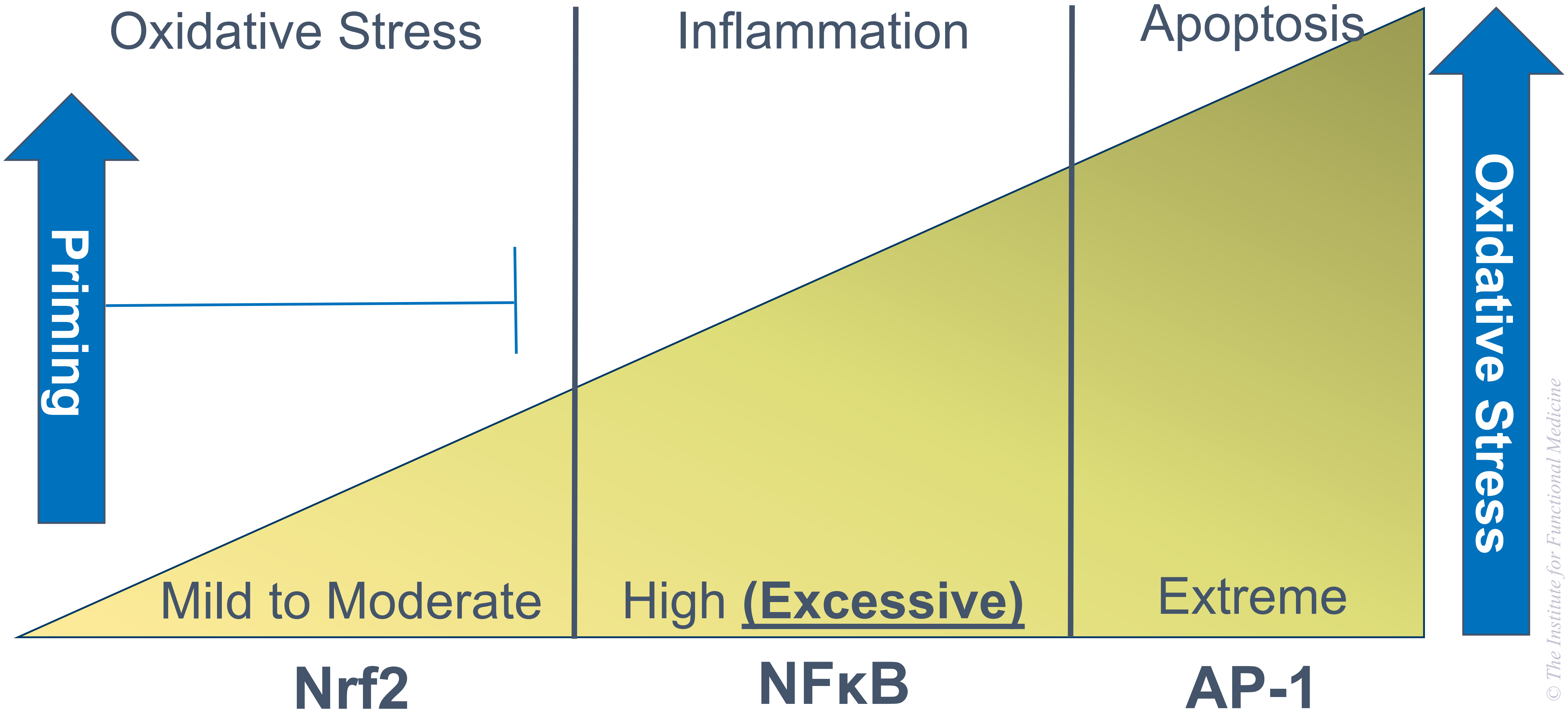


Mukherjee A, Ghosh KK, Chakrabortty S, Gulyás B, Padmanabhan P, Ball WB. Mitochondrial reactive oxygen species in infection and immunity. *Biomolecules*. 2024;14(6):670. doi:10.3390/biom14060670

Long Description
Reactive Oxygen Species as Outputs of the TCA Cycle

- This graphic explains that proteins, carbohydrates, and fats all feed into the tricarboxylic acid (T C A) cycle, and one of its outputs is reactive oxygen species (R O S).
- Proteins and carbohydrates are converted through glycolysis into pyruvate, while fats are broken down via acyl-Co A transport and beta-oxidation. Both pathways produce acetyl-Co A, which enters the T C A cycle in the mitochondria.
- The T C A cycle generates carbon dioxide, A T P, and high-energy electrons that feed into the electron transport chain, producing water and A T P.
- Alongside these beneficial outputs, the process also generates reactive oxygen species as a byproduct, which can contribute to oxidative stress if not regulated.

Differential Responses to Rising Oxidative Stress



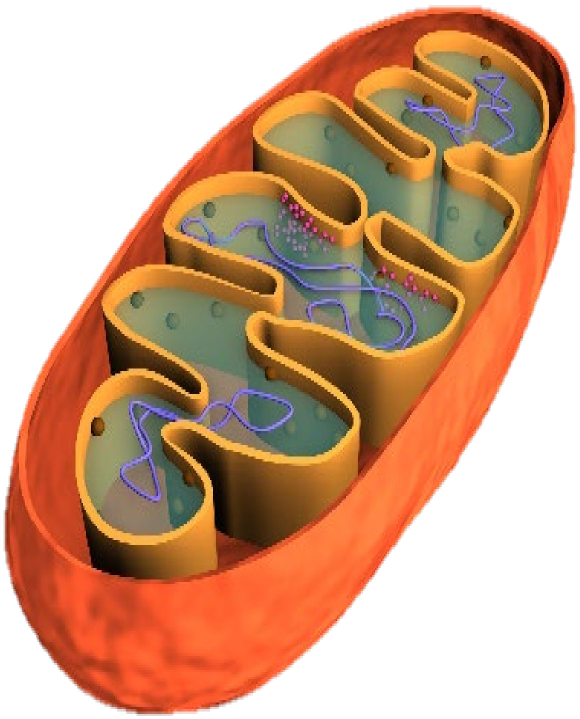
Adapted from: Stefanson AL, Bakovic M. Dietary regulation of Keap1/Nrf2/ARE pathway: focus on plant-derived compounds and trace minerals. *Nutrients*. 2014;6(9):3777-3801. doi:10.3390/nu6093777

Long Description for
Differential Responses to Rising Oxidative Stress

- This graphic illustrates how the body responds differently to increasing levels of oxidative stress.
- At mild to moderate oxidative stress, the N r f 2 pathway is activated, which supports protective and adaptive responses, a stage referred to as “priming.”
- When oxidative stress becomes high or excessive, the N F κ B pathway is triggered, leading to inflammation.
- At extreme oxidative stress levels, the AP-1 pathway is activated, resulting in apoptosis, or programmed cell death.
- The diagram shows oxidative stress increasing from left to right, with priming occurring at lower levels, inflammation at higher levels, and apoptosis at the highest levels.

Mitochondrial Functions:
More Than Just ATP and Energy Production

Mitochondria help maintain homeostasis.



Bioenergetics

Biosynthesis

Signaling/Metabolism

Mitochondrial Functions: Selected Examples

- **Bioenergetics**
  - ATP synthesis
  - Maintenance of ion gradients
- **Biosynthesis**
  - Phospholipid synthesis
  - Heme synthesis
- **Signaling/Metabolism**
  - Buffering calcium ion flux (from ER & plasma membrane)
  - Mitochondrial fragments activate immune response
  - ATP and adenosine are neurotransmitters
  - Regulation of cell growth, cell cycle, metabolism

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Mitochondrial Biogenesis

- Regulates mitochondrial mass based on energy needs of cell
- Generates new, healthy organelles
- Replaces damaged ones (mitophagy)
- Coordinated interaction between nuclear & mitochondrial genomes
- Mediated by a hierarchy of nuclear transcription factors, most of which are dependent on PGC-1 proteins, especially PGC-1α

Pickles S, Vigié P, Youle RJ. Mitophagy and quality control mechanisms in mitochondrial maintenance. *Curr Biol*. 2018;28(4):R170-R185. doi:10.1016/j.cub.2018.01.004

PGC-1α
(PPAR-γ co-activator 1α)

- Master regulator of mitochondrial biogenesis and energy homeostasis
- Most nuclear genes coding for mitochondrial proteins have binding sites for PCG-1α on their regulatory regions
- Predominantly expressed in mitochondrial-rich tissues: heart, skeletal muscle, brown adipose tissue, and liver (to some extent)

***PGC**-1α - peroxisome proliferator-activated receptor gamma coactivator 1-alpha*

Abu Shelbayeh O, Arroum T, Morris S, Busch KB. PGC-1α is a master regulator of mitochondrial lifecycle and ROS stress response. *Antioxidants (Basel)*. 2023;12(5):1075. doi:10.3390/antiox12051075

PGC-1α
(PPAR-γ co-activator 1α)

Activity decreases with:

- Caloric overload
- Saturated fats
- Refined carbohydrates, fructose
- Inflammatory mediators (e.g., TNFα)
- Pro-oxidants
- Inactivity
- Aging

Concept = Less Energy Needed

Abu Shelbayeh O, Arroum T, Morris S, Busch KB. PGC-1α is a master regulator of mitochondrial lifecycle and ROS stress response. *Antioxidants (Basel)*. 2023;12(5):1075. doi:10.3390/antiox12051075

Bioenergetics and Mitochondropathy: Part 2

Mechanisms – Mediators – Manifestations

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Clinical findings of mitochondrial dysfunction.
- Manifold manifestations.
- Myriad signs, symptoms, and syndromes.

Myriad Signs, Symptoms, and Syndromes of Mitochondrial Dysfunction

Ears & Eyes	Kidneys	Heart	Endocrine	Liver
• Visual loss and blindness • Optic atrophy • Hearing loss and deafness • Acquired strabismus • Retinitis pigmentosa	• Renal tubular dysfunction • Nephrotic syndrome • Albuminuria	• Tachycardia • Cardiac conduction defects (heart blocks) • Cardiomyopathy • WPW syndrome	• Hypothyroidism • Parathyroid failure (low calcium) • Short stature	• Hypoglycemia (low blood sugar) • Fatty liver

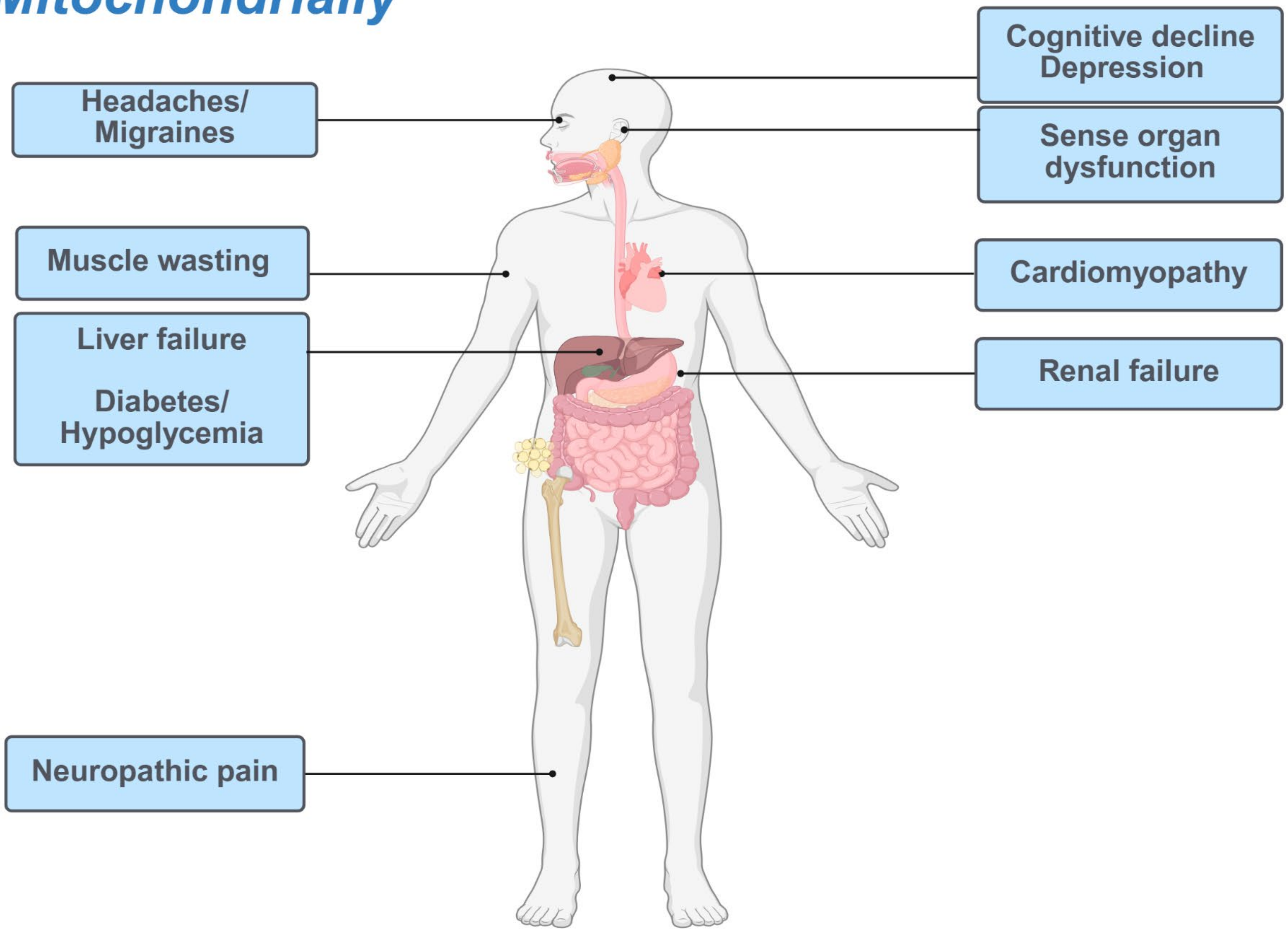
See: "References: Signs and Symptoms Suggesting Mitochondropathy"

Myriad Signs, Symptoms, and Syndromes of Mitochondrial Dysfunction

Brain	Nerves	Muscle	GI	Systemic
• Developmental delays • Dementia and Parkinson's • Neuro-psychiatric disturbances • Autism features • Intellectual disability • Seizures • Strokes	• Weakness • Dysautonomia	• Exercise intolerance out of proportion to weakness • Exercise-induced myalgia • Weakness • Cramping • Hypotonia • Unexplained rhabdomyolysis	• Exocrine pancreatic insufficiency • Pseudo-obstruction • GERD • Dysmotility, vomiting, or peristalsis • IBS, diarrhea, or constipation	• Failure to gain weight • IUGR • Microcephaly • Short stature • Respiratory problems • Sleep apnea

See: "References: Signs and Symptoms Suggesting Mitochondropathy"

Representative Patient Clinical Findings:
Think Mitochondrially



See: "References: Signs and Symptoms Suggesting Mitochondropathy"

Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description for
Representative Patient Clinical Findings: Think Mitochondrially

This graphic depicts a human body outline with labeled symptoms connected to specific body regions that may be related to mitochondrial dysfunction.

- Head - Headaches/Migraines, Cognitive decline, and depression.
- Eyes & Ears - Sense organ dysfunction
- Muscular system - Muscle wasting
- Heart - Cardiomyopathy
- Liver - Liver failure
- Pancreas - Diabetes/Hypoglycemia
- Kidneys - Renal failure
- Nervous system - Neuropathic pain

The overall meaning is that mitochondrial disorders can affect multiple organ systems, producing widespread symptoms across the nervous, muscular, cardiac, hepatic, endocrine, renal, and central and peripheral nervous systems.

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SIGNS AND SYMPTOMS SUGGESTING MITOCHONDROPATHY

Ears/Eyes:

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Kidneys:

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Heart:

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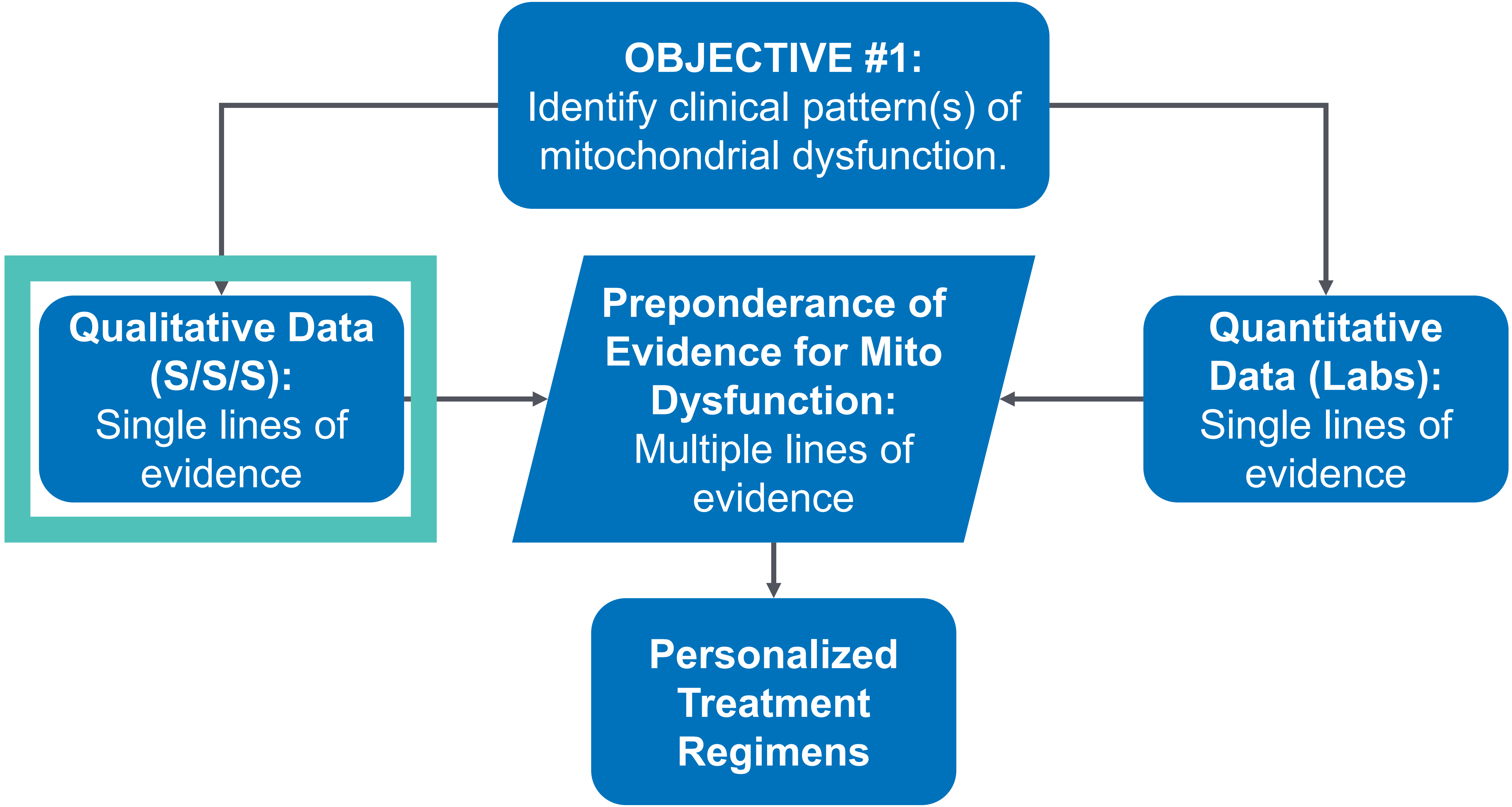
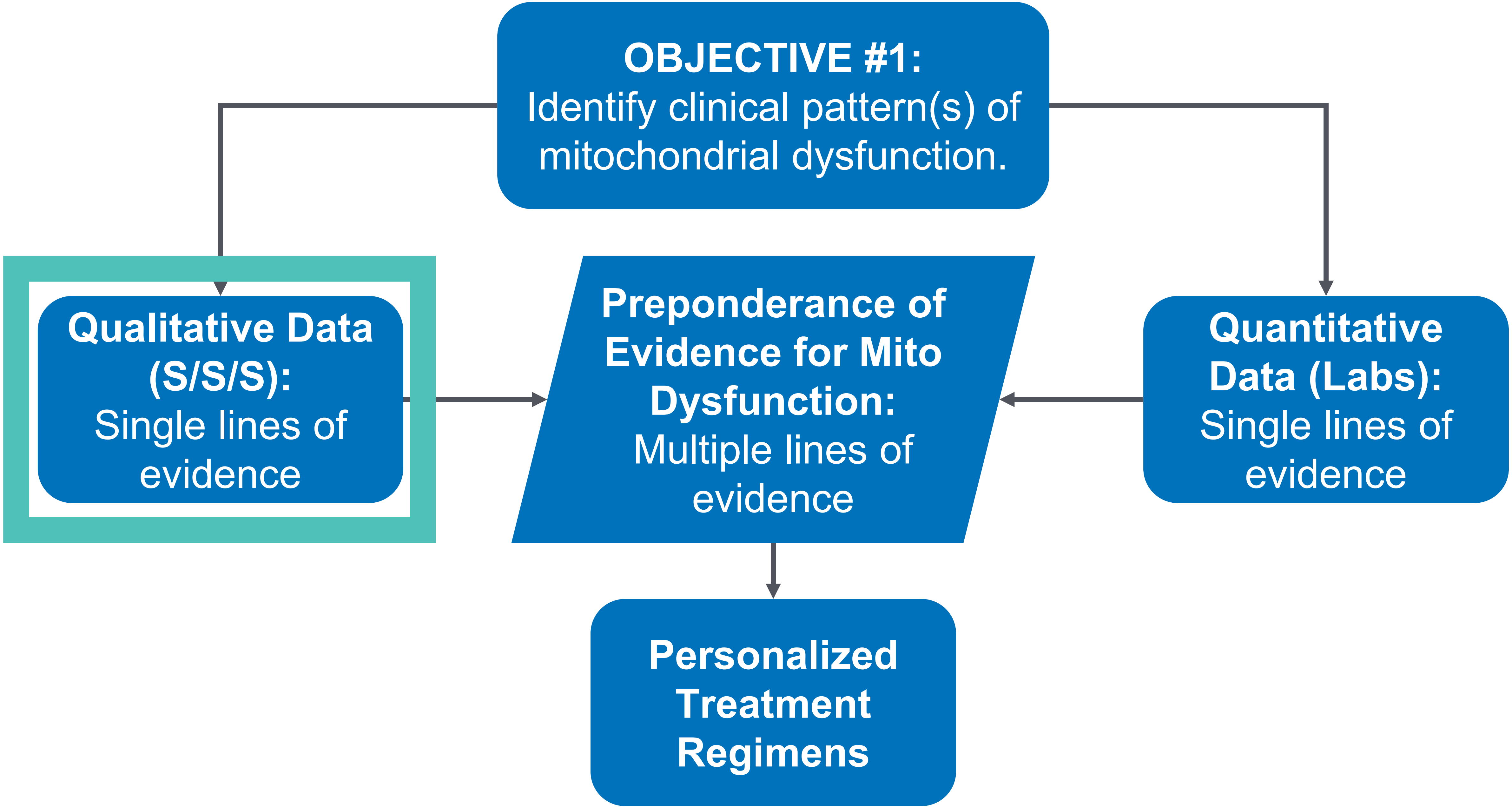

OBJECTIVE #1:
Identify clinical pattern(s) of mitochondrial dysfunction.
Qualitative Data (S/S/S):
Single lines of evidence
Preponderance of Evidence for Mito Dysfunction:
Multiple lines of evidence
Quantitative Data (Labs):
Single lines of evidence
Personalized Treatment Regimens


OBJECTIVE #1:
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Quantitative Data (Labs):
Single lines of evidence
Personalized Treatment Regimens

Long Description for
Objective number 1: Identify clinical patterns of mitochondrial dysfunction

This graphic illustrates a process for identifying mitochondrial dysfunction and guiding treatment.

- **Objective #1:** Identify clinical patterns of mitochondrial dysfunction.
- This involves gathering two types of single-line evidence:
  - **Qualitative data (S/S/S)** – likely referring to symptoms, signs, and syndromes from the patient.
  - **Quantitative data (labs)** – objective measurements from laboratory tests.
- When these single lines of evidence are combined, they create a **preponderance of evidence for mitochondrial dysfunction** through multiple lines of evidence.
- This comprehensive evidence then informs **personalized treatment regimens** tailored to the patient.
- On this slide, Qualitative Data is highlighted

The overall meaning of this graphic is that both qualitative and quantitative evidence must be integrated to confirm mitochondrial dysfunction and guide individualized therapy.

Qualitative Data: Single Lines of Evidence

Symptoms

Headaches
Migraines
Neuropathic pain
Chronic pain
Memory loss
Vision loss
Hearing loss
Fainting
Cognitive deficit
Neuropsych sx
Fatigue
Weakness
Exercise intolerance

Syndromes

Brain/Neuro:
Depression
Dementia
Parkinson’s
Autism
Dysautonomia
Neurodegenerative

Systemic:
Chronic fatigue
Fibromyalgia
Sleep apnea

Endocrine:
Diabetes/Met S
Hypothyroidism

Organs:
Cardiomyopathy
CHF
Dysrhythmias
NAFLD/NASH
Liver failure
Exocrine pancreas

GI:
IBS
GERD
Dysmotility
Chronic diarrhea
Chronic constipation
Cyclic vomiting

Signs

Wasting
Sarcopenia
Ptosis
Ophthalmoplegia
Hypotonia
Absent reflexes

See: “References: Clinical Findings of Mitochondropathy”

Long Description for
Qualitative Data: Single Lines of Evidence

The graphic outlines categories of qualitative data that serve as single lines of evidence for identifying mitochondrial dysfunction. It is divided into three main columns:

1. Symptoms – Patient-reported experiences that may suggest mitochondrial issues:

- Headaches
- Migraines
- Neuropathic pain
- Chronic pain
- Memory loss
- Vision loss
- Hearing loss
- Fainting
- Cognitive deficit
- Neuropsychiatric symptoms
- Fatigue
- Weakness
- Exercise intolerance

2. Syndromes – Recognized clinical patterns, grouped by system:

- **Brain/Neuro:** Depression, dementia, Parkinson's disease, autism, dysautonomia, neurodegenerative disorders
- **Systemic:** Chronic fatigue, fibromyalgia, sleep apnea
- **Endocrine:** Diabetes/metabolic syndrome, hypothyroidism
- **Organs:** Cardiomyopathy, congestive heart failure (CHF), dysrhythmias, non-alcoholic fatty liver disease/non-alcoholic steatohepatitis (NAFLD/NASH), liver failure, exocrine pancreatic insufficiency
- **Gastrointestinal (GI):** Irritable bowel syndrome (IBS), gastroesophageal reflux disease (GERD), dysmotility, chronic diarrhea, chronic constipation, cyclic vomiting

3. Signs – Clinically observed findings:

- Wasting
- Sarcopenia
- Ptosis
- Ophthalmoplegia
- Hypotonia
- Absent reflexes

Overall meaning: This chart presents a framework for recognizing possible mitochondrial dysfunction using qualitative clinical information, which includes patient symptoms, known syndromes, and observable signs.

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Functional Medicine Matrix

Mitochondrial Dysfunction: Manifold Manifestations

The Patient's Story

- Dysmotility
- GERD
- IBS
- Chronic diarrhea
- Chronic constipation
- Cyclic vomiting
- DIGIN dysfunctions

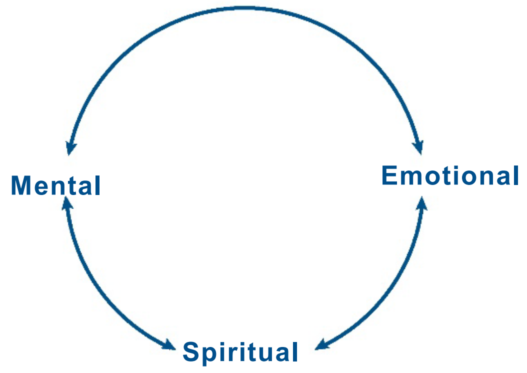
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ______________ CC: ______________

See: "References: Clinical Findings of Mitochondropathy"

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

Mitochondrial Dysfunction: Manifold Manifestations

The Patient's Story

Antecedents (Inherited & Acquired)

- Wasting
- Sarcopenia
- Hypotonia
- Absent reflexes

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ______________ CC: ______________

See: "References: Clinical Findings of Mitochondropathy"

Functional Medicine Matrix

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

- Diabetes
- Metabolic syndrome
- Hypothyroidism

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________ Date: ______________ CC: ______________

See: "References: Clinical Findings of Mitochondropathy"

Mitochondrial Dysfunction: Manifold Manifestations

Functional Medicine Matrix

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Mitochondrial Dysfunction: Manifold Manifestations

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

- Cardiomyopathy
- C H F
- Dysrhythmias
- Hypertension

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ____________ Date: ____________ CC: ____________

See: "References: Clinical Findings of Mitochondropathy"

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

Mitochondrial Dysfunction: Manifold Manifestations

- Liver failure

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ____________ CC: ____________

See: "References: Clinical Findings of Mitochondropathy"

Functional Medicine Matrix

Mitochondrial Dysfunction: Manifold Manifestations

- Chronic fatigue
- Sleep apnea
- Fibromyalgia

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

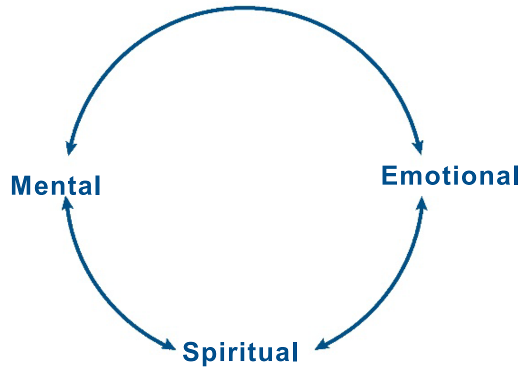
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ____________ CC: ____________

See: "References: Clinical Findings of Mitochondropathy"

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

Mitochondrial Dysfunction: Manifold Manifestations

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

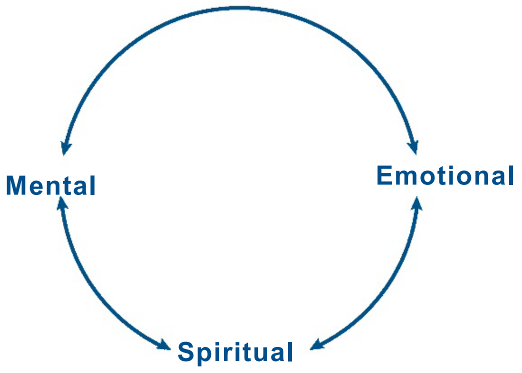
Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

- Chronic, systemic inflammation
- Autoimmunity

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ____________ CC: ____________

See: "References: Clinical Findings of Mitochondropathy"

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

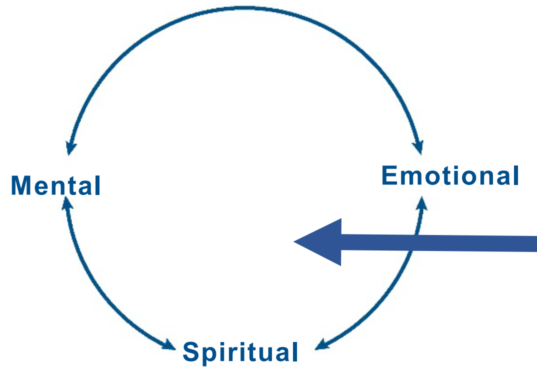
Mitochondrial Dysfunction: Manifold Manifestations

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

- Depressed mood
- Chronic pain
- Cognitive decline

Communication

Mitigation & Clearance

Transport

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ____________ CC: ____________

See: "References: Clinical Findings of Mitochondropathy"

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Secondary Mitochondropathy:
Clinical Hallmarks

- An atypical presentation of a “common disease”
- Recurrent setbacks or flare-ups in a chronic disease
- Worse with infection or stress or fasting
- Unpredictable in time, severity, features, & presentation
- Otherwise unexplained disease
- Clustered into “episodes”
- Affected relatives (consider shared environments)

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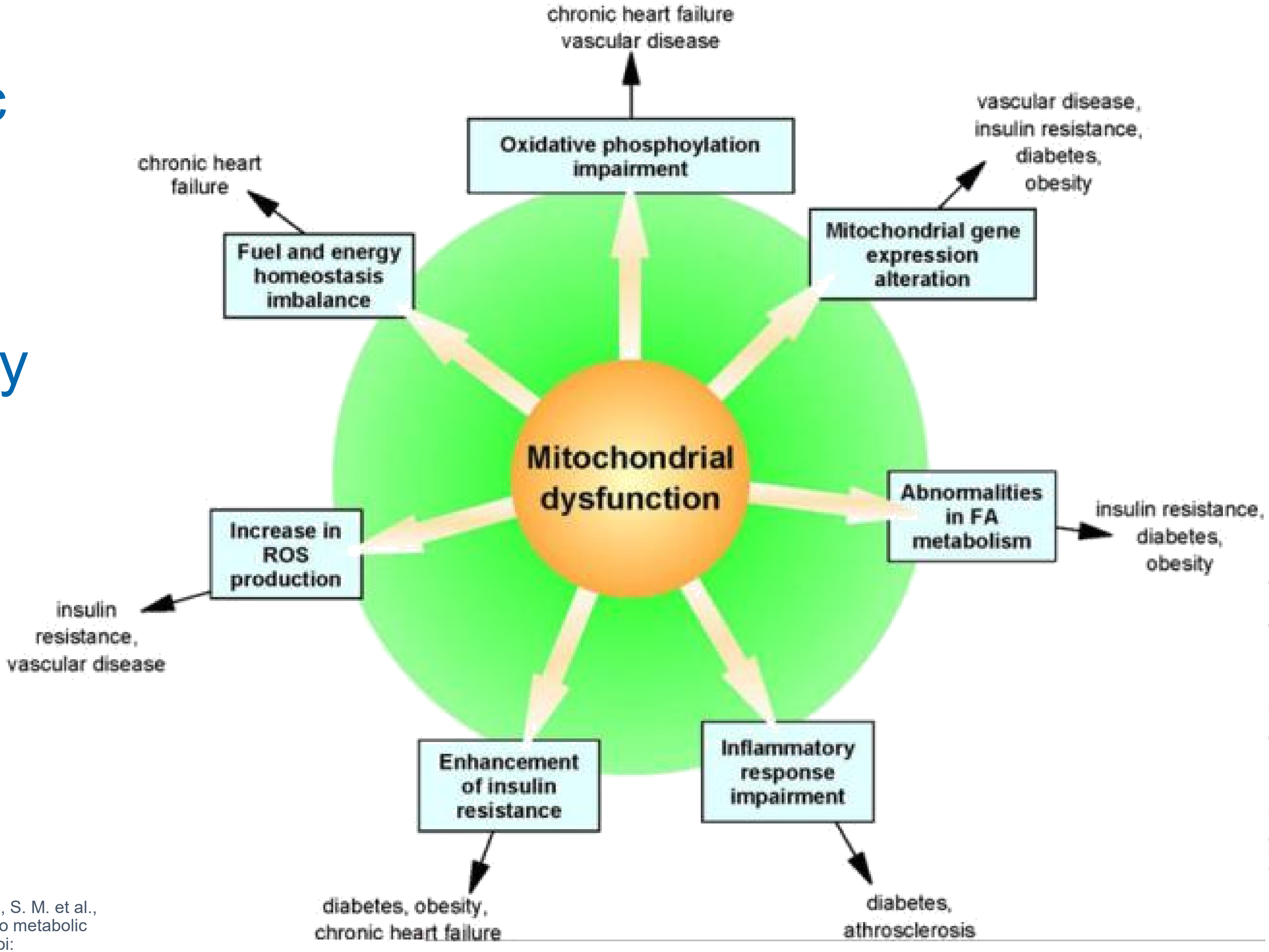
Cardiometabolic Syndrome as Representative Example of Mitochondropathy



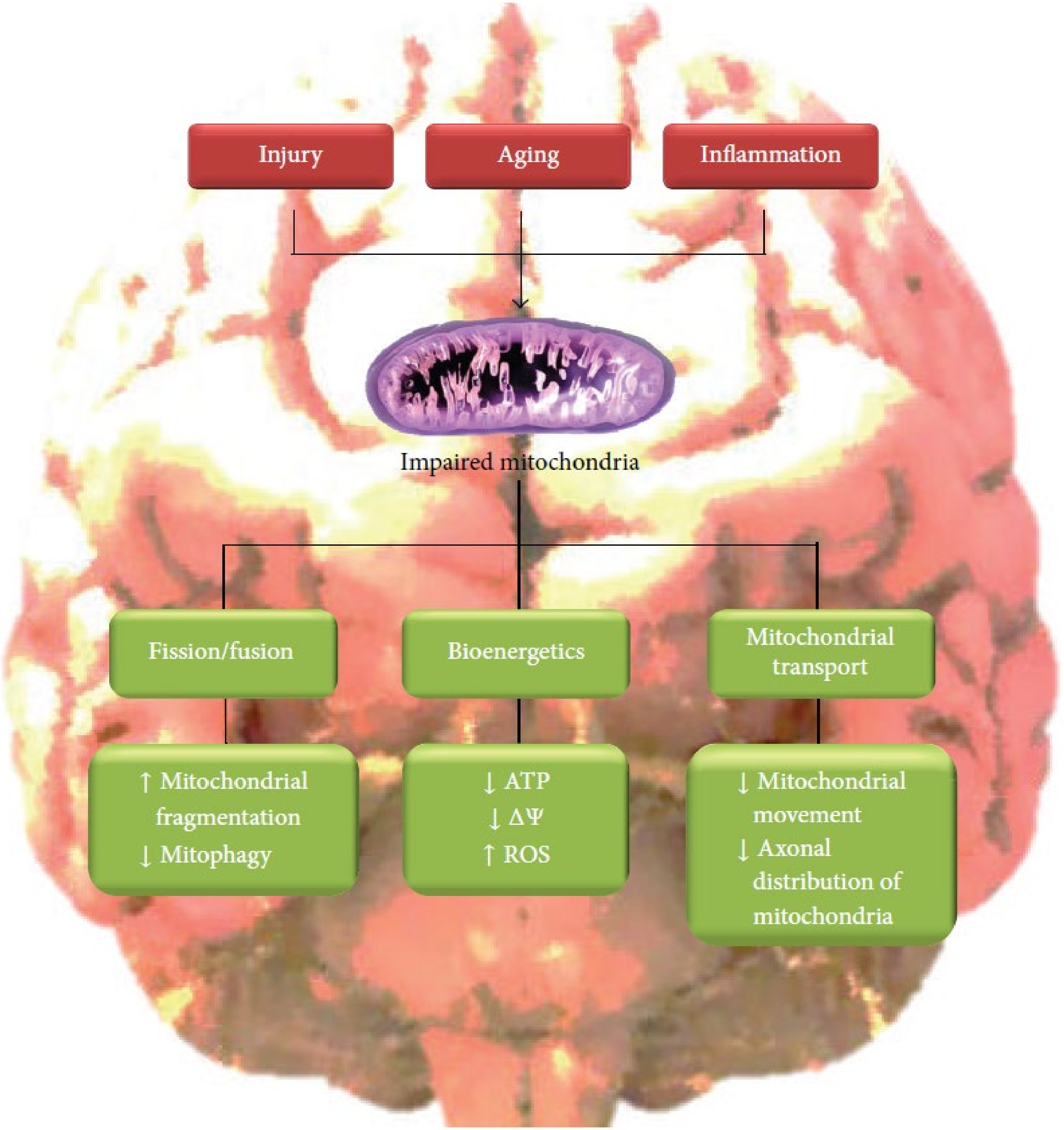
Reprinted from Translational Research, Vol. 161, Marcovina, S. M. et al., Translating the basic knowledge of mitochondrial functions to metabolic therapy: role of L-carnitine, Pages 73-84, Copyright 2013, doi: 10.1016/j.trsl.2012.10.006. with permission from Elsevier.

Long Description of Cardiometabolic Syndrome as Representative Example of Mitochondropathy

The graphic shows **mitochondrial dysfunction** at the center, with arrows pointing outward to seven interconnected consequences. Each consequence is linked to specific health conditions:

- **Oxidative phosphorylation impairment** – leads to chronic heart failure and vascular disease.
- **Mitochondrial gene expression alteration** – associated with vascular disease, insulin resistance, diabetes, and obesity.
- **Abnormalities in fatty acid metabolism** – linked to insulin resistance, diabetes, and obesity.
- **Inflammatory response impairment** – associated with diabetes and atherosclerosis.
- **Enhancement of insulin resistance** – leads to diabetes, obesity, and chronic heart failure.
- **Increase in reactive oxygen species (R O S) production** – linked to insulin resistance and vascular disease.
- **Fuel and energy homeostasis imbalance** – associated with chronic heart failure.

Overall meaning: The diagram emphasizes that mitochondrial dysfunction triggers multiple metabolic, genetic, and inflammatory disturbances, which contribute to a wide range of chronic diseases, especially cardiovascular and metabolic disorders.



Neurocognitive Dysfunction as Representative Example of Mitochondropathy

Cabezas-Opazo FA, Vergara-Pulgar K, Pérez MJ, Jara C, Osorio-Fuentealba C, Quintanilla RA. Mitochondrial dysfunction contributes to the pathogenesis of Alzheimer's disease. *Oxid Med Cell Longev*. 2015;2015:509654. doi:10.1155/2015/509654

Long Description for
Neurocognitive Dysfunction as Representative Example of Mitochondropathy

The graphic explains how injury, aging, and inflammation contribute to mitochondrial impairment and its effects.

- **Top level:** Three main causes — injury, aging, and inflammation — lead to impaired mitochondria.
- **Central point:** An illustration of a mitochondrion represents the impaired state.
- **Downstream effects** are shown in three functional categories:
  - **Fission/Fusion** –
    - Increased mitochondrial fragmentation
    - Decreased mitophagy (recycling of damaged mitochondria)
  - **Bioenergetics** –
    - Decreased A T P production
    - Decreased mitochondrial membrane potential (ΔΨ)
    - Increased reactive oxygen species (R O S)
  - **Mitochondrial Transport** –
    - Decreased mitochondrial movement
    - Reduced axonal distribution of mitochondria

Overall meaning: Damage from aging, injury, or inflammation can disrupt mitochondrial structure, energy production, and transport within cells, leading to impaired function and potentially contributing to neurodegenerative processes.

Common Clinical Conditions

See: “References: Common Clinical Conditions”

References

COMMON CLINICAL CONDITIONS

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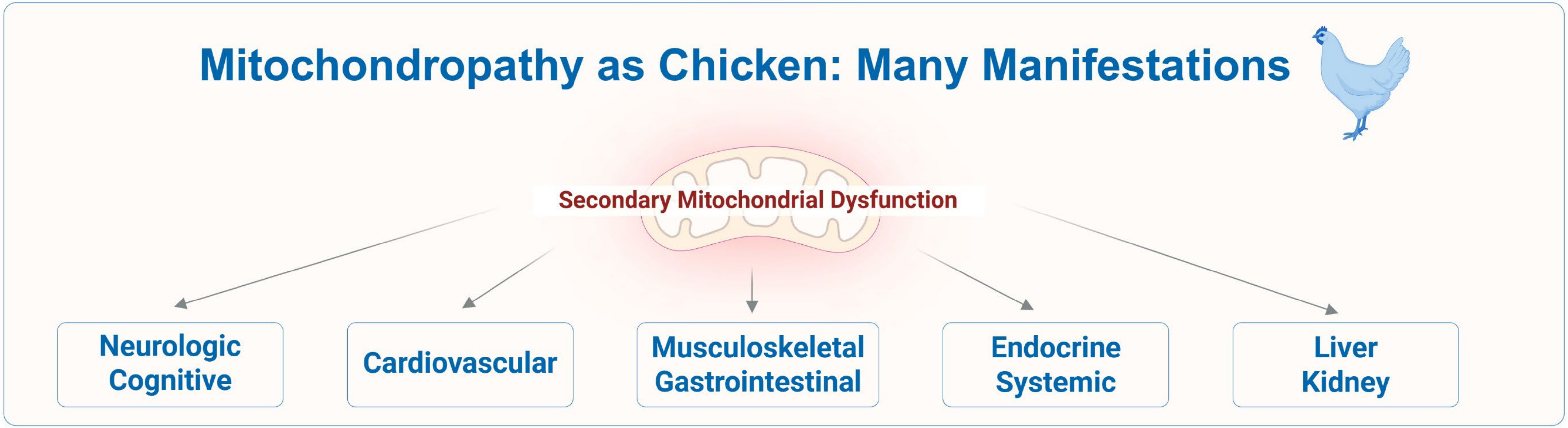

See: “References: Mitochondropathy: Many Root Causes and Manifestations”

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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

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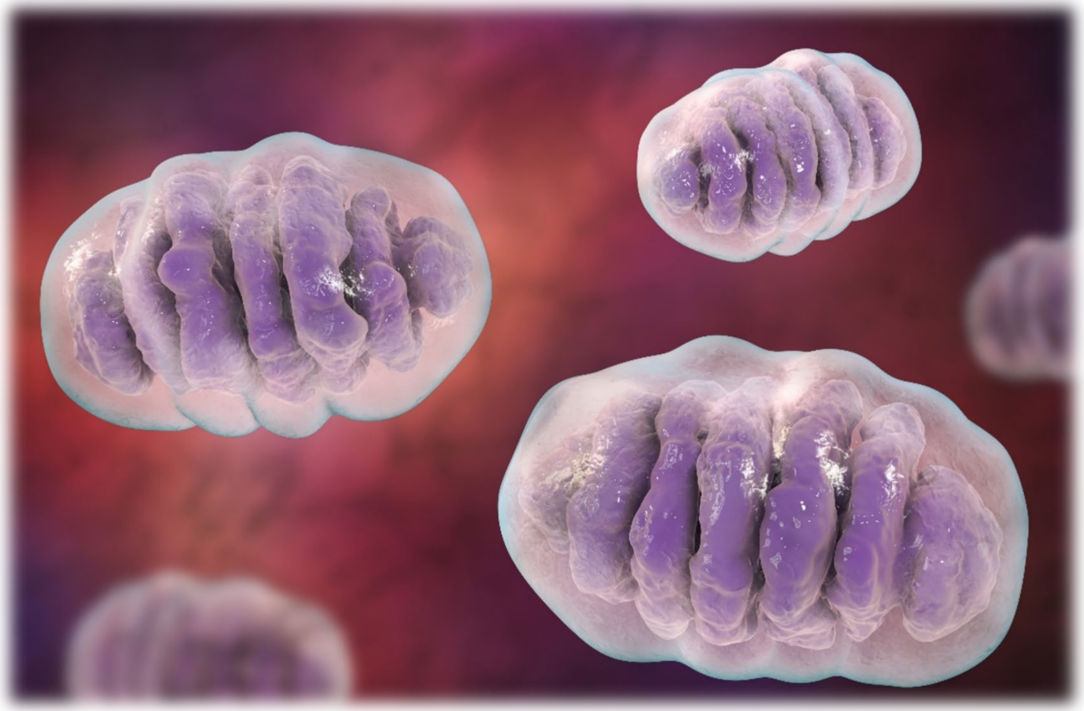
Clinical Focus

- Antecedents, triggers, and mediators of mitochondrial dysfunction.

Mitochondrial Dysfunction: Common Mediators

- Micronutrient deficiencies
- Dietary excesses
- Caloric overload
- Oxidative stress
- Glucotoxicity (glycation)
- Sedentarism
- Lipotoxicity
- Environmental toxins
- Pharmaceuticals



See: “References: Major Sources of Mitochondrial Dysfunction”

References

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Mitochondrial Dysfunction: Associated Micronutrient Deficiencies

- Magnesium
- Calcium
- Selenium
- Vitamin B12
- Vitamin K2 (MK4)
- Folate
- Thiamine
- Zinc
- Iron
- Copper
- Pantothenate (B5)
- Riboflavin (B2)
- Pyridoxine (B6)
- Biotin
- Carnitine
- CoQ10

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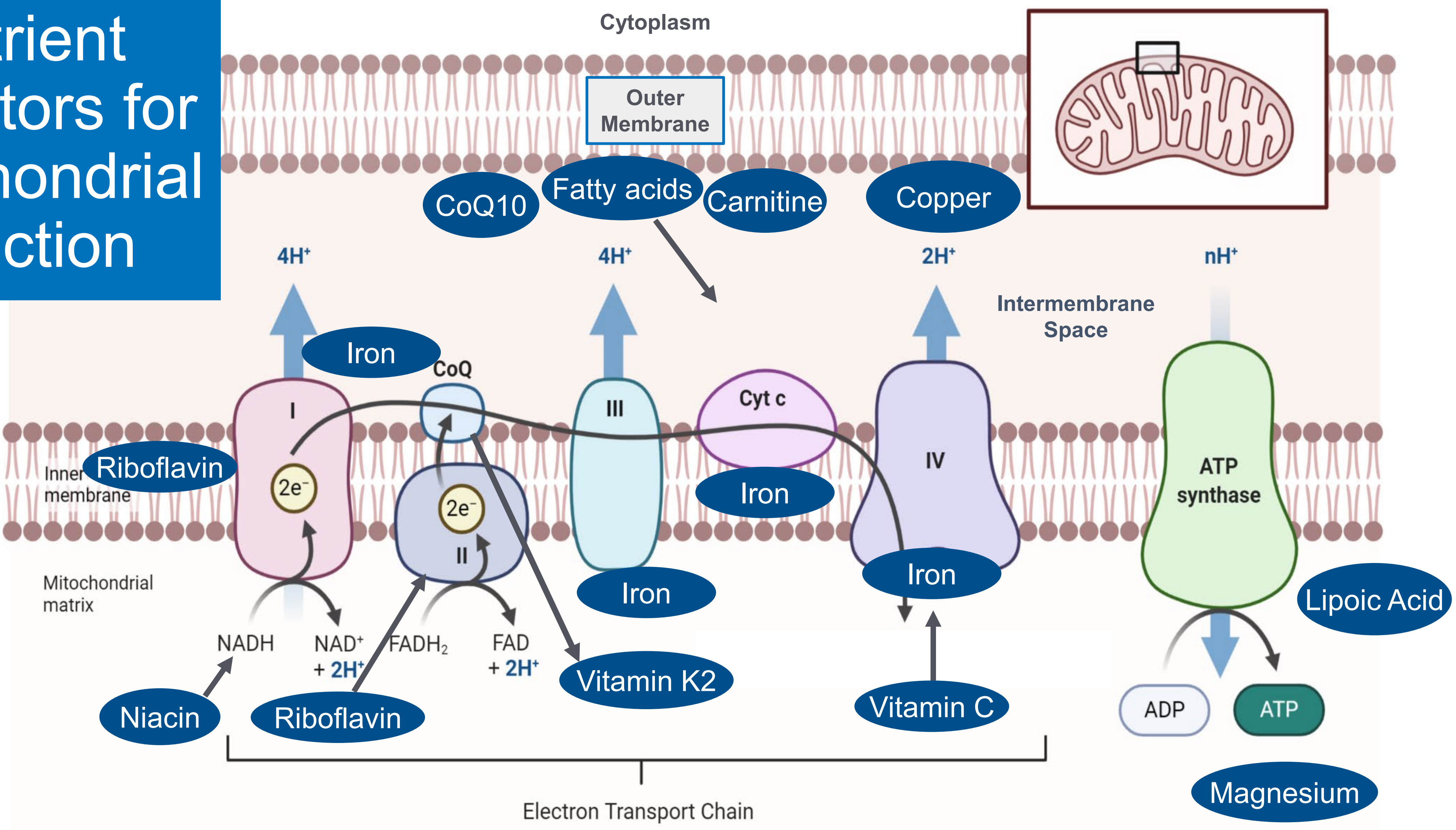
Nutrient Cofactors for Mitochondrial Function



Wesselink E, Koekkoek WAC, Grefte S, Witkamp RF, van Zanten ARH. Feeding mitochondria: potential role of nutritional components to improve critical illness convalescence. *Clin Nutr*. 2019;38(3):982-995. doi:10.1016/j.clnu.2018.08.032

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Long Description

Nutrient Cofactors for Mitochondrial Function

This diagram shows the key vitamins, minerals, and other nutrients required for proper mitochondrial energy production through the electron transport chain.

- It depicts the inner mitochondrial membrane with protein complexes 1 through 4 and A T P synthase.
- The nutrients and nutrient cofactors needed are as follows:
  - Niacin and riboflavin are needed for N A D H and F A D H_2 production.
  - Riboflavin also supports Complex 2.
  - Iron is required at multiple points in Complexes 1, 2, 3, and 4, while Coenzyme Q 10 carries electrons between Complexes 1 or 2 and 3.
  - Vitamin K 2, carnitine, and fatty acids are linked to Complex 3 function.
  - Copper and iron are needed for Complex 4.
  - Vitamin C supports iron function and electron transfer. Lipoic acid and magnesium are needed for A T P synthase, which converts A D P to A T P.

The graphic emphasizes that these micronutrients are essential cofactors for mitochondrial oxidative phosphorylation and energy generation.

Mitochondrial Dysfunction: Associated Dietary Factors

- **Caloric excess**
  - Numerous studies
- **Fructose**
- **Alcohol**
  - Cellular and mitochondrial effects of alcohol consumption
- **Trans fats**
  - Toxicological aspects of trans fat consumption over two sequential generations of rats: oxidative damage and preference for amphetamine
- **Advanced glycation end products**
  - Pathological significance of mitochondrial glycation

See: "References: Dietary Factors Associated With Mitochondrial Dysfunction"

References

DIETARY FACTORS ASSOCIATED WITH MITOCHONDRIAL DYSFUNCTION

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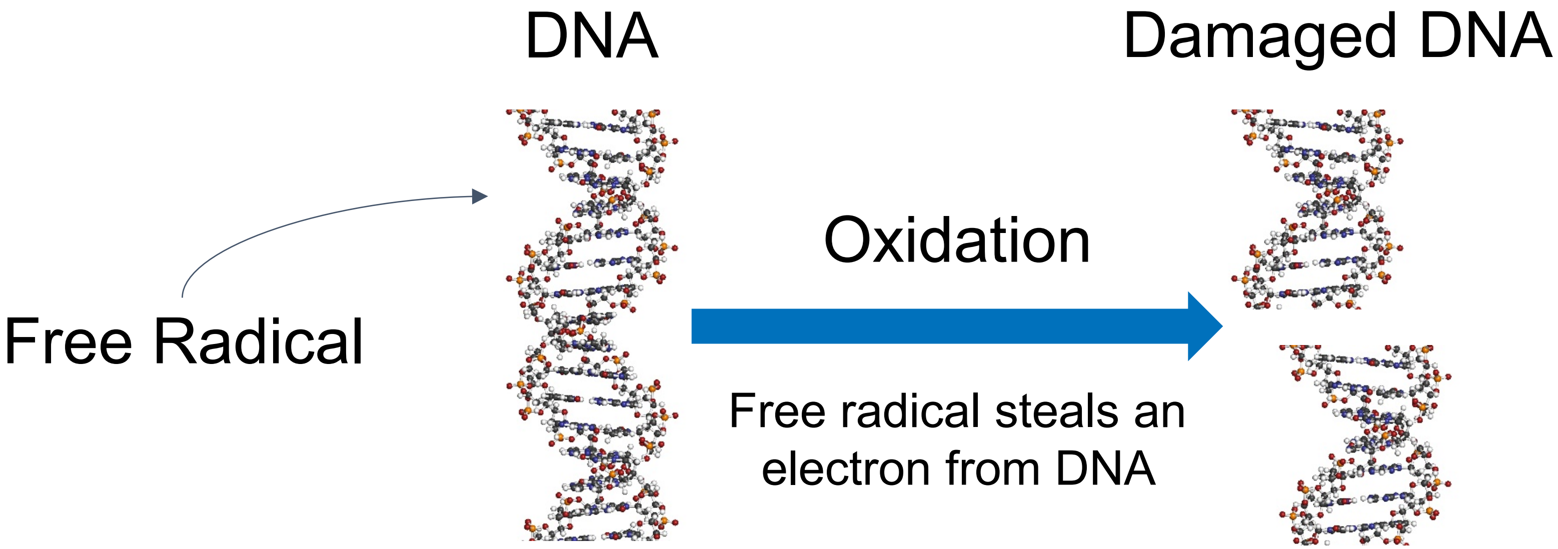
- Pun PB, Murphy MP. Pathological significance of mitochondrial glycation. *Int J Cell Biol*. 2012;2012:843505. doi:10.1155/2012/843505

Mitochondrial Oxidative Stress
Mediators of Excessive ROS

- **Dietary factors** (excess calories, fructose, alcohol)
- **Hyperglycemia** (impact on glycocalyx)
- **Inflammatory mediators** (TNF-α)
- **Hypoxia** (acute and chronic)



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- Manzo-Avalos S, Saavedra-Molina A. Cellular and mitochondrial effects of alcohol consumption. *Int J Environ Res Public Health*. 2010;7(12):4281-4304. doi:10.3390/ijerph7124281

Long Description

Mitochondrial Oxidative Stress: Mediators of Excessive R O S

The graphic illustrates how free radicals can damage D N A through oxidation.

- A **free radical** approaches a normal D N A molecule.
- During **oxidation**, the free radical steals an electron from the D N A strand.
- This results in **damaged D N A**, shown as an altered D N A structure.

Overall meaning: Oxidative stress from free radicals can directly harm D N A by electron theft, potentially leading to genetic mutations and cellular dysfunction.

Mitochondrial Oxidative Stress
Mediators of Excessive ROS

Hydroxide ion $[H-\ddot{O}:]^-$

Hydroxyl radical $H-\ddot{O}\cdot$

$$\textbf{Radiation} + H_2O \longrightarrow H_2O^+ + e^-$$

$$e^- + H_2O \longrightarrow H_2O^-$$

$$H_2O^+ \longrightarrow H^+ + OH\bullet$$

$$H_2O^- \longrightarrow H\bullet + OH^-$$

Ionizing Radiation

Long Description for
Mitochondrial Oxidative Stress - Mediators of Excessive ROS

The diagram explains how ionizing radiation generates reactive oxygen species (R O S) through water molecule breakdown, contributing to mitochondrial oxidative stress.

- Radiation interacts with water, producing positively charged water and a free electron.
- The positively charged water splits into a hydrogen ion and a hydroxyl radical
- The free electron can react with water to form negatively charged water, which then breaks into a hydrogen atom and a hydroxide ion.
- These reactive molecules are mediators of excessive R O S, leading to oxidative stress.

Mitochondrial Oxidative Stress
Mediators of Excessive ROS

- **Environmental chemical pollutants and toxicants**

- **Toxic metals and metalloids:**
  - Mercury
  - Cadmium
  - Arsenic
  - Lead

See: "References: Causes of Excessive Mitochondrial ROS"

References

CAUSES OF EXCESSIVE MITOCHONDRIAL ROS

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- Zolkipli-Cunningham Z, Falk MJ. Clinical effects of chemical exposures on mitochondrial function. *Toxicology*. 2017;391:90-99. doi:10.1016/j.tox.2017.07.009

Mitochondrial Toxins: Toxic Metals

- Iron (overload)
- Manganese (overload)
- Mercury (brain → neurotoxicity)
- Lead (brain → neurotoxicity)
- Arsenic (liver → hepatotoxicity)

- Belyaeva EA, Sokolova TV, Emelyanova LV, Zakharova IO. Mitochondrial electron transport chain in heavy metal-induced neurotoxicity: effects of cadmium, mercury, and copper. *ScientificWorldJournal*. 2012;2012:136063. doi:10.1100/2012/136063
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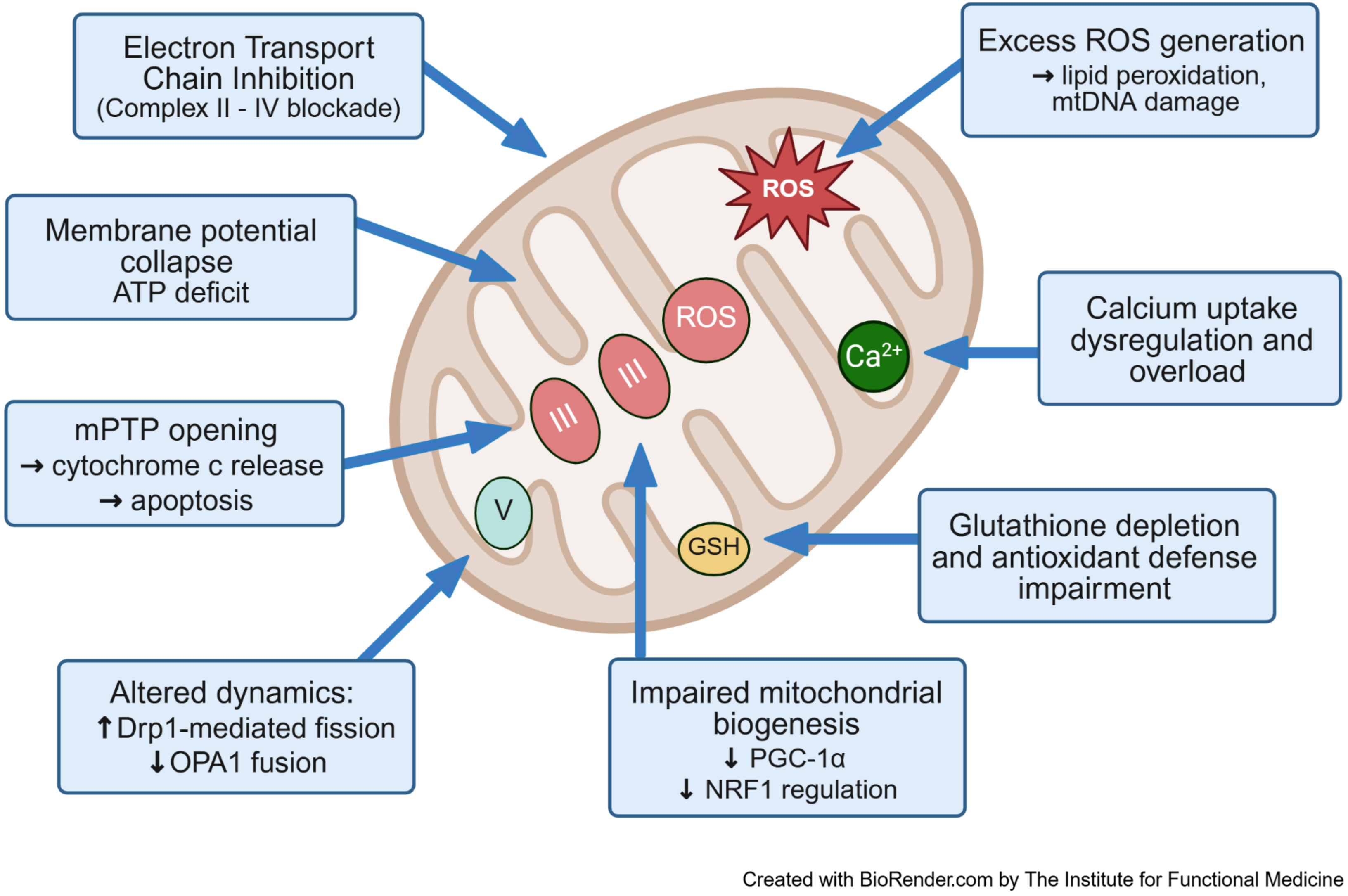
Methylmercury-Induced Mitochondrial Dysfunction



See: “References: Methylmercury-Induced Mitochondrial Dysfunction”

Long Description

Methylmercury-Induced Mitochondrial Dysfunction

The diagram explains how methylmercury causes mitochondrial dysfunction. Inside a mitochondrion, several harmful processes occur:

- **Electron transport chain inhibition** between Complex II and Complex IV, reducing A T P production.
- **Membrane potential collapse**, leading to an A T P energy deficit.
- **Opening of the mitochondrial permeability transition pore (m P T P)**, which releases cytochrome c and triggers apoptosis.
- **Altered mitochondrial dynamics**, with increased D r p 1-mediated fission and decreased O P A 1-mediated fusion.
- **Impaired mitochondrial biogenesis**, shown by reduced P G C-1 α activity and decreased N R F 1 regulation.
- **Excess reactive oxygen species (R O S) generation**, causing lipid peroxidation and mitochondrial D N A damage.
- **Calcium uptake dysregulation and overload**, disrupting mitochondrial function.
- **Glutathione (G S H) depletion and impaired antioxidant defenses**, reducing the cell's ability to neutralize oxidative stress.

Overall, methylmercury disrupts energy production, increases oxidative stress, damages mitochondrial structures, and promotes cell death pathways.

Key Pathways Illustrated

Electron-transport chain blockade (complex II → IV)

- Impairs proton pumping and downstream complex V activity.

$\Delta\Psi m$ collapse → ATP deficit

- Membrane depolarization hampers oxidative phosphorylation efficiency.

mPTP opening + cytochrome c efflux → apoptosis

- Cyclosporin-A–sensitive pore activation triggers caspase cascade.

ROS surge → lipid peroxidation & mtDNA lesions

- Superoxide and downstream radicals amplify damage.

Ca^{2+} dys-homeostasis

- Me/Hg perturbs uniporter fluxes, fostering mitochondrial Ca^{2+} overload.

GSH depletion & antioxidant exhaustion

- Seleno-/thiol binding decreases redox buffering capacity.

Suppressed biogenesis (↓PGC-1α/NRF1)

- Transcriptional silencing limits renewal of the organelle pool.

Fusion-fission imbalance (↑Drp1, ↓OPA1)

- Fragmented network exacerbates functional decline.

See: "References: Methylmercury-Induced Mitochondrial Dysfunction"

References

METHYLMERCURY-INDUCED MITOCHONDRIAL DYSFUNCTION

Pathway 1 (Animal Study):

- Tiernan CT, Edwin EA, Hawong HY, et al. Methylmercury impairs canonical dopamine metabolism in rat undifferentiated pheochromocytoma (PC12) cells by indirect inhibition of aldehyde dehydrogenase. *Toxicol Sci*. 2015;144(2):347-356. doi:10.1093/toxsci/kfv001

Pathways 2 and 7:

- Wang X, Yan M, Zhao L, et al. Low-dose methylmercury-induced apoptosis and mitochondrial DNA mutation in human embryonic neural progenitor cells. *Oxid Med Cell Longev*. 2016;2016:5137042. doi:10.1155/2016/5137042

Pathway 3 (Animal Study):

- Bragadin M, Marton D, Manente S, Grasso M, Toninello A. Methylmercury induces the opening of the permeability transition pore in rat liver mitochondria. *J Inorg Biochem*. 2002;89(1-2):159-162. doi:10.1016/s0162-0134(01)00366-x

Pathways 4 and 6:

- Farina M, Aschner M, Rocha JB. Oxidative stress in MeHg-induced neurotoxicity. *Toxicol Appl Pharmacol*. 2011;256(3):405-417. doi:10.1016/j.taap.2011.05.001

Pathway 5:

- Polunas M, Halladay A, Tjalkens RB, Philbert MA, Lowndes H, Reuhl K. Role of oxidative stress and the mitochondrial permeability transition in methylmercury cytotoxicity. *Neurotoxicology.* 2011;32(5):526-534. doi:10.1016/j.neuro.2011.07.006

Pathway 8:

- Li X, Pan J, Wei Y, et al. Mechanisms of oxidative stress in methylmercury-induced neurodevelopmental toxicity. *Neurotoxicology*. 2021;85:33-46. doi:10.1016/j.neuro.2021.05.002

Mitochondrial Toxins: Persistent Organic Pollutants (POPs)

- Epidemiologic & experimental studies have associated insulin resistance or type 2 diabetes mellitus with elevated body burdens of persistent organic pollutants, which can damage mtDNA.

- **Mitochondrial DNA abnormalities are known to cause pancreatic beta cell damage, insulin resistance, and diabetes mellitus.**

- Common POPs include:
  - Organochlorines (herbicides, pesticides)
  - Polychlorinated biphenyls (PCBs)
  - Hexachlorobenzene (HCB)
  - Polychlorinated dibenzo-p-dioxins (PCDDs)
  - Polychlorinated dibenzofurans (PCDFs)

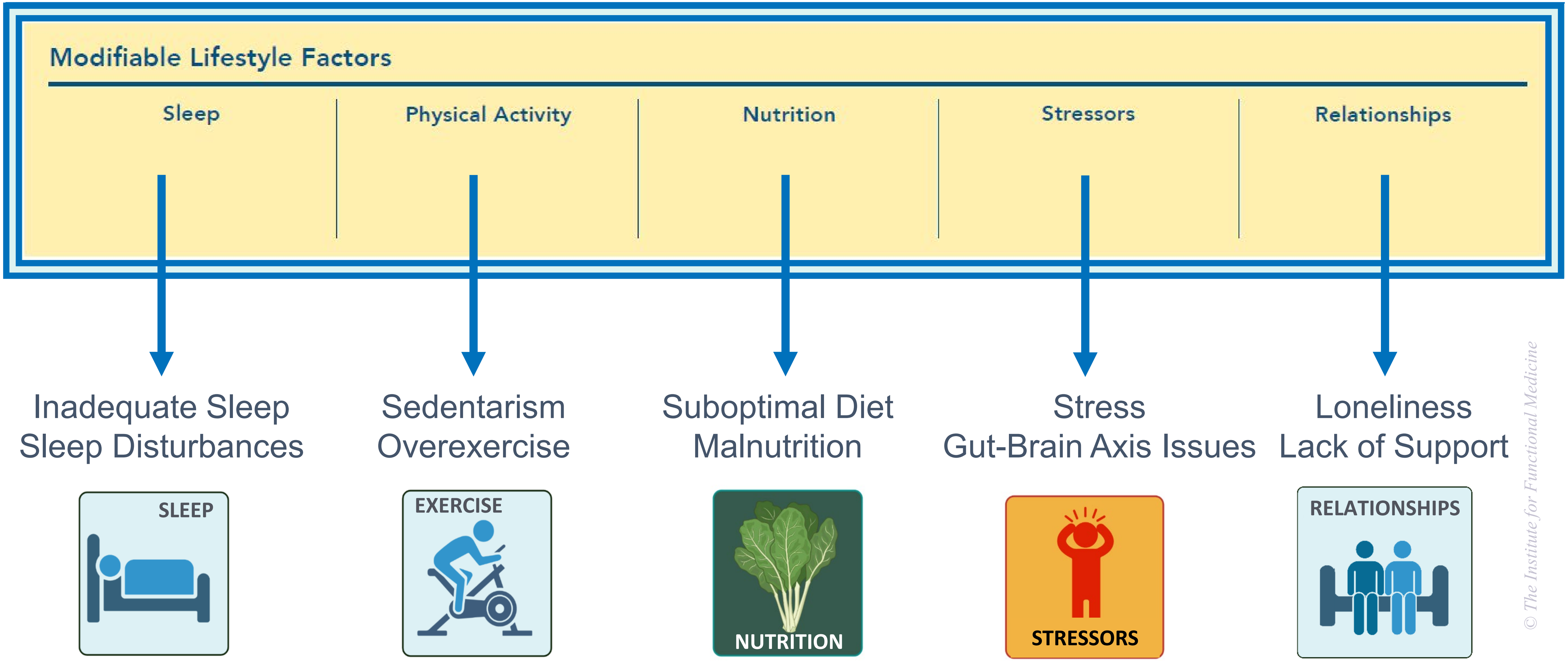
Lim S, Cho YM, Park KS, Lee HK. Persistent organic pollutants, mitochondrial dysfunction, and metabolic syndrome. *Ann N Y Acad Sci*. 2010;1201:166-176. doi:10.1111/j.1749-6632.2010.05622.x

Mitochondrial Toxins: Pharmaceuticals

- **Acetaminophen**: Irreversibly inhibits B-oxidation
- **Aminoglycoside antibiotics**
- **Anti-retroviral drugs**
- **Aspirin**: Inhibits and uncouples oxidative phosphorylation
- **Statins**: Multiple mechanisms
- **Cancer chemotherapy agents**: Platinum compounds
- **Metformin**: Complex I inhibitor
- **Tamoxifen**: Inhibits complexes III and IV
- **Valproic acid**: Inhibits complex IV

- Parikh S, Saneto R, Falk MJ, et al. A modern approach to the treatment of mitochondrial disease. *Curr Treat Options Neurol*. 2009;11(6):414-430. doi:10.1007/s11940-009-0046-0
- Begriche K, Massart J, Robin MA, Borgne-Sanchez A, Fromenty B. Drug-induced toxicity on mitochondria and lipid metabolism: mechanistic diversity and deleterious consequences for the liver. *J Hepatol*. 2011;54(4):773-794. doi:10.1016/j.jhep.2010.11.006

Mediators of Mitochondrial Dysfunction



See: "References: Mediators of Mitochondrial Dysfunction"

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References

MEDIATORS OF MITOCHONDRIAL DYSFUNCTION

Sleep & Relaxation:

- Haynes PR, Pyfrom ES, Li Y, et al. A neuron-glia lipid metabolic cycle couples daily sleep to mitochondrial homeostasis. *Nat Neurosci.* 2024;27(4):666-678. doi:10.1038/s41593-023-01568-1
- Richardson RB, Mailloux RJ. Mitochondria need their sleep: redox, bioenergetics, and temperature regulation of circadian rhythms and the role of cysteine-mediated redox signaling, uncoupling proteins, and substrate cycles. *Antioxidants (Basel*). 2023;12(3):674. doi:10.3390/antiox12030674

Exercise & Movement:

- Ostojic SM, Ratgeber L, Olah A, Betlehem J, Pongras A. What do over-trained athletes and patients with neurodegenerative diseases have in common? Mitochondrial dysfunction. *Exp Biol Med (Maywood)*. 2021;246(11):1241-1243. doi:10.1177/1535370221990619
- Raffin J, de Souto Barreto P, Le Traon AP, Vellas B, Aubertin-Leheudre M, Rolland Y. Sedentary behavior and the biological hallmarks of aging. *Ageing Res Rev*. 2023;83:101807. doi:10.1016/j.arr.2022.101807

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MEDIATORS OF MITOCHONDRIAL DYSFUNCTION

Nutrition:

- Bomer N, Pavez-Giani MG, Grote Beverborg N, Cleland JGF, van Veldhuisen DJ, van der Meer P. Micronutrient deficiencies in heart failure: mitochondrial dysfunction as a common pathophysiological mechanism? *J Intern Med*. 2022;291(6):713-731. doi:10.1111/joim.13456
- Engin A. The unrestrained overeating behavior and clinical perspective. *Adv Exp Med Biol*. 2024;1460:167-198. doi:10.1007/978-3-031-63657-8_6
- Izadi M, Sadri N, Abdi A, et al. Harnessing the fundamental roles of vitamins: the potent anti-oxidants in longevity. *Biogerontology*. 2025;26(2):58. doi:10.1007/s10522-025-10202-5

Stress:

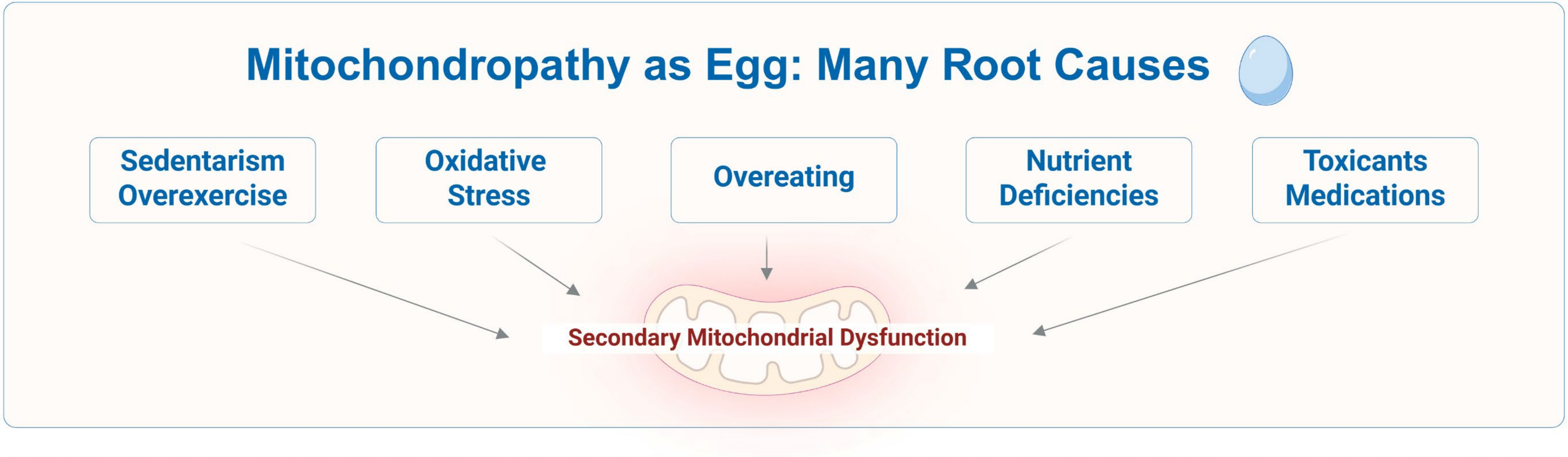
- Daniels TE, Olsen EM, Tyrka AR. Stress and psychiatric disorders: the role of mitochondria. *Annu Rev Clin Psychol*. 2020;16:165-186. doi:10.1146/annurev-clinpsy-082719-104030
- Lin J, Epel E. Stress and telomere shortening: insights from cellular mechanisms. *Ageing Res Rev*. 2022;73:101507. doi:10.1016/j.arr.2021.101507
- Picard M, McEwen BS. Psychological stress and mitochondria: a systematic review. *Psychosom Med*. 2018;80(2):141-153. doi:10.1097/PSY.0000000000000545

References

MEDIATORS OF MITOCHONDRIAL DYSFUNCTION

Relationships:

- Ahmed M, Cerda I, Maloof M. Breaking the vicious cycle: the interplay between loneliness, metabolic illness, and mental health. *Front Psychiatry*. 2023;14:1134865. doi:10.3389/fpsyt.2023.1134865
- Picard M, McEwen BS. Psychological stress and mitochondria: a conceptual framework. *Psychosom Med.* 2018;80(2):126-140. doi:10.1097/PSY.0000000000000544



See: “References: Mitochondropathy: Many Root Causes and Manifestations”

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Sedentarism/Overexercise:

- Ostojic SM, Ratgeber L, Olah A, Betlehem J, Pongras A. What do over-trained athletes and patients with neurodegenerative diseases have in common? Mitochondrial dysfunction. *Exp Biol Med (Maywood)*. 2021;246(11):1241-1243. doi:10.1177/1535370221990619
- Raffin J, de Souto Barreto P, Le Traon AP, Vellas B, Aubertin-Leheudre M, Rolland Y. Sedentary behavior and the biological hallmarks of aging. *Ageing Res Rev*. 2023;83:101807. doi:10.1016/j.arr.2022.101807

Oxidative Stress:

- Kowalczyk P, Krych S, Kramkowski K, Jęczmyk A, Hrapkowicz T. Effect of oxidative stress on mitochondrial damage and repair in heart disease and ischemic events. *Int J Mol Sci*. 2024;25(22):12467. doi:10.3390/ijms252212467

Overeating:

- Engin A. The unrestrained overeating behavior and clinical perspective. *Adv Exp Med Biol*. 2024;1460:167-198. doi:10.1007/978-3-031-63657-8_6

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Nutrient Deficiencies:

- Bomer N, Pavez-Giani MG, Grote Beverborg N, Cleland JGF, van Veldhuisen DJ, van der Meer P. Micronutrient deficiencies in heart failure: mitochondrial dysfunction as a common pathophysiological mechanism? *J Intern Med*. 2022;291(6):713-731. doi:10.1111/joim.13456
- Izadi M, Sadri N, Abdi A, et al. Harnessing the fundamental roles of vitamins: the potent anti-oxidants in longevity. *Biogerontology*. 2025;26(2):58. doi:10.1007/s10522-025-10202-5

Toxicants/Medications:

- Duarte-Hospital C, Tête A, Brial F, et al. Mitochondrial dysfunction as a hallmark of environmental injury. *Cells*. 2021;11(1):110. doi:10.3390/cells11010110
- Ryu DY, Pang WK, Adegoke EO, Rahman MS, Park YJ, Pang MG. Bisphenol-A disturbs hormonal levels and testis mitochondrial activity, reducing male fertility. *Hum Reprod Open*. 2023;2023(4):hoad044. doi:10.1093/hropen/hoad044

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

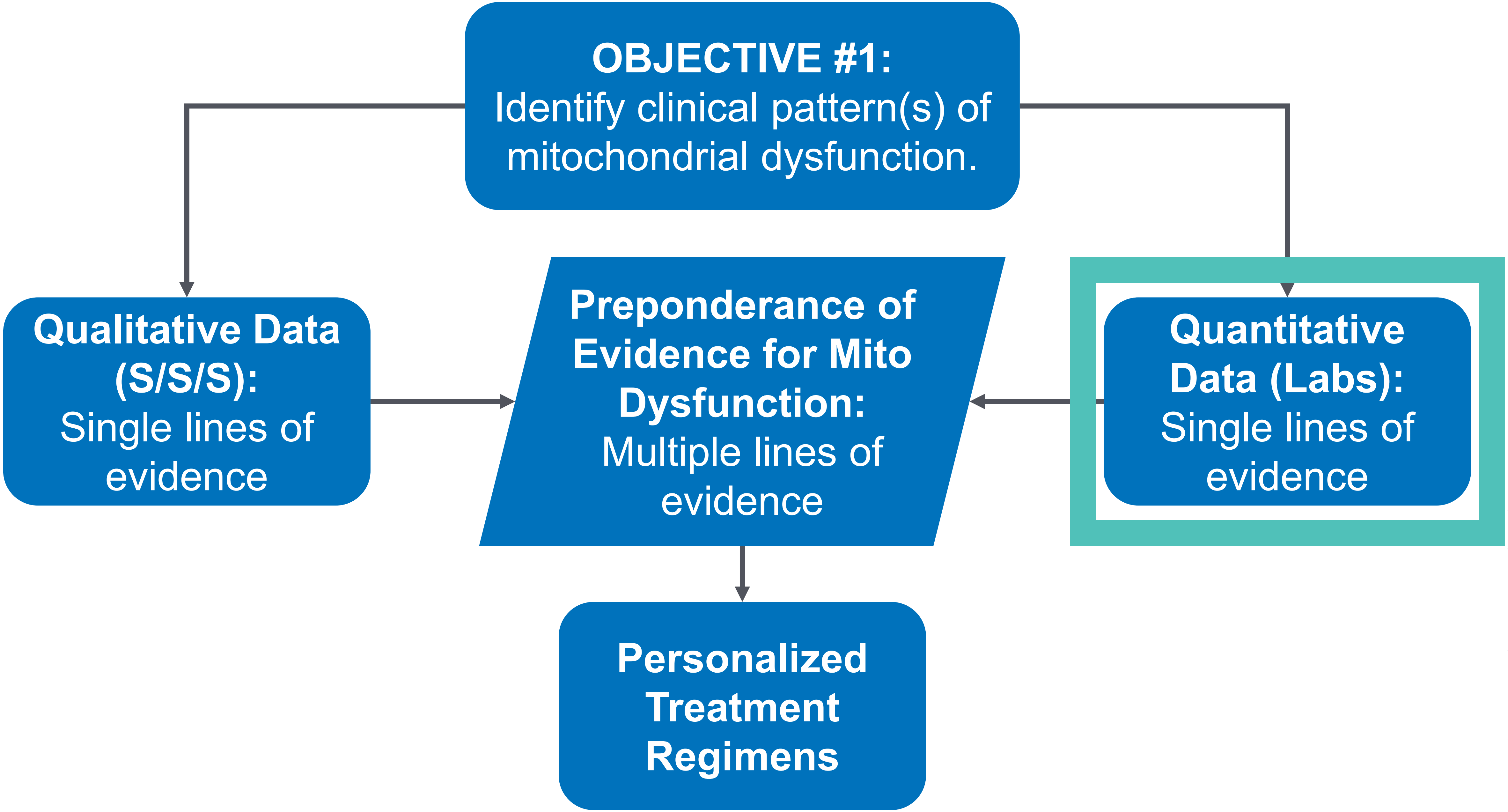
Bioenergetics and Mitochondropathy: Part 3

Mechanisms – Mediators – Manifestations

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Laboratory evaluation of mitochondrial dysfunction.
- Biomarkers of susceptibility.
- Biomarkers of effect.


OBJECTIVE #1:
Identify clinical pattern(s) of mitochondrial dysfunction.
Qualitative Data (S/S/S):
Single lines of evidence
Preponderance of Evidence for Mito Dysfunction:
Multiple lines of evidence
Quantitative Data (Labs):
Single lines of evidence
Personalized Treatment Regimens

Long Description for
Objective number 1: Identify clinical patterns of mitochondrial dysfunction

This graphic illustrates a process for identifying mitochondrial dysfunction and guiding treatment.

- **Objective #1:** Identify clinical patterns of mitochondrial dysfunction.
- This involves gathering two types of single-line evidence:
    - **Qualitative data (S/S/S)** – likely referring to symptoms, signs, and syndromes from the patient.
    - **Quantitative data (labs)** – objective measurements from laboratory tests.
- When these single lines of evidence are combined, they create a **preponderance of evidence for mitochondrial dysfunction** through multiple lines of evidence.
- This comprehensive evidence then informs **personalized treatment regimens** tailored to the patient.
- On this slide, **Quantitative Data** is highlighted

The overall meaning of this graphic is that both qualitative and quantitative evidence must be integrated to confirm mitochondrial dysfunction and guide individualized therapy.

There are no diagnostic laboratory tests, BUT…

There are lines of evidence, including laboratory test results, that can support your clinical suspicion of mitochondropathy and warrant therapeutic interventions.

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (CBC & Iron Status)	Biomarker of Susceptibility	Biomarker of Effect	Comment
Hemoglobin (Low)	✓		May affect oxygen availability for oxidative phosphorylation
WBC (Low)		✓	Line of evidence: high oxidative stress states
Neutrophils (Low)		✓	Line of evidence: high oxidative stress states
Platelets (Low)		✓	Line of evidence: high oxidative stress states
Ferritin (Low)	✓		Iron (Fe) is a cofactor in the mitochondrial electron transport chain

See: “References: Laboratory Tests”

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (CBC & Iron Status)	Biomarker of Susceptibility	Biomarker of Effect	Comment

See: “References: Laboratory Tests”

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (Basic Chemistries)	Biomarker of Susceptibility	Biomarker of Effect	Comment
Carbon Dioxide (CO_2) (High)	✓	✓	Line of evidence for acidosis; complex associations with mito function
Glucose (HbA1c) (High)		✓	Mito dysfunction impairs insulin signaling and mediates insulin resistance
Albumin (Low)	✓		Albumin is the most abundant antioxidant in the blood
Total Protein (Low)		✓	Line of evidence for impaired mitochondrial biosynthetic function
Creatinine (High)		✓	Line of evidence for kidney disease mediated by mitochondrial dysfunction

See: “References: Laboratory Tests”

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (Basic Chemistries)	Biomarker of Susceptibility	Biomarker of Effect	Comment
Carbon Dioxide (CO_2) (High)			Line of evidence for acidosis; complex associations with mito function
Glucose (HbA1c) (High)			Mito dysfunction impairs insulin signaling and mediates insulin resistance
Albumin (Low)			Albumin is the most abundant antioxidant in the blood
Total Protein (Low)			Line of evidence for impaired mitochondrial biosynthetic function
Creatinine (High)			Line of evidence for kidney disease mediated by mitochondrial dysfunction

See: “References: Laboratory Tests”

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (Basic Chemistries)	Biomarker of Susceptibility	Biomarker of Effect	Comment
ALT and AST (High)		✓	Enzymes catalyzing alanine and aspartate entry into TCA cycle; damage if in blood
GGT (High)		✓	Indicator of low glutathione and increased oxidative stress
RBC Magnesium (Low)	✓		Cofactor in mitochondrial electron transport chain
B Vitamins (Low)	✓		Cofactor in mitochondrial electron transport chain
RBC Copper (Low)	✓		Cofactor in mitochondrial electron transport chain

See: "References: Laboratory Tests"

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (Basic Chemistries)	Biomarker of Susceptibility	Biomarker of Effect	Comment
ALT and AST (High)			Enzymes catalyzing alanine and aspartate entry into TCA cycle; damage if in blood
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RBC Magnesium (Low)			Cofactor in mitochondrial electron transport chain
B Vitamins (Low)			Cofactor in mitochondrial electron transport chain
RBC Copper (Low)			Cofactor in mitochondrial electron transport chain

See: "References: Laboratory Tests"

References

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Hemoglobin:

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Ferritin:

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Glucose (HbA1c):

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References

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RBC Magnesium, RBC Copper, B Vitamins:

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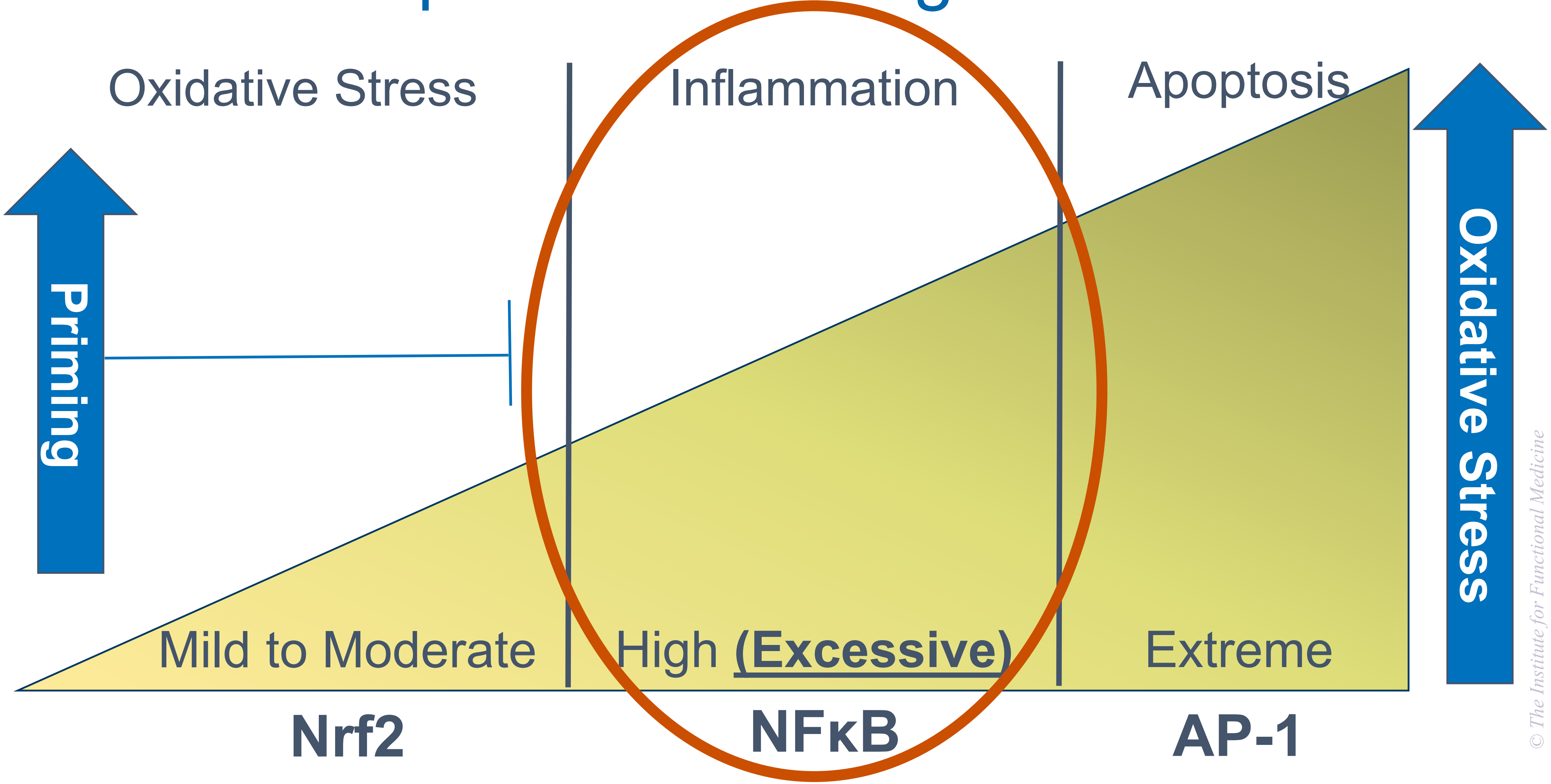
Clinical Focus

- Pathophysiology of mitochondrial dysfunction.

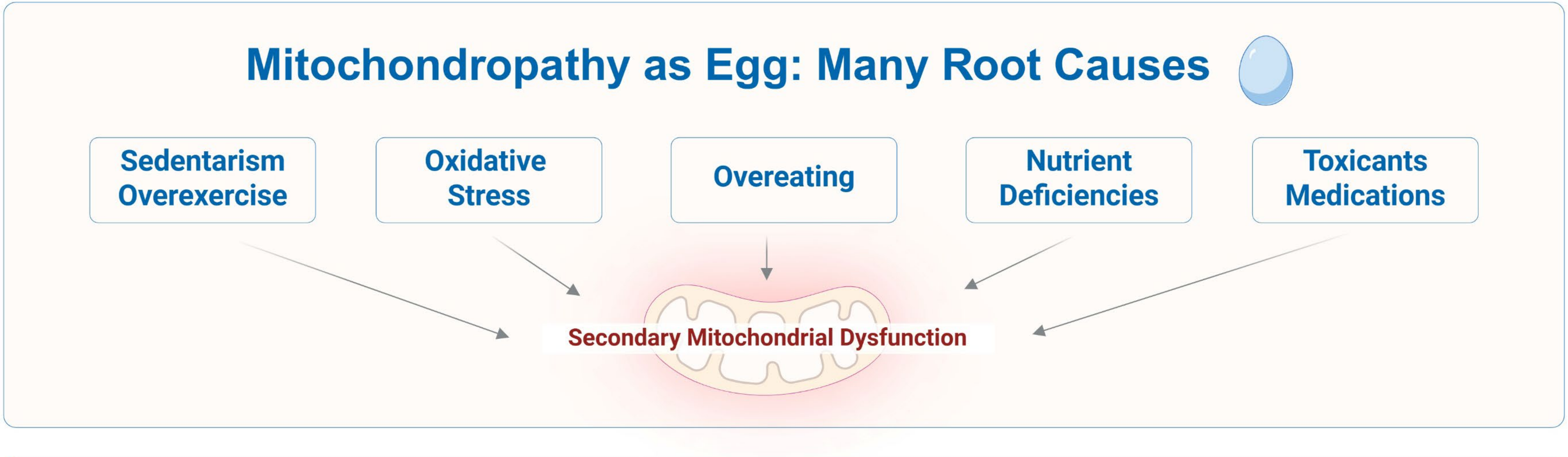
Differential Responses to Rising Oxidative Stress



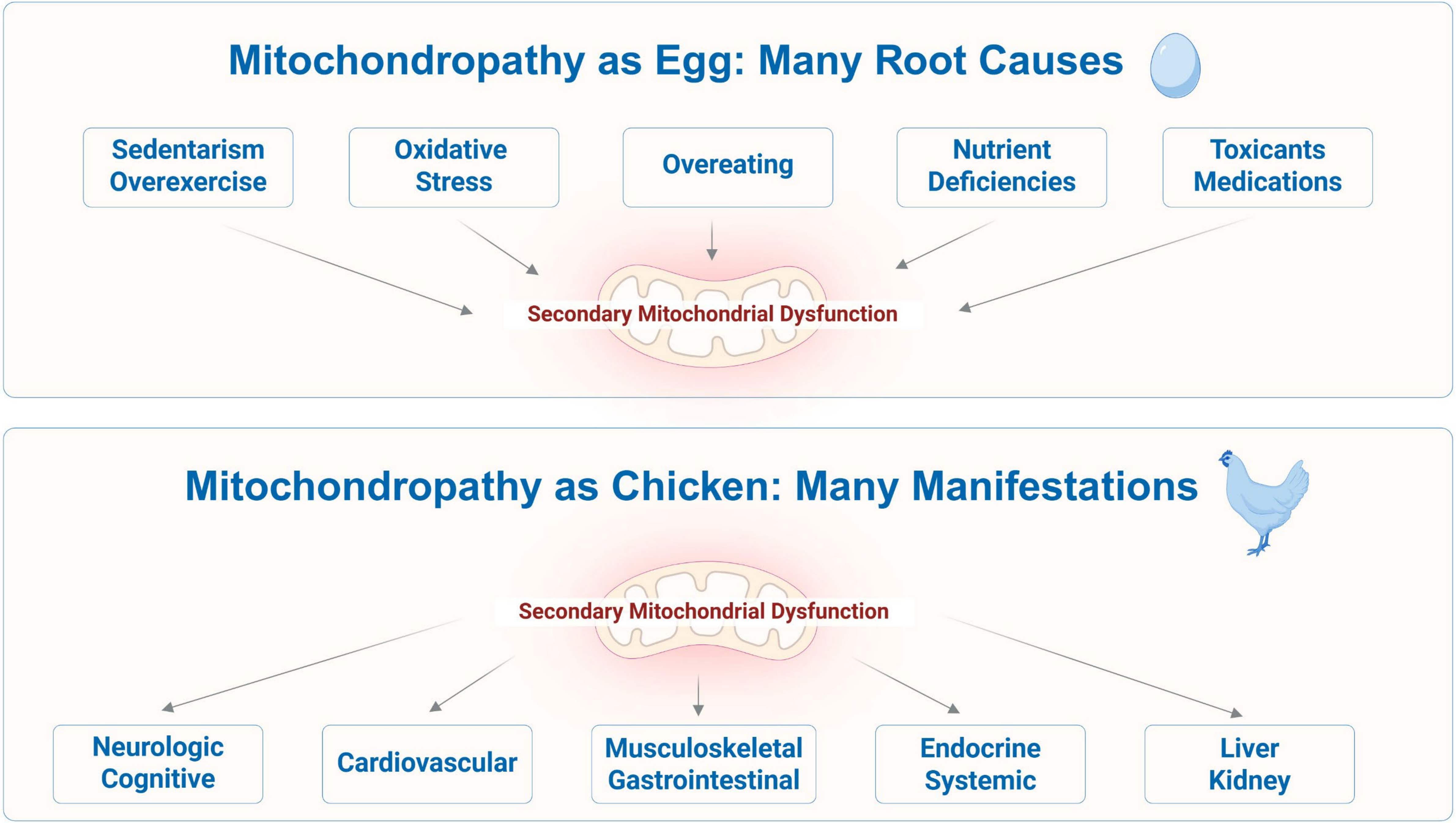
Adapted from: Stefanson AL, Bakovic M. Dietary regulation of Keap1/Nrf2/ARE pathway: focus on plant-derived compounds and trace minerals. *Nutrients*. 2014;6(9):3777-3801. doi:10.3390/nu6093777

Long Description for
Differential Responses to Rising Oxidative Stress

- This graphic illustrates how the body responds differently to increasing levels of oxidative stress.
- At mild to moderate oxidative stress, the N r f 2 pathway is activated, which supports protective and adaptive responses, a stage referred to as “priming.”
- When oxidative stress becomes high or excessive, the N F κ B pathway is triggered, leading to inflammation.
- At extreme oxidative stress levels, the AP-1 pathway is activated, resulting in apoptosis, or programmed cell death.
- The diagram shows oxidative stress increasing from left to right, with priming occurring at lower levels, inflammation at higher levels, and apoptosis at the highest levels.
- The inflammation section in the center is emphasized



See: “References: Mitochondropathy: Many Root Causes and Manifestations”



See: “References: Mitochondropathy: Many Root Causes and Manifestations”

Long Description –
Mitochondropathy: Many Root Causes and Manifestations

This graphic uses an "egg and chicken" metaphor to explain the two-way relationship between secondary mitochondrial dysfunction and chronic disease. It is divided into two panels:

Top Panel: "Mitochondropathy as Egg: Many Root Causes“

- This panel explains that secondary mitochondrial dysfunction can be caused by various lifestyle and environmental factors. These root causes include:
- Sedentary lifestyle or overexercise
- Oxidative stress
- Overeating
- Nutrient deficiencies
- Exposure to toxicants or medications
- These factors lead to **secondary mitochondrial dysfunction**, depicted as the “egg” in the metaphor.

Bottom Panel: "Mitochondropathy as Chicken: Many Manifestations"

- This panel illustrates that secondary mitochondrial dysfunction can also be a contributor to a wide range of health issues, referred to as the “chicken” in the metaphor. The potential disease manifestations include:
- Neurologic and cognitive symptoms
- Cardiovascular disease
- Musculoskeletal and gastrointestinal issues
- Endocrine and systemic dysfunction
- Liver and kidney disorders

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Sedentarism/Overexercise:

- Ostojic SM, Ratgeber L, Olah A, Betlehem J, Pongras A. What do over-trained athletes and patients with neurodegenerative diseases have in common? Mitochondrial dysfunction. *Exp Biol Med (Maywood)*. 2021;246(11):1241-1243. doi:10.1177/1535370221990619
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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Neurologic/Cognitive:

- Monzio Compagnoni G, Di Fonzo A, Corti S, Comi GP, Bresolin N, Masliah E. The role of mitochondria in neurodegenerative diseases: the lesson from Alzheimer's disease and Parkinson's disease. *Mol Neurobiol*. 2020;57(7):2959-2980. doi:10.1007/s12035-020-01926-1
- Qiao L, Yang G, Wang P, Xu C. The potential role of mitochondria in the microbiota-gut-brain axis: implications for brain health. *Pharmacol Res*. 2024;209:107434. doi:10.1016/j.phrs.2024.107434

Cardiovascular:

- Rocca C, Soda T, De Francesco EM, et al. Mitochondrial dysfunction at the crossroad of cardiovascular diseases and cancer. *J Transl Med*. 2023;21(1):635. doi:10.1186/s12967-023-04498-5
- Wang X, Yu Q, Liao X, et al. Mitochondrial dysfunction in arrhythmia and cardiac hypertrophy. *Rev Cardiovasc Med*. 2023;24(12):364. doi:10.31083/j.rcm2412364

Musculoskeletal:

- Macchi C, Giachi A, Fichtner I, et al. Mitochondrial function in patients affected with fibromyalgia syndrome is impaired and correlates with disease severity. *Sci Rep*. 2024;14(1):30247. doi:10.1038/s41598-024-81298-x
- Qiu B, Xie X, Xi Y. Mitochondrial quality control: the real dawn of intervertebral disc degeneration? *J Transl Med*. 2024;22(1):1126. doi:10.1186/s12967-024-05943-9

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MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Gastrointestinal:

- Xiao Y, Liu X, Xie K, et al. Mitochondrial dysfunction induced by HIF-1α under hypoxia contributes to the development of gastric mucosal lesions. *Clin Transl Med*. 2024;14(4):e1653. doi:10.1002/ctm2.1653
- Zhang YF, Fan MY, Bai QR, et al. Precision therapy for ulcerative colitis: insights from mitochondrial dysfunction interacting with the immune microenvironment. *Front Immunol*. 2024;15:1396221. doi:10.3389/fimmu.2024.1396221

Endocrine:

- Krako Jakovljevic N, Pavlovic K, Jotic A, et al. Targeting mitochondria in diabetes. *Int J Mol Sci*. 2021;22(12):6642. doi:10.3390/ijms22126642
- Siemers KM, Klein AK, Baack ML. Mitochondrial dysfunction in PCOS: insights into reproductive organ pathophysiology. *Int J Mol Sci*. 2023;24(17):13123. doi:10.3390/ijms241713123

Liver/Kidney:

- Diep TN, Liu H, Wang Y, Wang Y, Hoogewijs D, Yan LJ. Redox imbalance and mitochondrial abnormalities in kidney disease—volume II. *Biomolecules*. 2024;14(8):973. doi:10.3390/biom14080973
- Zhang R, Yan Z, Zhong H, et al. Gut microbial metabolites in MASLD: implications of mitochondrial dysfunction in the pathogenesis and treatment. *Hepatol Commun*. 2024;8(7):e0484. doi:10.1097/HC9.0000000000000484

Excessive Mitochondrial ROS

- H_2O_2 can react with iron (from O_2- damaged TCA cycle enzymes) to form hydroxyl radical OH- – short-lived but highly toxic
- Superoxide interacts with NO (from mtNOS) to form peroxynitrite (ONOO-): extremely toxic free radical – damages protein
- **Hydroxyl & peroxynitrite radicals damage mitochondrial structures, especially the ETC, leading to inefficient energy production and more oxidative stress (feed forward cycle)**

See: "References: Excessive ROS and Mitochondrial Dysfunction"

Excessive Oxidative Stress (ROS) Can Cause Mitochondrial Dysfunction

Oxidative stress from free radicals, ROS, and RNS affects many cellular components:

- Damaged Fats
- Damaged Sugars
- Damaged Proteins
- Damaged DNA

Accumulation of the end products of oxidative damage:

- Advanced Glycosylated End Products (AGEs)
- Protein Oxidation (NitroTyrosine)
- Oxidized LDL, Isoprostane F2, Lipid Peroxides, MDA
- DNA Damage (8-OH dG)

See: "References: Excessive ROS and Mitochondrial Dysfunction"

References

EXCESSIVE ROS AND MITOCHONDRIAL DYSFUNCTION

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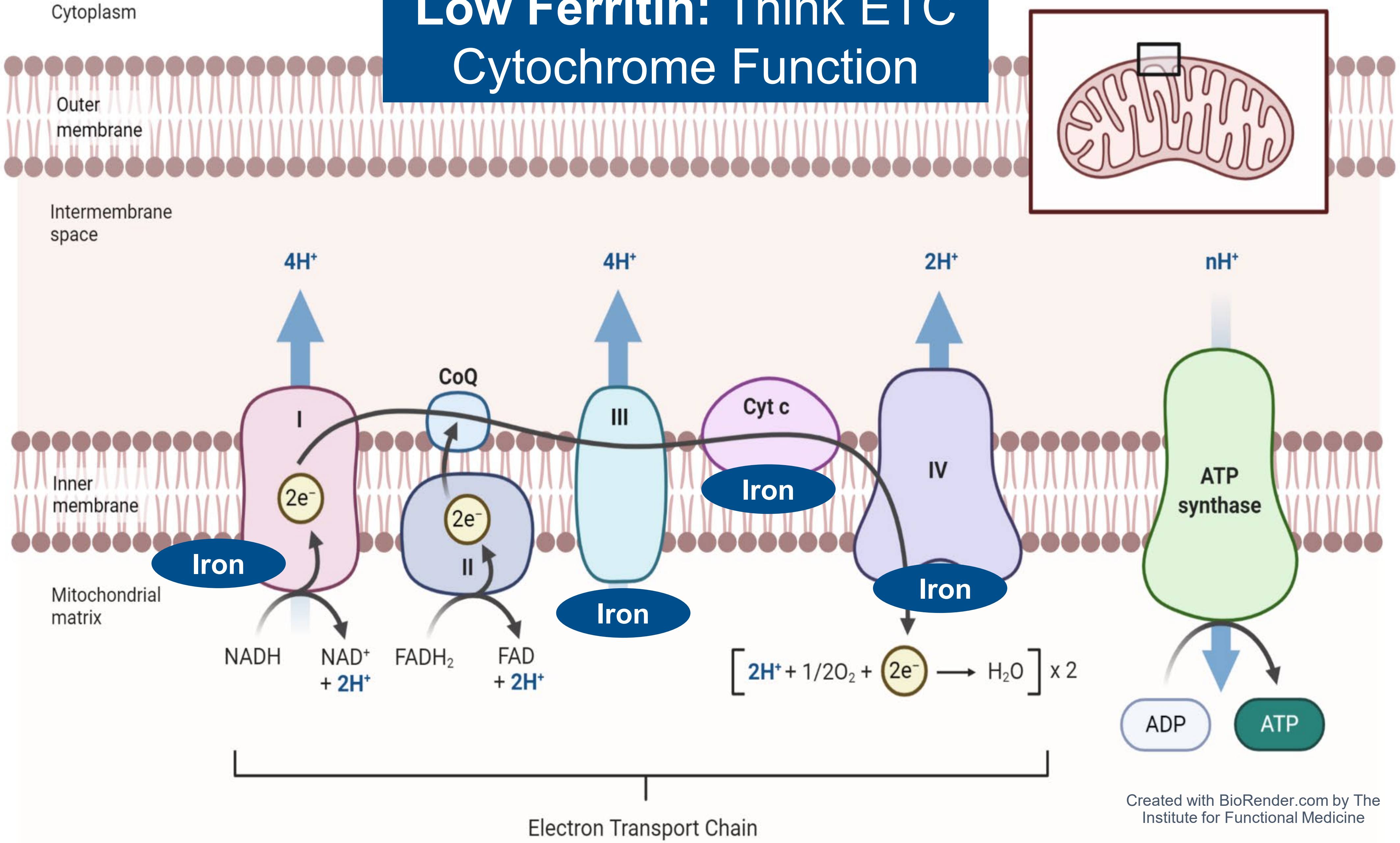
Low Ferritin: Think ETC Cytochrome Function



Created with BioRender.com by The Institute for Functional Medicine

Wesselink E, Koekkoek WAC, Grefte S, Witkamp RF, van Zanten ARH. Feeding mitochondria: potential role of nutritional components to improve critical illness convalescence. *Clin Nutr*. 2019;38(3):982-995. doi:10.1016/j.clnu.2018.08.032

Two Deadly Mitochondrial Poisons

- **Cyanide**
- **Carbon dioxide**

- First identified as mitochondrial toxins in 1940s
- Block mitochondrial energy production by displacing oxygen from heme (hemoglobin, cytochrome c)

Both are found in cigarette smoke!

Mitochondrial Toxins: Environmental Chemicals

- **MPP+ ($C_{12}H_{12}N^+$)**: binds & inhibits complex I (MPTP metabolite via monoamine oxidase)
- **Rotenone**: blocks ubiquinone binding to complex I
- **Pyridaben**: inhibits complex I
- **Paraquat**: increases ROS from complex I
- **Maneb (Mn containing fungicide)**: complex III inhibitor

See: "References: Mitochondrial Toxins"

Mitochondrial Toxins: Environmental Chemicals

- **2,4 Dinitrophenol**: used for weight loss, uncouples oxidative phosphorylation – potentially fatal
- **Atrazine**: inhibits complexes I & II
- **Organochlorines (e.g., dioxin)**: disrupts signaling
- **Bisphenol A**: inhibits complex II
- **Organophosphates**: neurotoxins

See: "References: Mitochondrial Toxins"

Mitochondrial Toxins: Pharmaceuticals

- **Acetaminophen**: irreversibly inhibits β-oxidation
- **Aminoglycoside antibiotics**
- **Anti-retroviral drugs**
- **Aspirin**: inhibits & uncouples oxidative phosphorylation
- **Statins**: Multiple mechanisms

See: "References: Mitochondrial Toxins"

Mitochondrial Toxins: Pharmaceuticals

- **Cancer chemotherapy agents (platinum compounds)**
- **Metformin**: complex I inhibitor
- **Tamoxifen**: inhibits complexes III & IV
- **Valproic acid**: inhibits complex IV

See: "References: Mitochondrial Toxins"

Metformin:
Mechanisms of Dysfunction

Metformin mildly inhibits complex I. This increases ratio of AMP to ATP, which activates AMPK (energy sensor), which inhibits hepatic gluconeogenesis, lowering blood sugar.

See: “References: Mitochondrial Toxins”

References

MITOCHONDRIAL TOXINS

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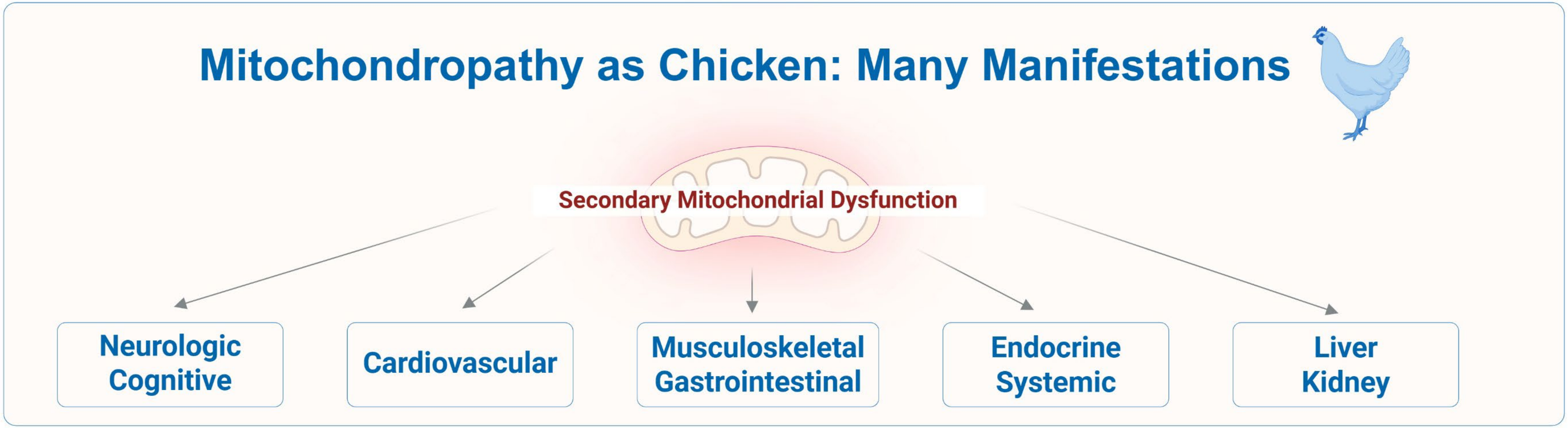
Clinical Focus

- Mechanisms of action of mitochondrial dysfunction on other bodily processes and diseases.



See: “References: Mitochondropathy: Many Root Causes and Manifestations”

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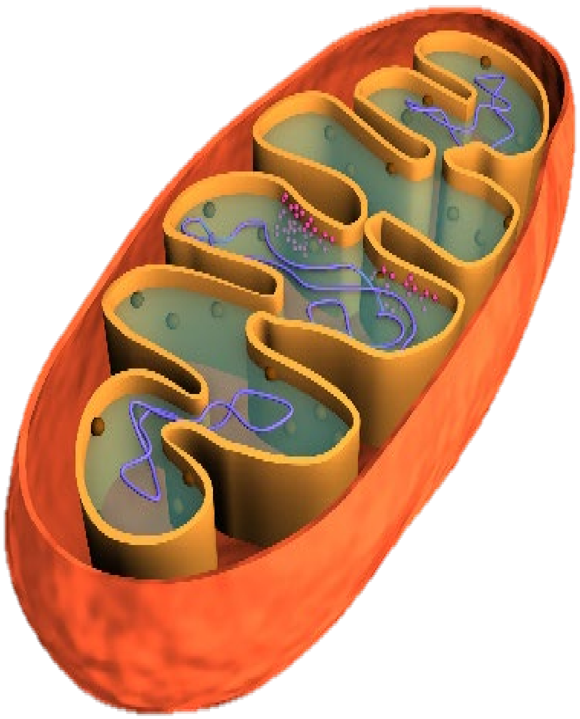
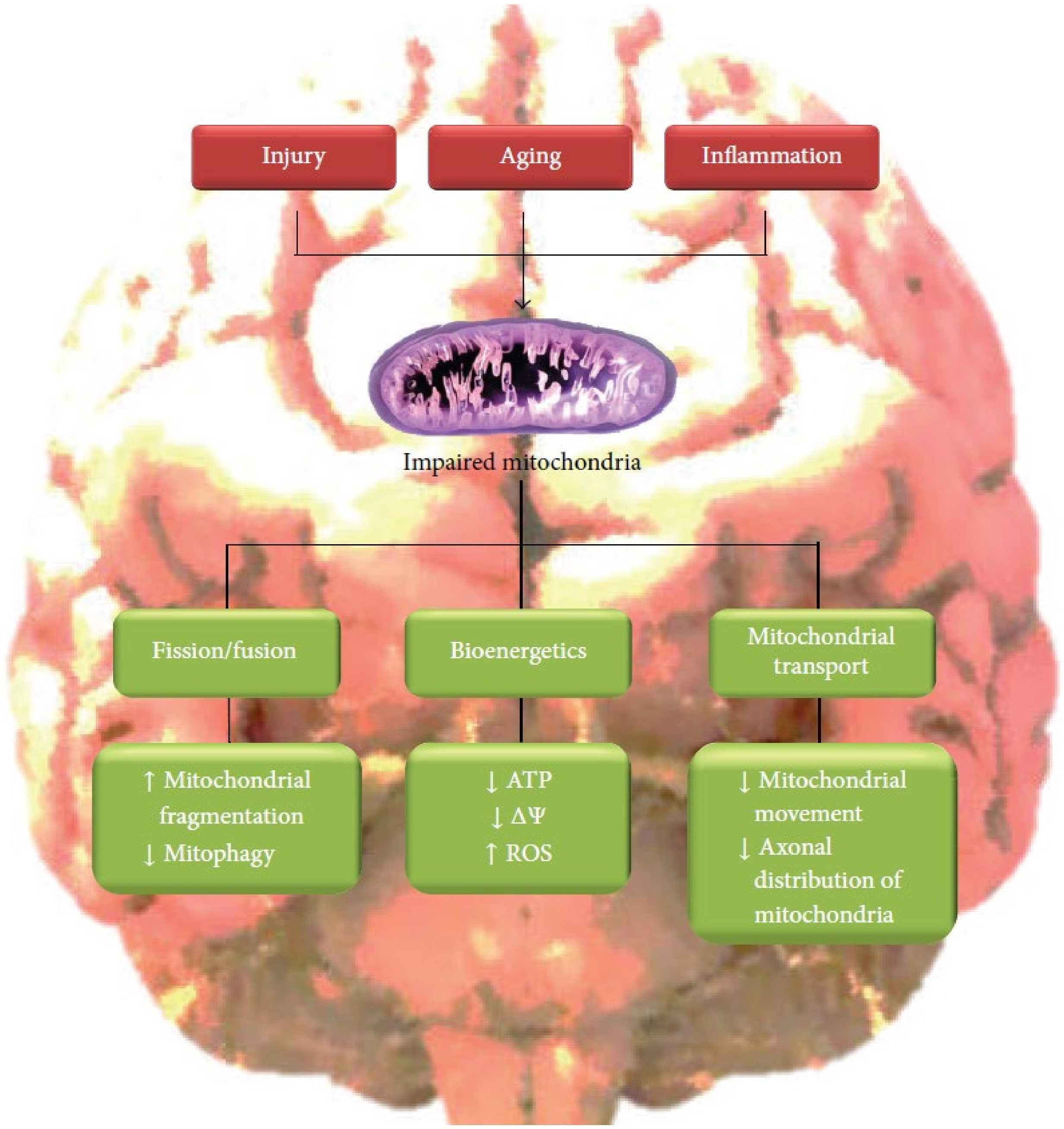
Mitochondrial Functions: Energy Deficit Disorders

Mitochondria help maintain homeostasis.



- **Bioenergetics**
- **Biosynthesis**
- **Signaling/Metabolism**



Neurocognitive Dysfunction as Representative Example of Mitochondropathy

Cabezas-Opazo FA, Vergara-Pulgar K, Pérez MJ, Jara C, Osorio-Fuentealba C, Quintanilla RA. Mitochondrial dysfunction contributes to the pathogenesis of Alzheimer's disease. *Oxid Med Cell Longev*. 2015;2015:509654. doi:10.1155/2015/509654

Long Description for **Neurocognitive Dysfunction as Representative Example of Mitochondropathy**

The graphic explains how injury, aging, and inflammation contribute to mitochondrial impairment and its effects.

- **Top level:** Three main causes — injury, aging, and inflammation — lead to impaired mitochondria.
- **Central point:** An illustration of a mitochondrion represents the impaired state.
- **Downstream effects** are shown in three functional categories:
    - **Fission/Fusion** –
        - Increased mitochondrial fragmentation
        - Decreased mitophagy (recycling of damaged mitochondria)
    - **Bioenergetics** –
        - Decreased A T P production
        - Decreased mitochondrial membrane potential (ΔΨ)
        - Increased reactive oxygen species (R O S)
    - **Mitochondrial Transport** –
        - Decreased mitochondrial movement
        - Reduced axonal distribution of mitochondria

Overall meaning: Damage from aging, injury, or inflammation can disrupt mitochondrial structure, energy production, and transport within cells, leading to impaired function and potentially contributing to neurodegenerative processes.

The Brain Is Uniquely Vulnerable to Oxidative Damage

- Multiple sources of ROS generation (e.g., NOx, complex I, p450s, neurotrophic factor withdrawal)
- Redox active metal-rich (catalytic iron)
- Auto-oxidation of monoamines
- Glutamate excitotoxicity
- Limited antioxidant and repair capacity (low catalase, mitochondria lack catalase)
- Resident immune cells (microglia) produce ROS and cytokines

See: "References: The Brain Is Uniquely Vulnerable to Oxidative Damage"

References

THE BRAIN IS UNIQUELY VULNERABLE TO OXIDATIVE DAMAGE

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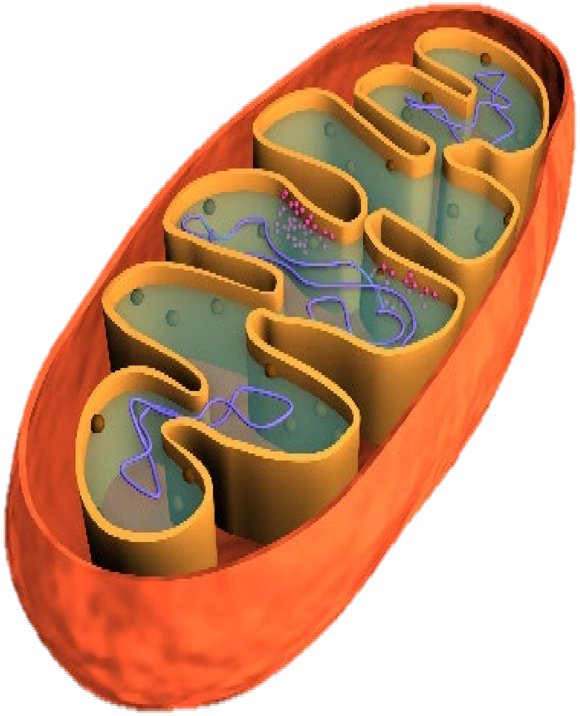
Mitochondrial Functions: Biosynthesis Disorders

Mitochondria help maintain homeostasis.



Bioenergetics

Biosynthesis

Signaling/Metabolism

Phospholipid Biosynthesis

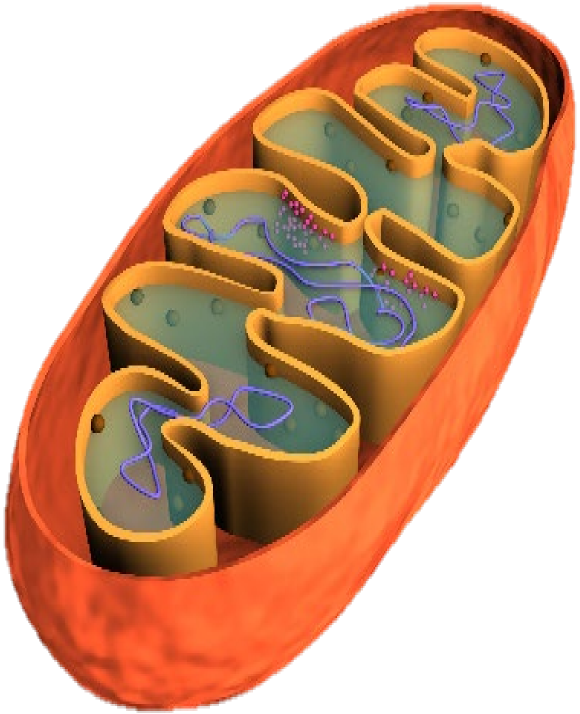
Phospholipid biosynthesis and transport is necessary for mitochondrial integrity and bioenergetics.

There is a complex transmembrane flow of phospholipids to various mitochondrial components.

- Acoba MG, Senoo N, Claypool SM. Phospholipid ebb and flow makes mitochondria go. *J Cell Biol*. 2020;219(8):e202003131. doi:10.1083/jcb.202003131
- Zhu Y, Tong X, Xue J, et al. Phospholipid biosynthesis modulates nucleotide metabolism and reductive capacity. *Nat Chem Biol*. 2025;21(1):35-46. doi:10.1038/s41589-024-01689-z

Mitochondrial Functions:
Insulin Resistance as a Cellular Signaling Disorder

Mitochondria help maintain homeostasis.



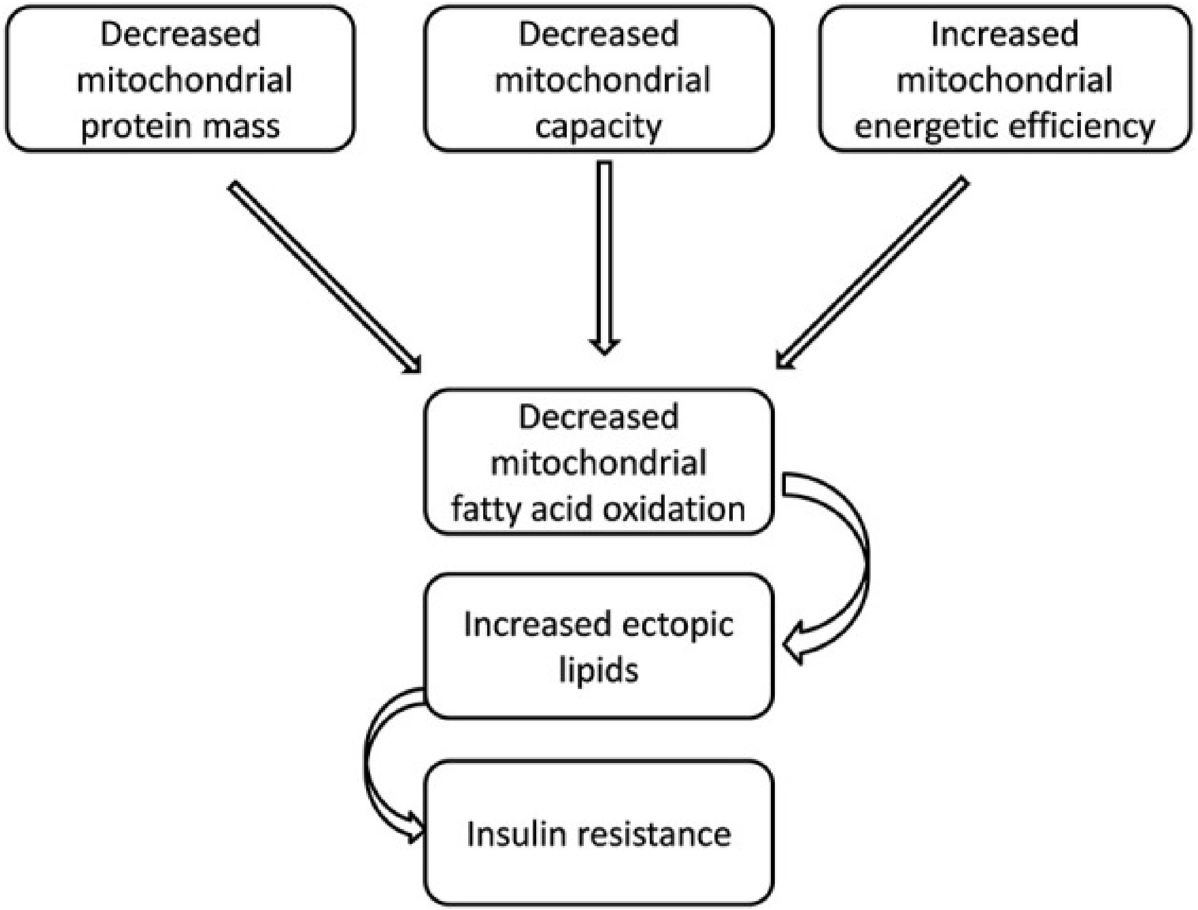
Bioenergetics

Biosynthesis

Signaling/Metabolism

Mitochondrial Theory of Insulin Resistance

- **Depressed PGC-1α levels, oxidative damage, and/or environmental toxins** impair mitochondrial β-oxidation & OXPHOS in skeletal muscle & liver
- **Intracellular build up of lipid metabolites** (fatty acyl-CoA, ceramide, diacylglycerol) & lipid peroxides cause lipotoxicity, which…
- **Impairs intracellular insulin signaling pathways**



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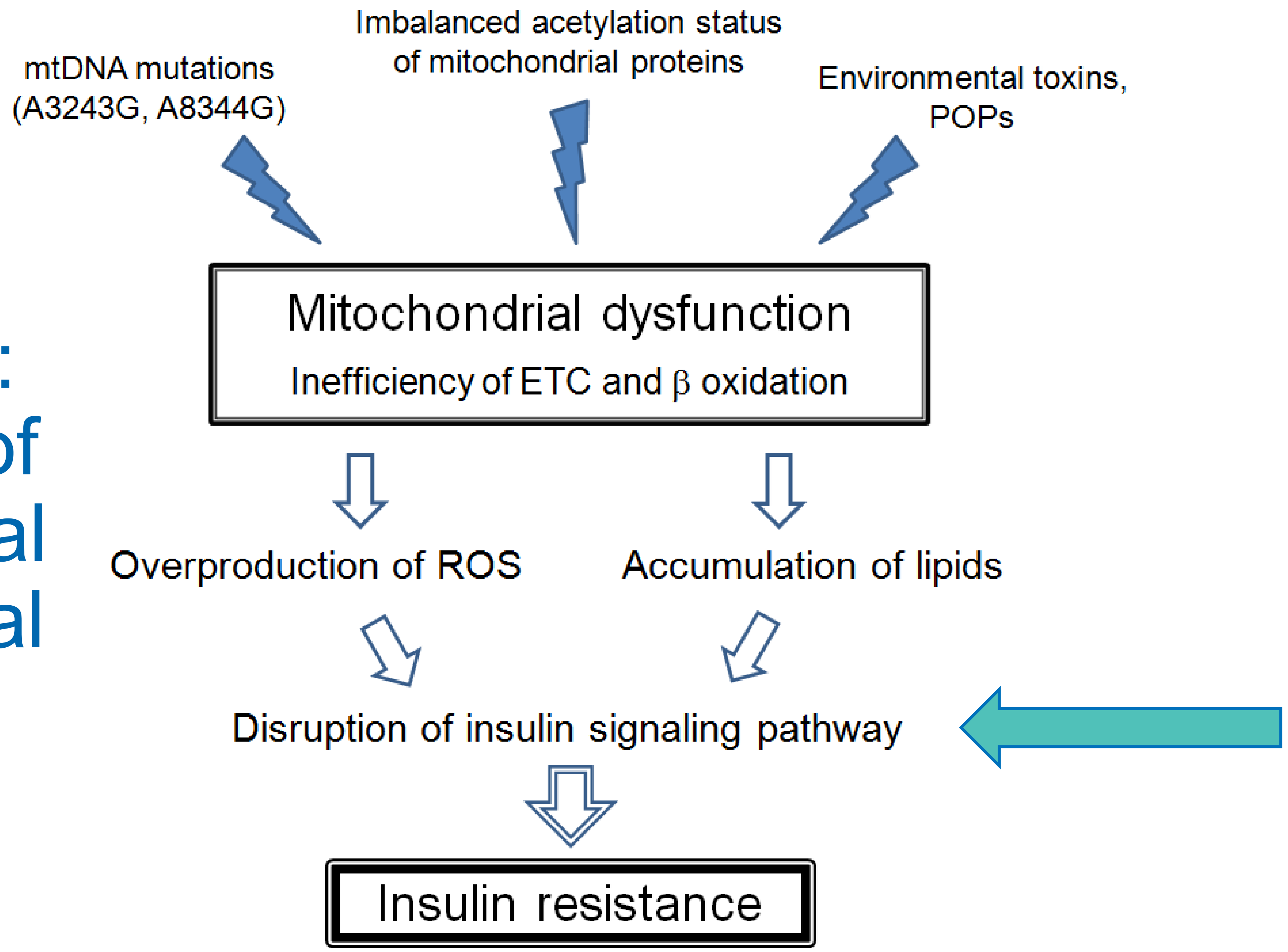
- Figure: Crescenzo R, Bianco F, Mazzoli A, Giacco A, Liverini G, Iossa S. Mitochondrial efficiency and insulin resistance. *Front Physiol*. 2015;5:512. doi:10.3389/fphys.2014.00512. Licensed under CC BY 4.0.

Insulin Resistance: A Disorder of Dysfunctional Mitochondrial Signaling



- Wang M, Wang XC, Zhang ZY, Mou B, Hu RM. Impaired mitochondrial oxidative phosphorylation in multiple insulin-sensitive tissues of humans with type 2 diabetes mellitus. *J Int Med Res*. 2010;38(3):769-781.
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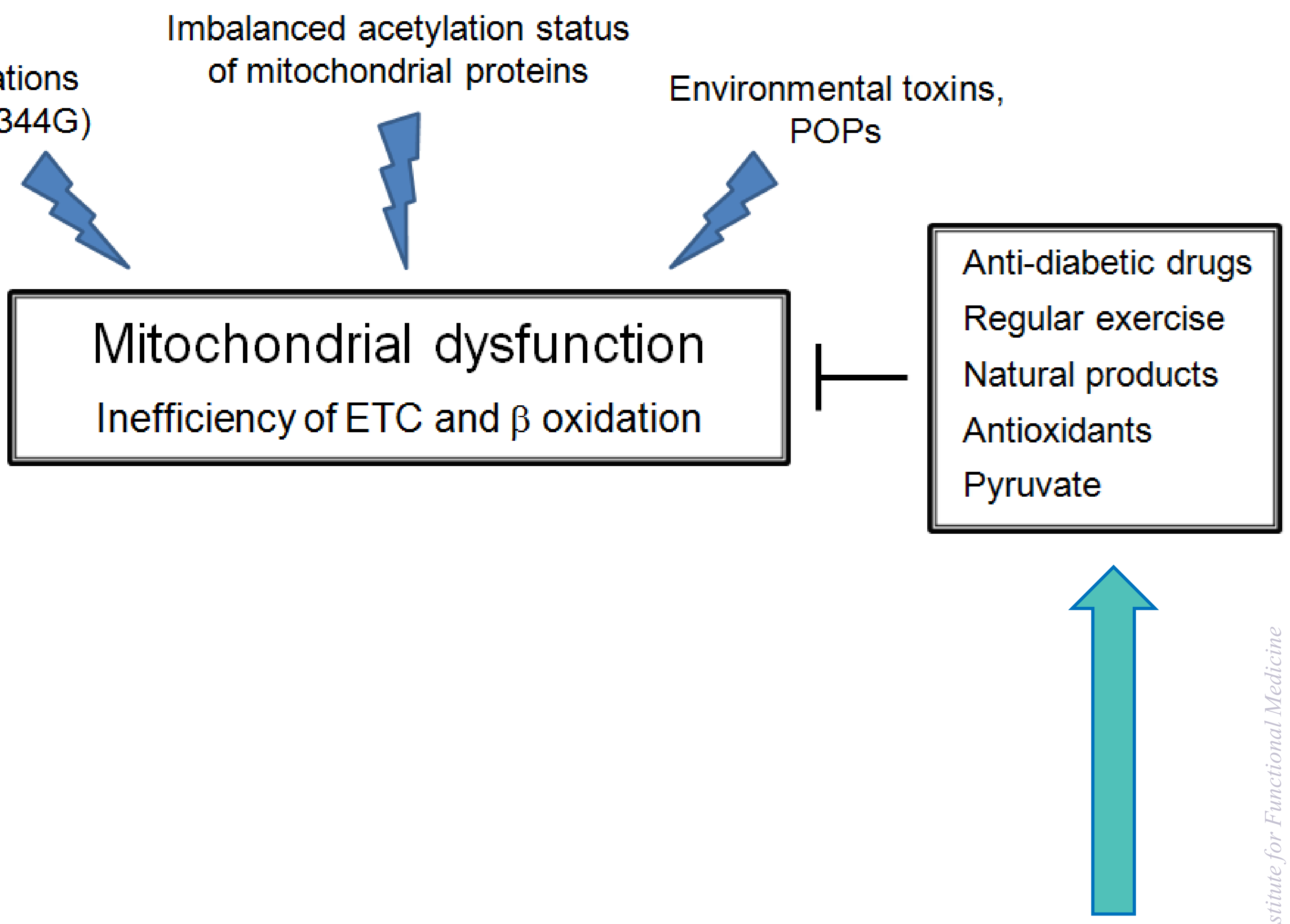
Long Description for
Insulin Resistance:A Disorder of Dysfunctional Mitochondrial Signaling

The diagram explains how mitochondrial dysfunction contributes to insulin resistance.

- Mitochondrial dysfunction—caused by factors such as mitochondrial D N A mutations, imbalanced acetylation of mitochondrial proteins, and environmental toxins like persistent organic pollutants (POPs)—leads to inefficient electron transport chain (E T C) activity and beta oxidation.
- This dysfunction causes overproduction of reactive oxygen species (R O S) and accumulation of lipids.
- These changes disrupt the insulin signaling pathway, ultimately resulting in insulin resistance.

Insulin Resistance: A Disorder of Dysfunctional Mitochondrial Signaling



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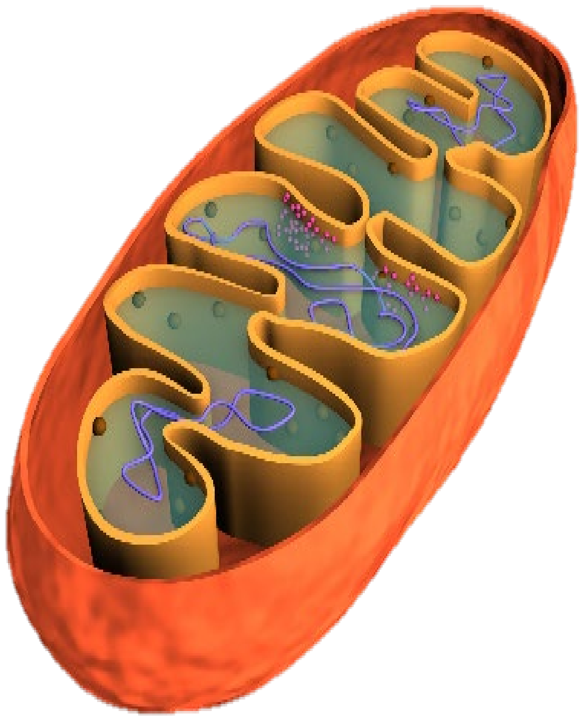
Long Description for **Insulin Resistance: A Disorder of Dysfunctional Mitochondrial Signaling** (with inhibitors shown)

- This is the same diagram as the previous slide, but inhibitors of Mitochondrial Dysfunction are now listed
- Mitochondrial dysfunction, defined as inefficiency of the electron transport chain (E T C) and beta oxidation, can be triggered by mitochondrial D N A mutations, imbalanced acetylation status of mitochondrial proteins, and environmental toxins such as persistent organic pollutants (POPs).
- This dysfunction can be counteracted by several interventions: anti-diabetic drugs, regular exercise, natural products, antioxidants, and pyruvate. These interventions block or reduce the negative effects of mitochondrial dysfunction.

Mitochondrial Functions:
Energy Deficit Aspects of Metabolic Syndrome

Mitochondria help maintain homeostasis.



- **Bioenergetics**
- **Biosynthesis**
- **Signaling/Metabolism**

Insulin Resistance

- Mitochondrial-induced insulin resistance
- Glucose less able to enter insulin-sensitive cells
- β-oxidation is inhibited, leading to lipid accumulation in skeletal muscle, liver, & heart
- Gluconeogenesis is inhibited
- Krebs cycle intermediates are depleted
- Blood glucose elevates → uptake by insulin-sensitive adipocytes
- Metabolic needs met by protein catabolism

- **All of these conditions are intracellular energy deficits:** Obesity, CHF, cachexia, diabetes, fatty liver

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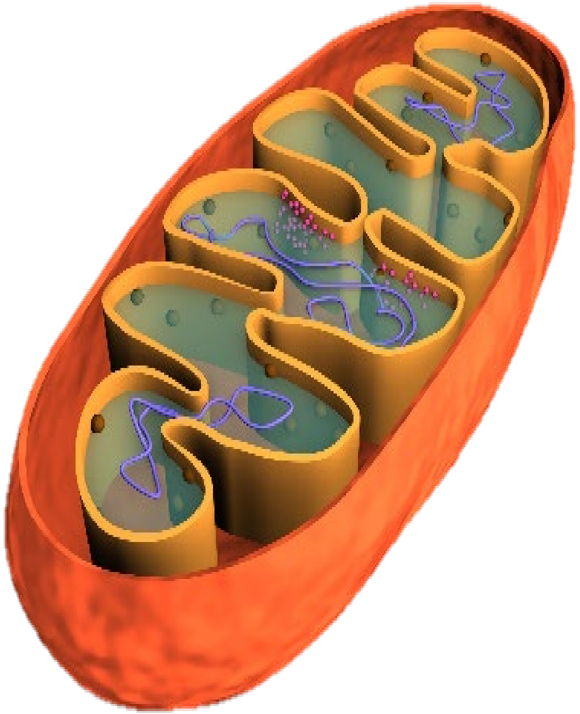
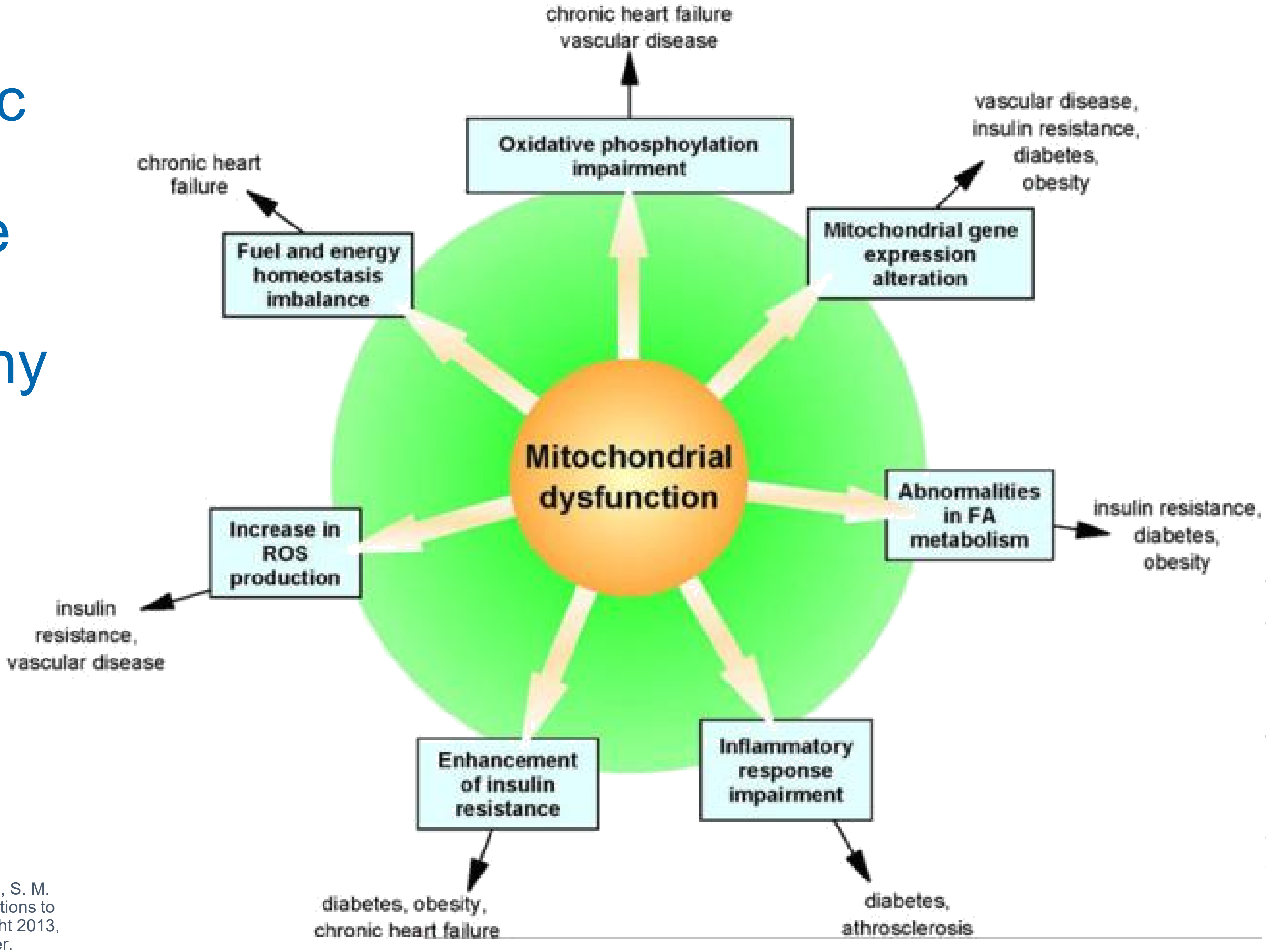
Mitochondrial Functions:
Metabolic Syndrome Encompasses All Three Facets

Mitochondria help maintain homeostasis.



- Bioenergetics
- Biosynthesis
- Signaling/Metabolism

Cardiometabolic Syndrome as Representative Example of Mitochondropathy



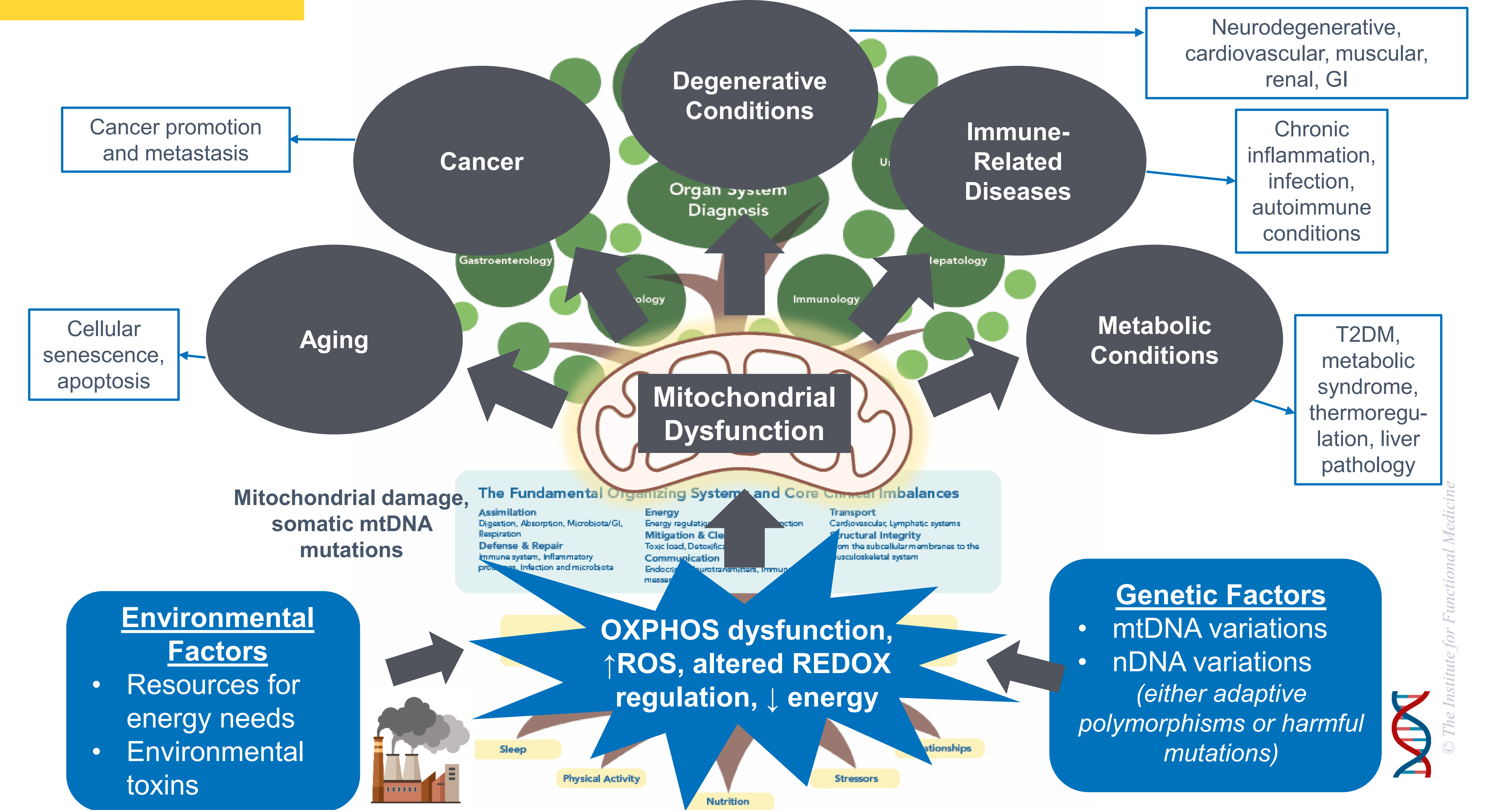
Reprinted from Translational Research, Vol. 161, Marcovina, S. M. et al., Translating the basic knowledge of mitochondrial functions to metabolic therapy: role of L-carnitine, Pages 73-84, Copyright 2013, doi: 10.1016/j.trsl.2012.10.006. with permission from Elsevier.

Long Description of
Cardiometabolic Syndrome as Representative Example of Mitochondropathy

The graphic shows **mitochondrial dysfunction** at the center, with arrows pointing outward to seven interconnected consequences. Each consequence is linked to specific health conditions:

- **Oxidative phosphorylation impairment** – leads to chronic heart failure and vascular disease.
- **Mitochondrial gene expression alteration** – associated with vascular disease, insulin resistance, diabetes, and obesity.
- **Abnormalities in fatty acid metabolism** – linked to insulin resistance, diabetes, and obesity.
- **Inflammatory response impairment** – associated with diabetes and atherosclerosis.
- **Enhancement of insulin resistance** – leads to diabetes, obesity, and chronic heart failure.
- **Increase in reactive oxygen species (R O S) production** – linked to insulin resistance and vascular disease.
- **Fuel and energy homeostasis imbalance** – associated with chronic heart failure.

Overall meaning: The diagram emphasizes that mitochondrial dysfunction triggers multiple metabolic, genetic, and inflammatory disturbances, which contribute to a wide range of chronic diseases, especially cardiovascular and metabolic disorders.



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Icons: Created with BioRender.com by The Institute for Functional Medicine.

Long Description for
Mitochondrial Dysfunction organized like the functional medicine tree

- The functional medicine tree. The leaves at the top show organ system diagnoses.
- The trunk, labeled “Clinical Findings,” connects the organ systems to deeper root causes.
- The root section lists “The Fundamental Organizing Systems and Core Clinical Imbalances,” which are the nodes of the functional medicine matrix.
- Feeding into the core clinical imbalances are Antecedents, Triggers, and Mediators as well as Modifiable Lifestyle factors, shown as feeding into the roots of the tree.
- Factors Contributing to Mitochondrial Dysfunction organized like the functional medicine tree. Mitochondrial dysfunction is like the tree trunk and is at the center, leading to organ system dysfunctions above like leaves, and include:
  - **Cancer** (promotion and metastasis)
  - **Aging** (cellular senescence, apoptosis)
  - **Degenerative Conditions** (neurodegenerative, cardiovascular, muscular, renal, gastrointestinal)
  - **Immune-Related Diseases** (chronic inflammation, infection, autoimmune conditions)
  - **Metabolic Conditions** (type 2 diabetes, metabolic syndrome, thermoregulation problems, liver pathology)
- Below, mitochondrial damage and somatic m t D N A mutations are noted as contributors.
- Key dysfunctions feeding into the mitochondrial dysfunction tree trunk include oxidative phosphorylation (OXPHOS) impairment, increased R O S, altered redox regulation, and reduced energy.
- Like A T Ms, factors that feed at the root level include environmental factors such as resources for energy needs and environmental toxins. Also feeding in at this level are genetic factors, which include mitochondrial D N A and nuclear D N A variations – these may be adaptive or harmful.

Cancer:
A Disorder of Mitochondrial Metabolism?

- Excessive caloric intake is associated with increased risk for cancer, while caloric restriction is protective.
- Metabolism generates ROS, which contribute to oncogenic mutations (reprogramming of genes).
- Glycolytic switch: inhibits pyruvate entering TCA cycle, inducing aerobic glycolysis (Warburg effect).

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Cancer:
A Disorder of Mitochondrial Metabolism?

- Cancer cells undergo metabolic reprogramming:
  - Increased aerobic glycolysis (Warburg effect) – rates up to 200x higher than in normal tissues (basis for PET scanning)
  - Glutamine utilization as alternate fuel for citric acid cycle (via alpha-ketoglutarate)
  - Net effect is to rapidly produce amino acids, nucleotides, and lipids for rapid cell proliferation

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Cancer:
A Disorder of Mitochondrial Metabolism?

- Upregulation of glutaminase increases utilization of glutamine via the TCA cycle.

- Glucose & glutamine provide the metabolites, carbon skeletons, NADH, & ATP needed to build new cancer cells.

- Which is initiating event: environmental activation of aerobic glycolysis or oxidative damage to genes & proteins (epigenetic) that results in the Warburg effect?

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- See: "Additional References: Mitochondria and Cancer"

Additional References

MITOCHONDRIA AND CANCER

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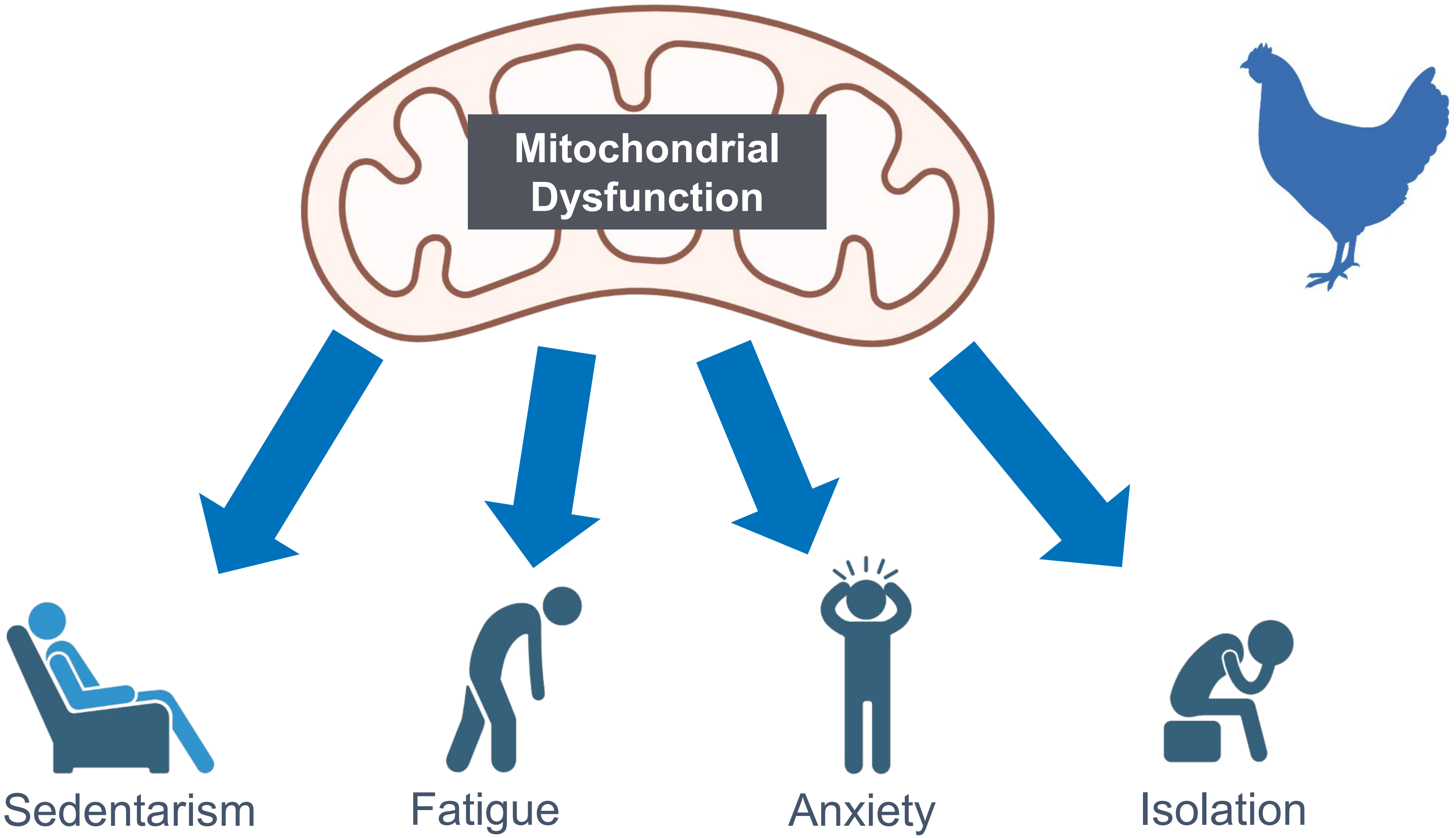
- Weinberg F, Chandel NS. Mitochondrial metabolism and cancer. *Ann N Y Acad Sci*. 2009;1177:66-73. doi:10.1111/j.1749-6632.2009.05039.x

Mitochondrial Dysfunction as Mediator



Created with BioRender.com by The Institute for Functional Medicine

See: "References: Mitochondrial Dysfunction as Mediator"

References

MITOCHONDRIAL DYSFUNCTION AS MEDIATOR

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Representative Patient Clinical Findings: *Think Mitochondrially*

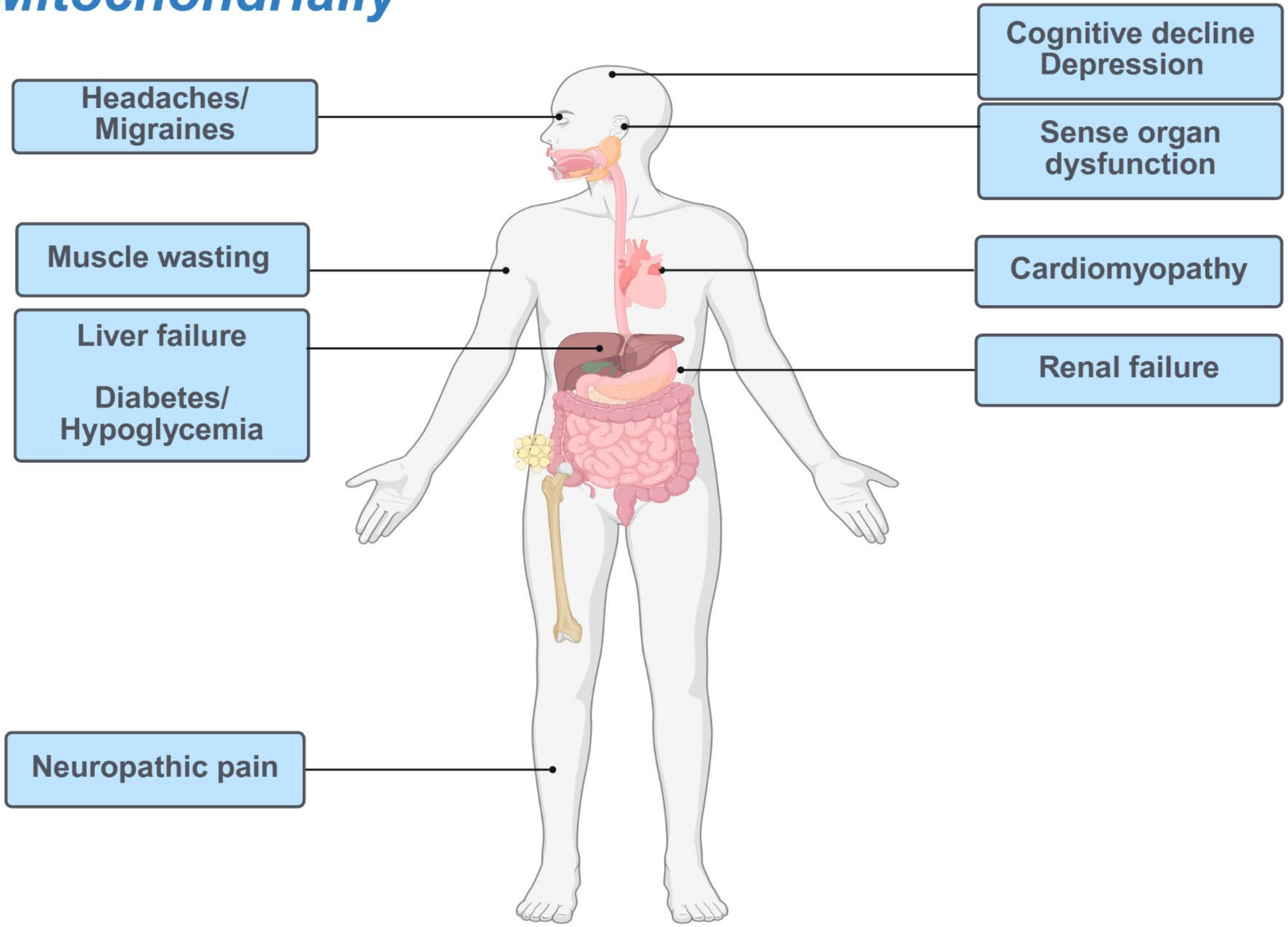


Image: Created with BioRender.com by The Institute for Functional Medicine

See: “References: Signs and Symptoms Suggesting Mitochondropathy”

Long Description for
Representative Patient Clinical Findings: Think Mitochondrially

This graphic depicts a human body outline with labeled symptoms connected to specific body regions that may be related to mitochondrial dysfunction.

- Head - Headaches/Migraines, Cognitive decline, and depression.
- Eyes & Ears - Sense organ dysfunction
- Muscular system - Muscle wasting
- Heart - Cardiomyopathy
- Liver - Liver failure
- Pancreas - Diabetes/Hypoglycemia
- Kidneys - Renal failure
- Nervous system - Neuropathic pain

The overall meaning is that mitochondrial disorders can affect multiple organ systems, producing widespread symptoms across the nervous, muscular, cardiac, hepatic, endocrine, renal, and central and peripheral nervous systems.

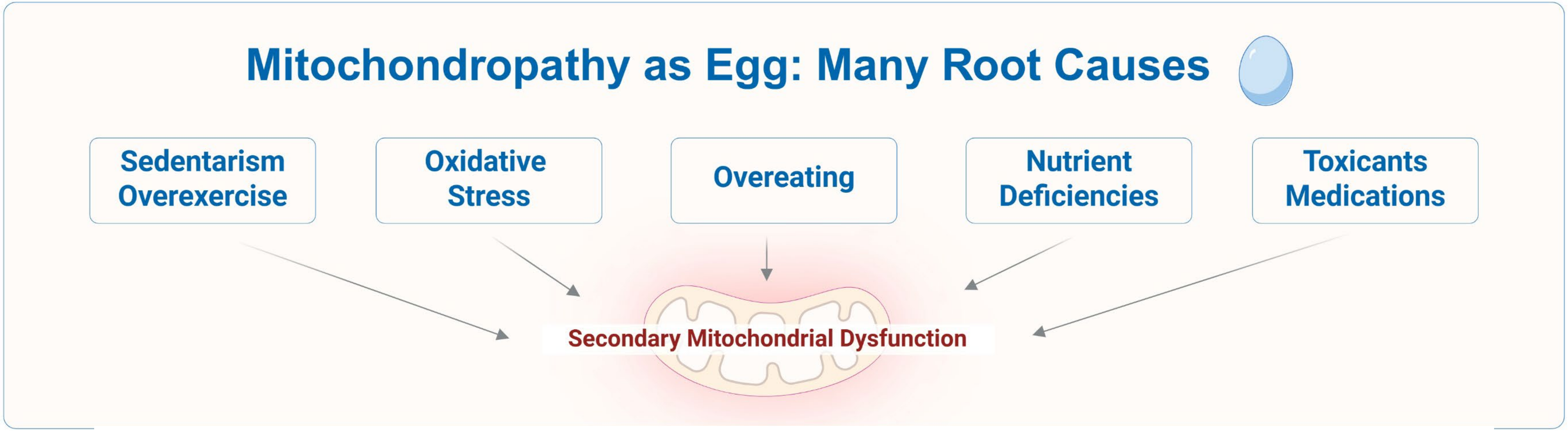
Bioenergetics and Mitochondropathy: Part 4

Mechanisms – Mediators – Manifestations

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Foundational therapeutic interventions for mitochondrial dysfunction.



When amelioration of mitochondrial dysfunction is identified as a necessary component of a therapeutic regimen, multiple root causes will need to be addressed.

At this stage of your training, the foundational interventions are diet, lifestyle, 5R gut interventions, and avoidance of toxic exposures.

See: "References: Mitochondropathy: Many Root Causes and Manifestations"

Getting Healthy Is a Team Sport

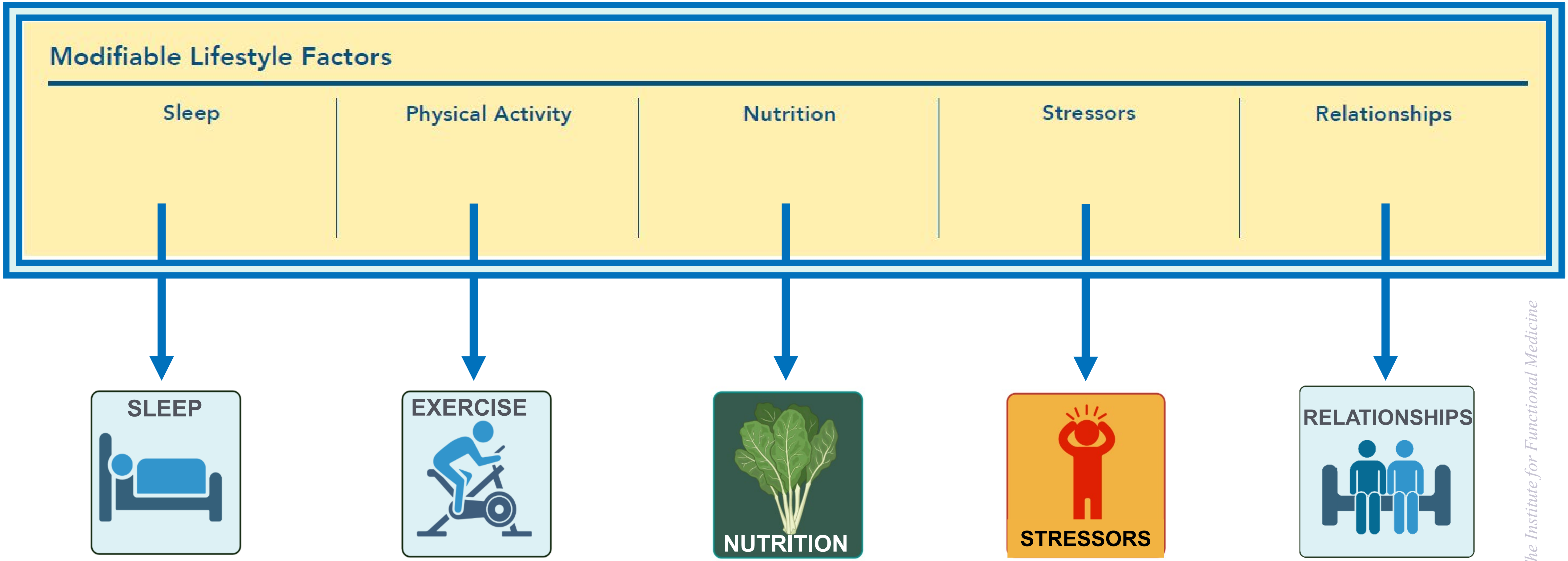
Foundational Therapeutic Approach: Mitochondrial Dysfunction

LIFESTYLE FIRST

FOOD FIRST

GUT FIRST

Mitochondria and Lifestyle Factors



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Sleep and Mitochondrial Health

- The circadian rhythm appears to be intricately connected to the process of mitophagy, impacting the structure, number, and function of mitochondria.
- Sleep deprivation has been shown to increase the production of ROS within the brain.
- Poor sleep quality, particularly long sleep latency, negatively impacts mitochondrial DNA copy number (mtDNAcn), which is a measure of mitochondrial function.

Improving sleep quantity and quality can enhance overall mitochondrial function.

See: “References: Sleep and Mitochondrial Health”

References

SLEEP AND MITOCHONDRIAL HEALTH

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Mitochondrial Biogenesis:
Exercise

AMPK → PGC-1alpha

SIRT1 → PGC-1alpha

PGC-1alpha
(muscle, brain, liver, brown adipose tissue)

Gene regulation
(numerous)

Mitochondrial biogenesis → ↑ATP ↑Fatty acid oxidation, ↑Energy

See: “References: Exercise and Mitochondrial Health”

Long Description for

Mitochondrial Biogenesis: Exercise

- The diagram shows how exercise stimulates mitochondrial biogenesis through two main pathways.
- Exercise activates **A M P K** and **SIRT1**, which both lead to increased activity of **P G C-1 alpha** in tissues such as muscle, brain, liver, and brown adipose tissue.
- PGC-1alpha regulates numerous genes involved in energy metabolism. This results in enhanced mitochondrial biogenesis, which increases ATP production, fatty acid oxidation, and overall energy availability.

References

EXERCISE AND MITOCHONDRIAL HEALTH

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Stressors and Mitochondrial Health

- The stress response is energetically demanding, requiring mitochondria to produce greater amounts of ATP and increasing their allostatic load.
  - Different types of stress, including psychological, economic, and social, may result in this increased need for energy; if chronic, this may lead to poor health outcomes.
- Acute stress has been found to increase serum cell-free mitochondrial DNA (cf-mtDNA), a marker that is often elevated in inflammatory diseases and trauma.

Decreasing allostatic load can enhance overall mitochondrial function.

See: "References: Stressors and Mitochondrial Health"

References

STRESSORS AND MITOCHONDRIAL HEALTH

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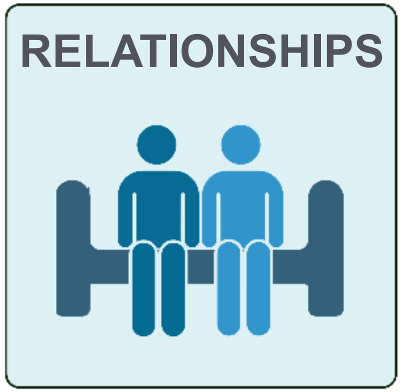
Relationships and Mitochondrial Health



- The way in which mitochondria communicate with one another throughout the body has been compared to social interactions between humans.
- Loneliness may be considered one of the greatest threats to overall health, as it increases mitochondrial allostatic load and negatively impacts mitochondrial function.
    - This, in turn, disrupts proper neuroendocrine and metabolic processes, leading to poor health outcomes and increased social isolation.

Humans are social beings, right down to our mitochondria.

See: "References: Relationships and Mitochondrial Health"

References

RELATIONSHIPS AND MITOCHONDRIAL HEALTH

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Foundational Therapeutic Approach:
Mitochondrial Dysfunction

LIFESTYLE FIRST

FOOD FIRST

GUT FIRST

Supporting Mitochondrial Function:
Use Food First!

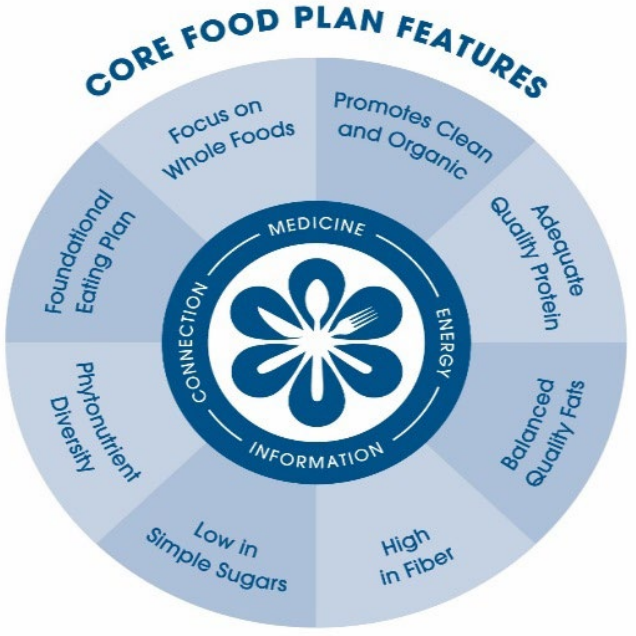
The Clinical Path to the Foundational Food Plans

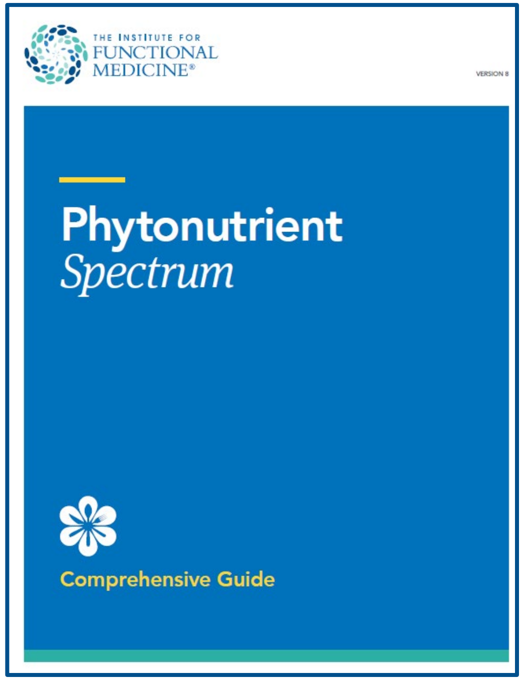
Medical History

- Chief Complaints
- Conditions
- Timeline and ATMs
- Medication Review

ABCDs of Nutrition Evaluation

- Anthropometrics
- Biomarkers and Labs
- Clinical Indications From NPE
- Diet and Lifestyle Review
- Matrix Review





Long Description for
The Clinical Path to the Foundational Food Plans

This diagram outlines the clinical process for choosing foundational IFM food plans.

- It begins with **Medical History**, which includes reviewing chief complaints, medical conditions, a health timeline and antecedents-triggers-mediators (A T Ms), and a medication review.
- This leads to the **ABCDs of Nutrition Evaluation**, covering anthropometrics, biomarkers and lab results, clinical indications from the Nutrition Physical Exam (N P E), diet and lifestyle review, and a matrix review.
- From there, one can choose the Core Food Plan or Phytonutrient Spectrum

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Nutrition Interventions

Functional Nutrition Fundamentals

Food Is...

- Energy
- Information
- Connection
- Medicine

Foundational Interventions

- Phytonutrient Spectrum Foods
- Phytonutrient Spectrum Checklist
- Core Food Plan
- Core Food Plan–Vegan
- Core Food Plan–Vegetarian

Therapeutic Interventions

- Cardiometabolic Food Plan
- Elimination Diet
- Detox Food Plan
- Mito Food Plan

Personalized Nutrition Plan

Long Description for
Nutrition Interventions

This diagram shows the process of creating a personalized nutrition plan through functional medicine nutrition interventions.

- On the left, **Functional Nutrition Fundamentals** describe food as energy, information, connection, and medicine.
- These fundamentals lead to **Foundational Interventions**, which include:
  - Phytonutrient Spectrum Foods
  - Phytonutrient Spectrum Checklist
  - Core Food Plan
  - Core Food Plan – Vegan
  - Core Food Plan – Vegetarian
- Foundational interventions, which are emphasized on this slide, connect both to **Therapeutic Interventions** and directly to the **Personalized Nutrition Plan**.
- **Therapeutic Interventions** listed are:
  - Cardiometabolic Food Plan
  - Elimination Diet
  - Detox Food Plan
  - Mito Food Plan

The pathway shows that foundational nutrition strategies form the base for more targeted therapeutic plans, which together lead to an individualized nutrition approach.

Phytochemicals: Mechanisms of Action

Curcumin, Quercetin, Green Tea Polyphenols

AMPK

SIRT1

PGC-1alpha
(muscle, brain, liver, brown adipose tissue)

Gene regulation
(numerous)

Mitochondrial biogenesis → ↑ATP ↑Fatty acid oxidation, ↑Energy

Long Description for

Phytochemicals: Mechanisms of Action Curcumin, Quercetin, Green Tea Polyphenols

The diagram explains how certain phytochemicals—specifically curcumin, quercetin, and green tea polyphenols—promote mitochondrial biogenesis.

- These compounds activate two pathways: **A M P K** and **SIRT 1**.
- Both pathways stimulate **P G C-1 alpha** in tissues such as muscle, brain, liver, and brown adipose tissue.
- PGC-1alpha regulates numerous genes involved in energy metabolism.
- The result is increased mitochondrial biogenesis, leading to higher ATP production, greater fatty acid oxidation, and increased overall energy.

Foundational Food Interventions:

Phytonutrient Food Plan

Core Food Plan

Recommended "Core" First Step for Most Patients: A more nutrient-dense diet

- Inflammatory elements
- Inconsistent behaviors
- Imbalanced fats
- High glycemic impact
- Ultra-processed

Nutrient-Depleted Diet

Core Food Plan

- Nutrient-dense
- Healthy, whole foods
- Phytonutrient rainbow
- Balanced dietary choices

Energy

Features that support energy regulation and mitochondrial function

Long Description for

Recommended “Core” First Step for Most Patients: A more nutrient-dense diet

The graphic recommends a “core” first step for most patients—transitioning to a more nutrient-dense diet.

- It shows moving from a **Nutrient-Depleted Diet**, which contains inflammatory elements, inconsistent behaviors, imbalanced fats, high glycemic impact, and ultra-processed foods, toward a **Core Food Plan**.
- The Core Food Plan is nutrient-dense, based on healthy whole foods, includes a variety of phytonutrients (“phytonutrient rainbow”), and emphasizes balanced dietary choices.
- This shift supports **energy** by providing features that help regulate energy and improve mitochondrial function.

5R Remove Interventions

- Caloric excess
- Fructose
- Alcohol
- Trans fat
- Refined carbohydrates, sugar, and advanced glycation end products (preformed and hyperglycemic)

See: “References: 5R Remove Interventions”

References

5R REMOVE INTERVENTIONS

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Dietary Choices: The Phytonutrient Spectrum

- Diversity of phytonutrient-rich fruits and vegetables is essential to establish a baseline healthy eating plan.

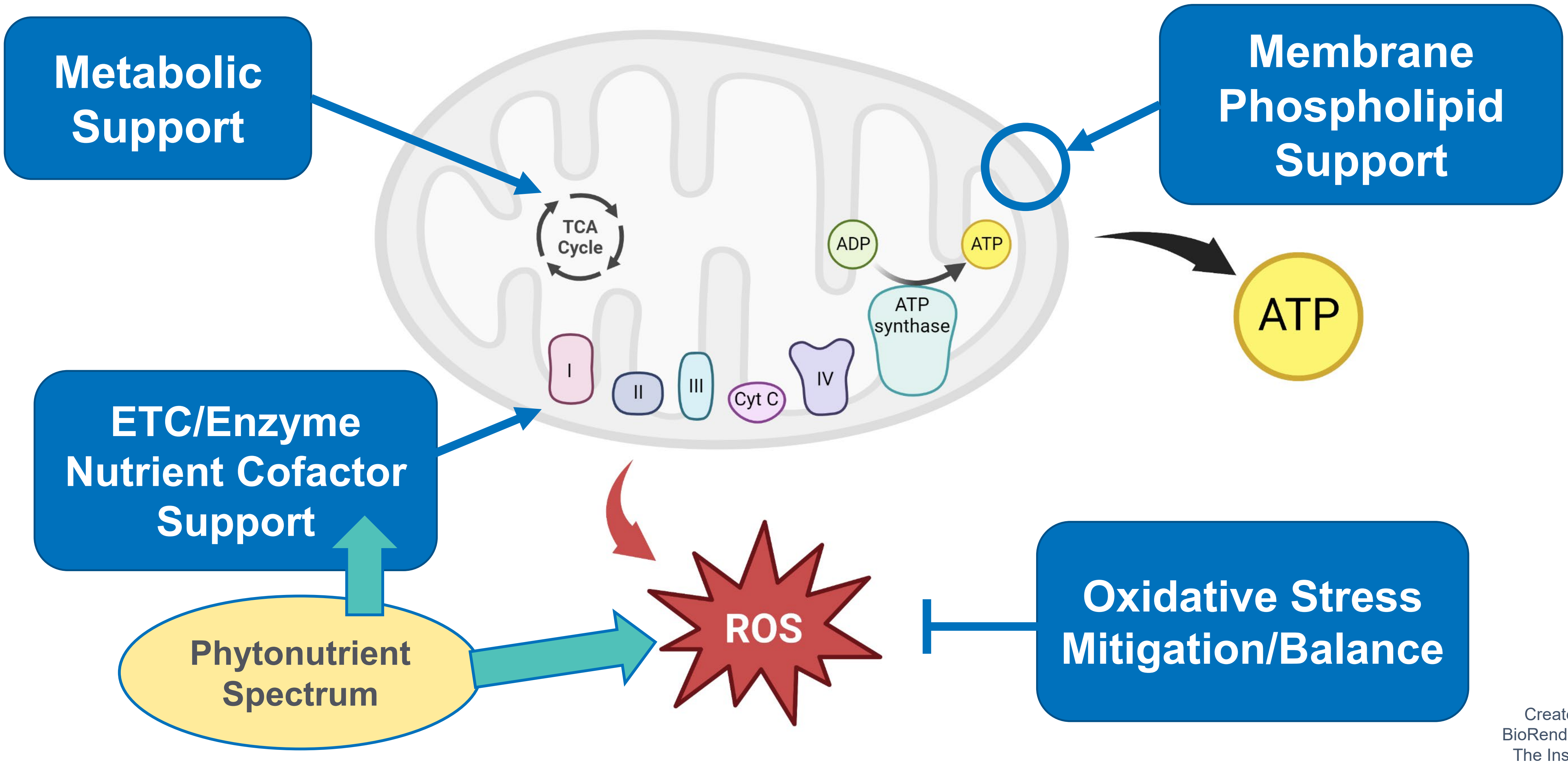
- Educates patients on importance of eating fruits and vegetables of varied color each day

The Problem, the Solution, and the Goal

- Symptoms of fatigue, pain, "brain fog," and other neurological conditions can be a sign of mitochondrial dysfunction.
- Anti-inflammatory, low-carbohydrate, high-quality fats diet with phytonutrient diversity supports energy production and vitality.

By supporting the mitochondria, the patient can fuel cellular energy production, decrease inflammation, and support blood sugar balance.

Multimodal Therapeutic Regimen to Address All Aspects of Mitochondrial Dysfunction



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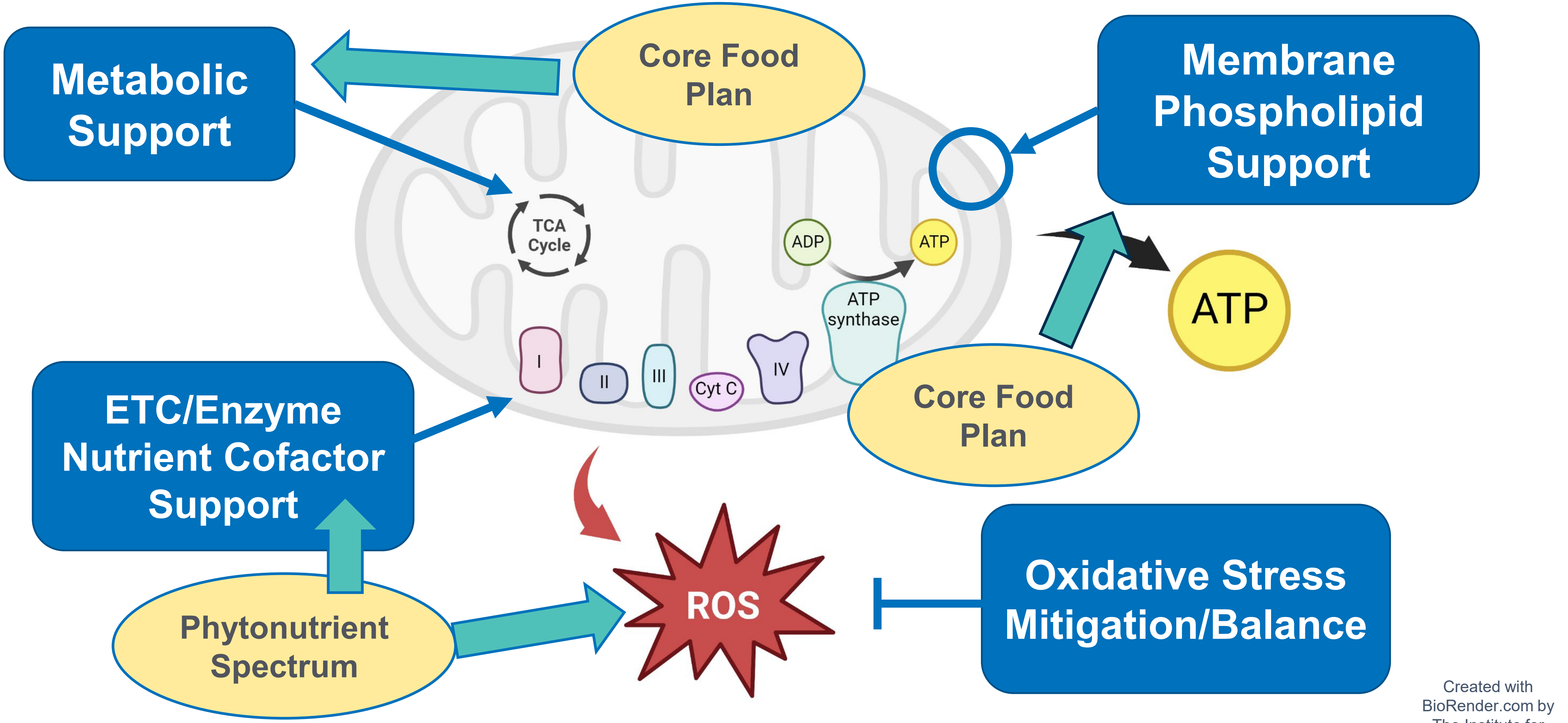
Long Description for **Multimodal Therapeutic Regimen to Address All Aspects of Mitochondrial Dysfunction**

The diagram illustrates a multimodal therapeutic approach to address all aspects of mitochondrial dysfunction. A mitochondrion is shown with its major functions and processes. Key intervention areas are:

- **Metabolic Support** – supports the tricarboxylic acid (TCA) cycle.
- **Membrane Phospholipid Support** – maintains mitochondrial membrane structure and supports ATP production.
- **ETC/Enzyme Nutrient Cofactor Support** – supports the electron transport chain and enzyme activity.
- **Phytonutrient Spectrum** – contributes to reducing reactive oxygen species (ROS). The phytonutrient spectrum is a point of emphasis on this slide.
- **Oxidative Stress Mitigation/Balance** – helps control excess ROS to protect mitochondrial function.

ATP production is highlighted as a primary output, while the integration of these support strategies aims to optimize mitochondrial health and energy production.

Multimodal Therapeutic Regimen to Address All Aspects of Mitochondrial Dysfunction



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Long Description for **Multimodal Therapeutic Regimen to Address All Aspects of Mitochondrial Dysfunction (Core Food Plan included)**

The diagram presents a multimodal therapeutic strategy for addressing all aspects of mitochondrial dysfunction, with the **Core Food Plan** integrated into multiple support areas.

Key intervention areas are:

- **Metabolic Support** – assisted by the Core Food Plan to optimize energy metabolism.
- **Membrane Phospholipid Support** – maintains mitochondrial membrane health, supporting ATP production.
- **ETC/Enzyme Nutrient Cofactor Support** – aided by the phytonutrient spectrum to enhance electron transport chain function and enzyme activity.
- **Oxidative Stress Mitigation/Balance** – reduces excess reactive oxygen species (ROS) to protect mitochondrial structures and function.

- The Core Food Plan is shown as central to both metabolic support and overall mitochondrial function, with its role in promoting ATP generation and balancing oxidative stress through nutrient-rich dietary strategies.

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Nutrition Interventions

Functional Nutrition Fundamentals

Food Is…

- Energy
- Information
- Connection
- Medicine

Foundational Interventions

- Phytonutrient Spectrum Foods
- Phytonutrient Spectrum Checklist
- Core Food Plan
- Core Food Plan–Vegan
- Core Food Plan–Vegetarian

Therapeutic Interventions

- Cardiometabolic Food Plan
- Elimination Diet
- Detox Food Plan
- Mito Food Plan

Personalized Nutrition Plan

Long Description for
Nutrition Interventions

This diagram shows the process of creating a personalized nutrition plan through functional medicine nutrition interventions.

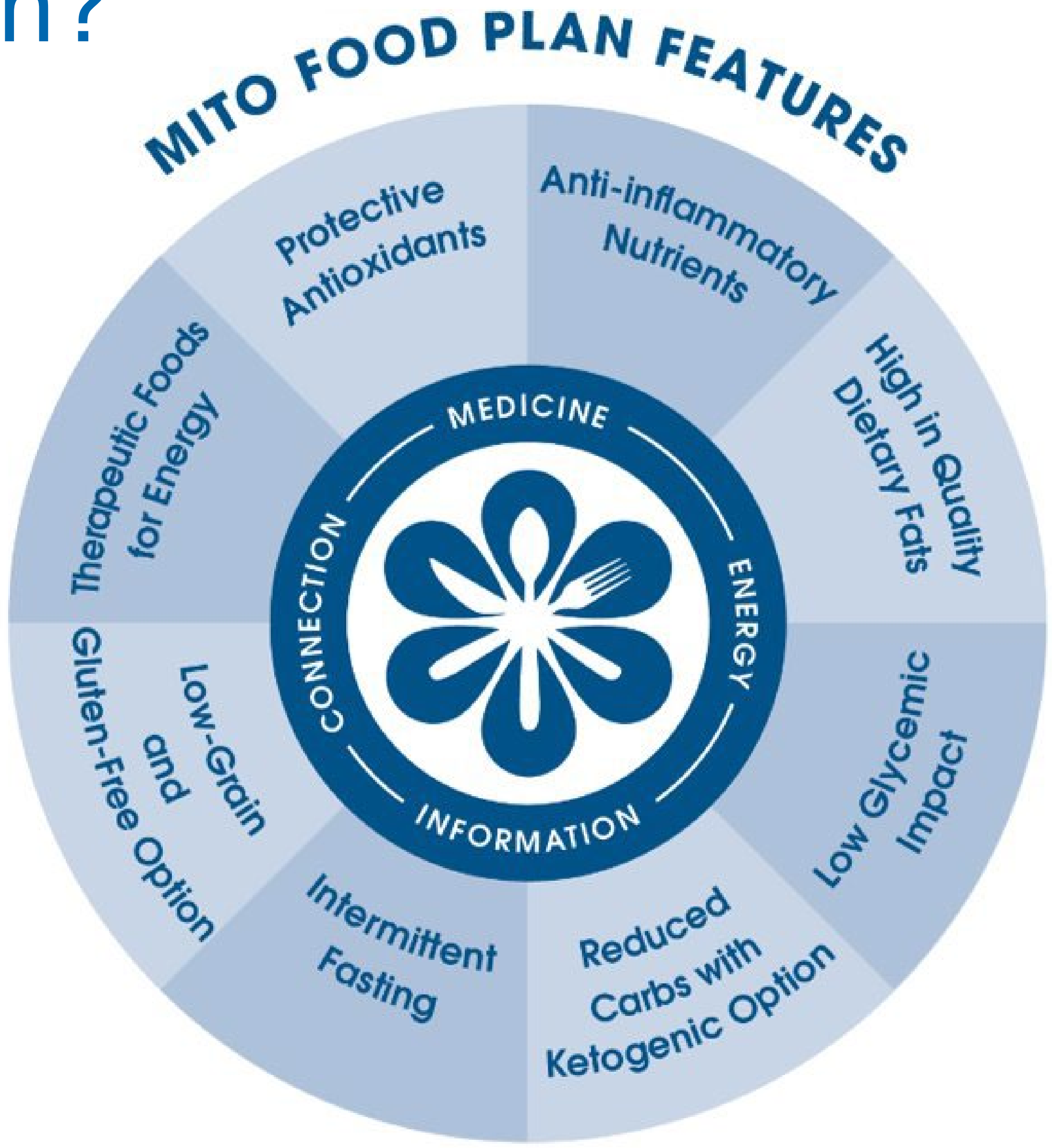
- On the left, **Functional Nutrition Fundamentals** describe food as energy, information, connection, and medicine.
- These fundamentals lead to **Foundational Interventions**, which include:
  - Phytonutrient Spectrum Foods
  - Phytonutrient Spectrum Checklist
  - Core Food Plan
  - Core Food Plan – Vegan
  - Core Food Plan – Vegetarian
- Foundational interventions connect both to **Therapeutic Interventions** and directly to the **Personalized Nutrition Plan**.
- **Therapeutic Interventions,** which are emphasized on this slide, include:
  - Cardiometabolic Food Plan
  - Elimination Diet
  - Detox Food Plan
  - Mito Food Plan

The pathway shows that foundational nutrition strategies form the base for more targeted therapeutic plans, which together lead to an individualized nutrition approach.

What Is the Mito Food Plan?

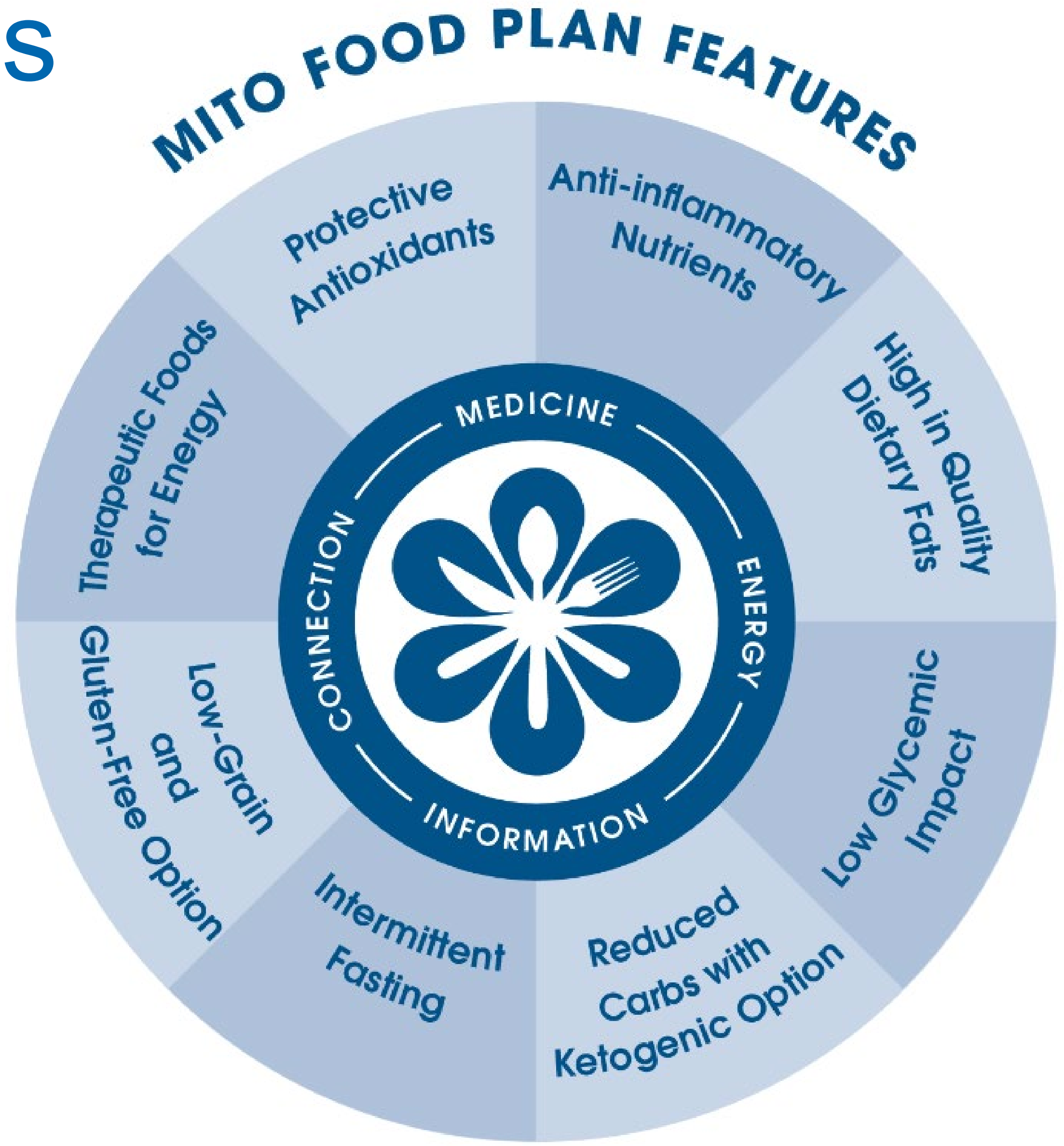
An anti-inflammatory, low-glycemic, gluten-free, low-grain, high-quality fats approach to:

- Support healthy mitochondria through foods that improve energy production



Mito Food Plan Features

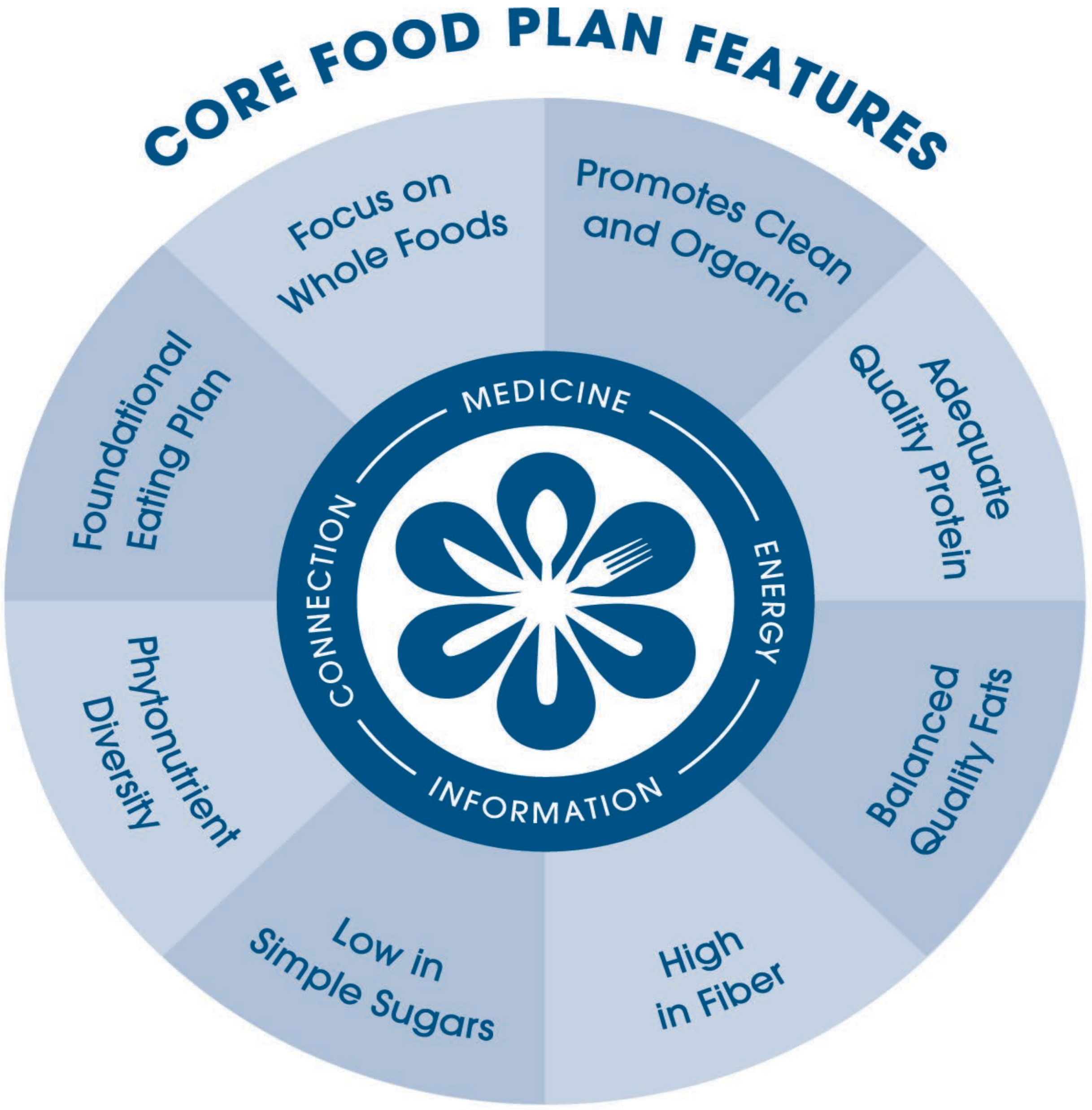
1. Anti-Inflammatory Nutrients
2. Protective Antioxidants
3. High in Quality Dietary Fats
4. Low Glycemic Impact
5. Reduced Carbs with Ketogenic Options
6. Intermittent Fasting and Caloric Restriction
7. Low-Grain and Gluten-Free Option
8. Therapeutic Foods for Energy

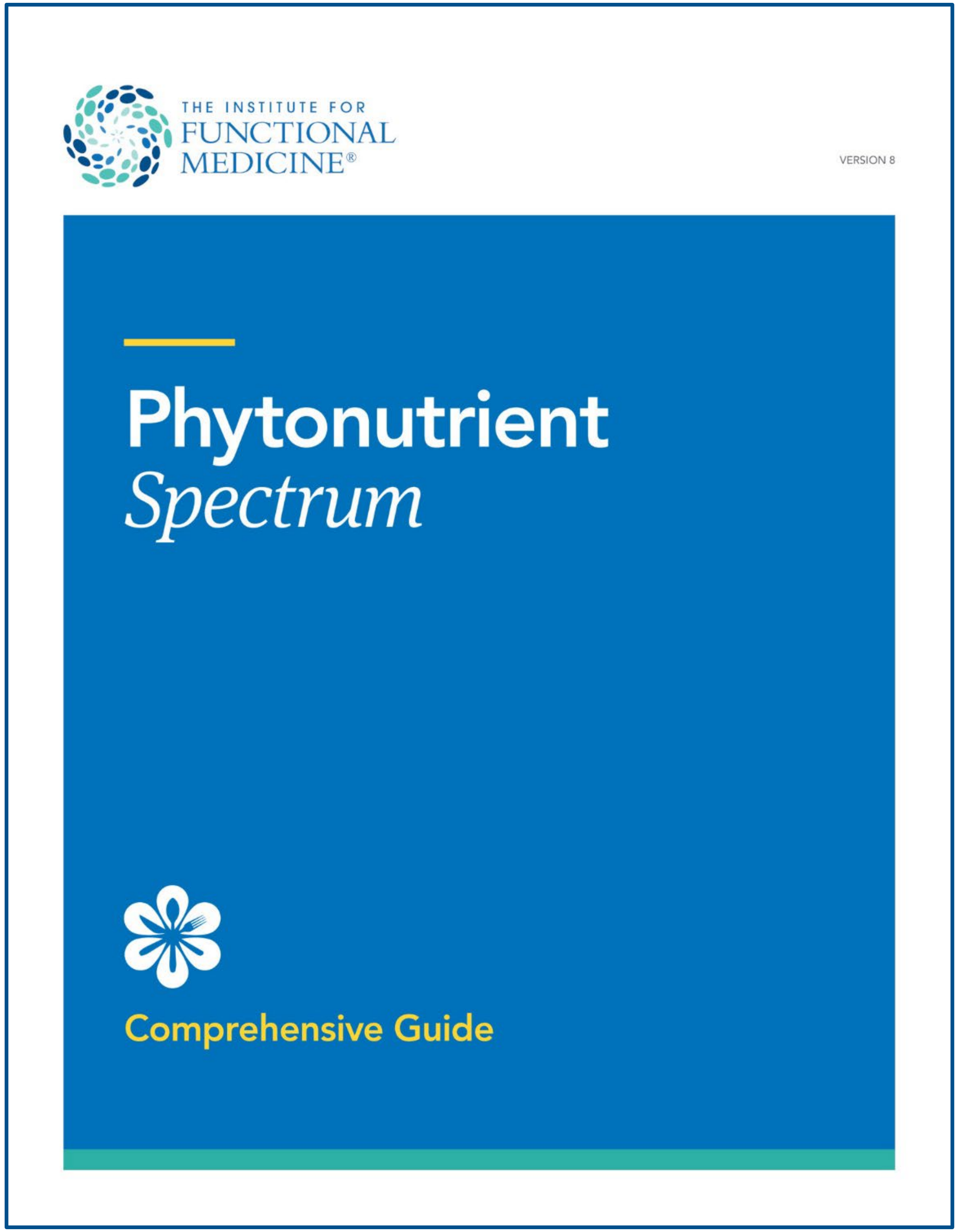


Core & Mito Food Plan Features

CONNECTION BETWEEN FOUNDATIONAL AND THERAPEUTIC FOOD PLANS

Core Food Plan	Mito Food Plan
Focus on Whole Foods & Phytonutrient Diversity	Condition-Specific Nutrients, Protective Antioxidants
Encourages Organic	Anti-Inflammatory Nutrients
Adequate Quality Protein	Energy Production, Low Glycemic Impact
Balanced Quality Fats	Energy Production, High in Quality Fats, Low Glycemic Impact
High Fiber	Low Glycemic Impact
Low in Simple Sugars	Low Glycemic Impact, Ketogenic Option
Gluten-Free (optional)	Gluten-Free
Targeted Calories (optional)	Intermittent Fasting and Calorie Restriction


CORE FOOD PLAN FEATURES
Focus on Whole Foods
Promotes Clean and Organic
Adequate Quality Protein
Balanced Quality Fats
High in Fiber
Low in Simple Sugars
Phytonutrient Diversity
Foundational Eating Plan
MEDICINE
ENERGY
INFORMATION
CONNECTION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VERSION 8
Phytonutrient
Spectrum
Comprehensive Guide

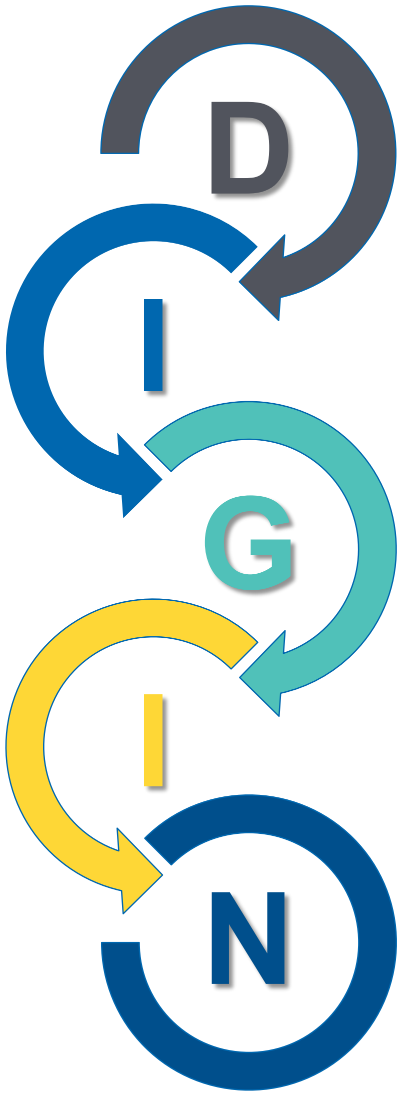
Foundational Therapeutic Approach: Mitochondrial Dysfunction

LIFESTYLE FIRST

FOOD FIRST

GUT FIRST

The DIGIN Mnemonic of Gut Dysfunction



***D**igestion/Absorption*

***I**ntestinal Permeability*

***G**ut Microbiota/Dysbiosis*

***I**nflammation/Immune*

***N**ervous System*

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Components of the 5R Approach to Gut Health

Remove

Replace

Reinoculate (Repopulate)

Repair

Rebalance

Second-Tier Support for Mitochondrial Function: Nutraceuticals That Reduce Oxidative Stress

Phytochemical	Dose
Curcumin	500-1,500 mg QD
Quercetin	250-500 mg TID
Green Tea Polyphenols	50-100 mg QD

See: "References: Second-tier Support for Mitochondrial Function"

Second-Tier Support for Mitochondrial Function: Nutraceuticals That Reduce Oxidative Stress

Nutrient	Dose
Coenzyme Q10	50-200 mg
N-Acetylcysteine	500-3,000 mg

See: "References: Second-tier Support for Mitochondrial Function"

References

SECOND-TIER SUPPORT FOR MITOCHONDRIAL FUNCTION

Curcumin to reduce oxidative stress (decreased malondialdehyde) (Strong Evidence)

- Qiu L, Gao C, Wang H, et al. Effects of dietary polyphenol curcumin supplementation on metabolic, inflammatory, and oxidative stress indices in patients with metabolic syndrome: a systematic review and meta-analysis of randomized controlled trials. *Front Endocrinol (Lausanne)*. 2023;14:1216708. Published 2023 Jul 14. doi:10.3389/fendo.2023.1216708

Curcumin to reduce inflammation (decreased CRP) (Strong Evidence)

- Dehzad MJ, Ghalandari H, Nouri M, Askarpour M. Antioxidant and anti-inflammatory effects of curcumin/turmeric supplementation in adults: A GRADE-assessed systematic review and dose-response meta-analysis of randomized controlled trials. *Cytokine*. 2023 Apr;164:156144. doi: 10.1016/j.cyto.2023.156144. Epub 2023 Feb 15.

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References

SECOND-TIER SUPPORT FOR MITOCHONDRIAL FUNCTION

Quercetin to reduce inflammation (CRP) (Moderate, Conflicting Evidence)

- Mohammadi-Sartang M, Mazloom Z, Sherafatmanesh S, Ghorbani M, Firoozi D. Effects of supplementation with quercetin on plasma C-reactive protein concentrations: a systematic review and meta-analysis of randomized controlled trials. *Eur J Clin Nutr.* 2017;71(9):1033-1039. doi:10.1038/ejcn.2017.55
- Ou Q, Zheng Z, Zhao Y, Lin W. Impact of quercetin on systemic levels of inflammation: a meta-analysis of randomised controlled human trials. *Int J Food Sci Nutr.* 2020;71(2):152-163. doi:10.1080/09637486.2019.1627515

Green tea polyphenols to improve oxidative stress (increased total antioxidant capacity and decreased malondialdehyde) (Moderate Evidence)

- Rasaei N, Asbaghi O, Samadi M, et al. Effect of Green Tea Supplementation on Antioxidant Status in Adults: A Systematic Review and Meta-Analysis of Randomized Clinical Trials. *Antioxidants (Basel).* 2021;10(11):1731. Published 2021 Oct 29. doi:10.3390/antiox10111731
- Martinez-Negrin G, Acton JP, Cocksedge SP, Bailey SJ, Clifford T. The effect of dietary (poly)phenols on exercise-induced physiological adaptations: A systematic review and meta-analysis of human intervention trials. *Crit Rev Food Sci Nutr.* 2022;62(11):2872-2887. doi:10.1080/10408398.2020.1860898

References

SECOND-TIER SUPPORT FOR MITOCHONDRIAL FUNCTION

Coenzyme Q10 to improve oxidative stress (reduced malondialdehyde, increased total antioxidant capacity) (Strong Evidence)

- Dabbaghi Varnousfaderani S, Musazadeh V, Ghalichi F, et al. Alleviating effects of coenzyme Q10 supplements on biomarkers of inflammation and oxidative stress: results from an umbrella meta-analysis. *Front Pharmacol.* 2023;14:1191290. Published 2023 Aug 8. doi:10.3389/fphar.2023.1191290
- Akbari A, Mobini GR, Agah S, et al. Coenzyme Q10 supplementation and oxidative stress parameters: a systematic review and meta-analysis of clinical trials. *Eur J Clin Pharmacol*. 2020;76(11):1483-1499. doi:10.1007/s00228-020-02919-8

N-Acetylcysteine to reduce inflammation (reduced CRP) (Strong Evidence)

- Askari M, Faryabi R, Mozaffari H, Darooghegi Mofrad M. The effects of N-Acetylcysteine on serum level of inflammatory biomarkers in adults. Findings from a systematic review and meta-analysis of randomized clinical trials. *Cytokine*. 2020;135:155239. doi:10.1016/j.cyto.2020.155239

Clinical Focus

- Summary, Conclusion, and Key Clinical Takeaways
- Clinical Flow Path for Mitochondrial Dysfunction

Mitochondrial Dysfunction: Key Clinical Takeaways

Secondary mitochondropathy is a highly prevalent condition mediated by dietary and lifestyle factors, as well as environmental agents and insults.

Secondary mitochondropathy contributes to many chronic diseases and syndromes and is often associated with atypical manifestations of these.

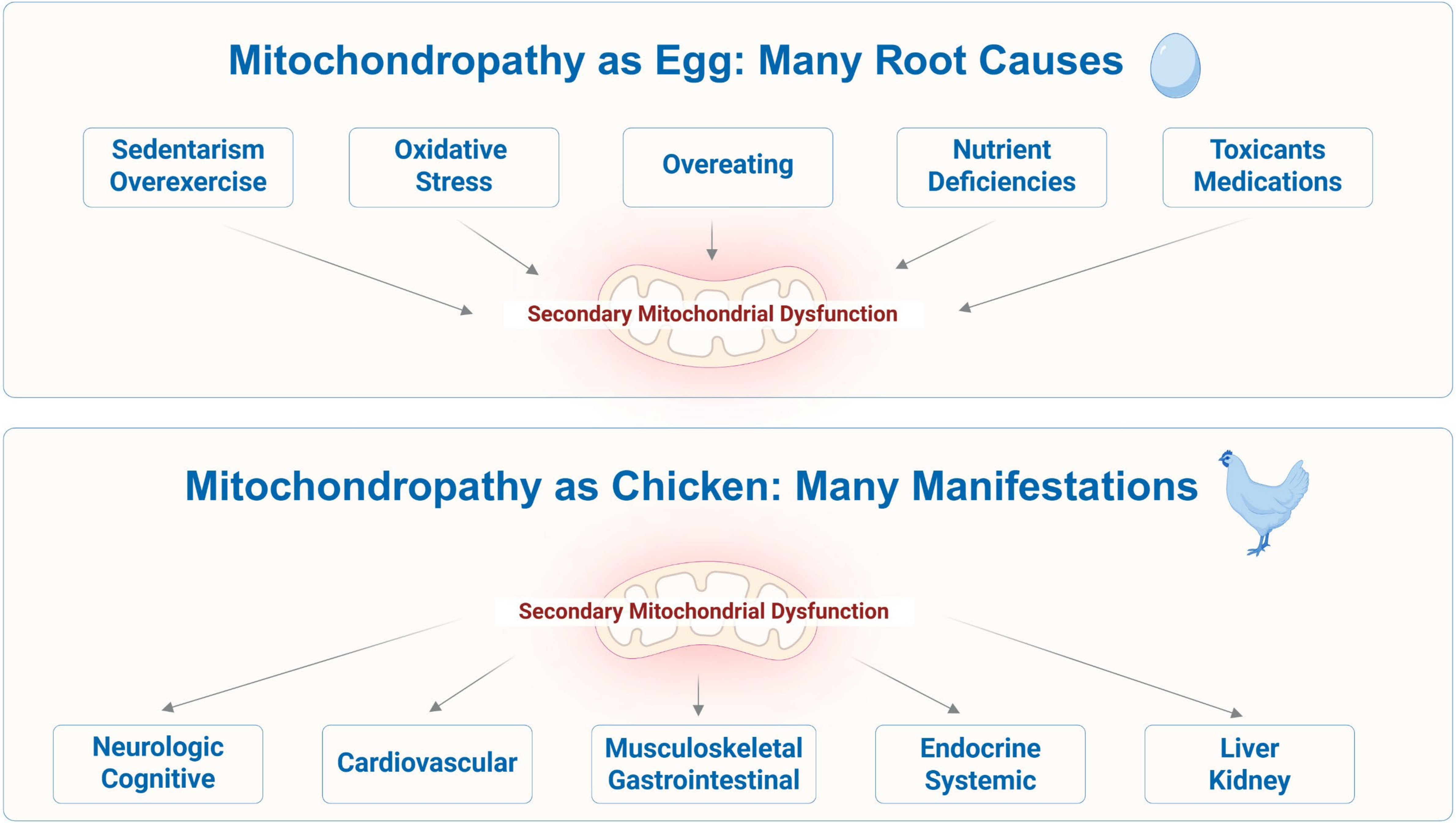
There are few pathognomonic and quantitative findings of mitochondrial dysfunction, but abnormal results of common laboratory analytes may provide single lines of evidence to support a clinical hypothesis of mitochondrial dysfunction.

Mitochondrial Dysfunction: Key Clinical Takeaways

Use the entire professional team approach to address mitochondrial dysfunction whenever possible.

Diet, lifestyle, gut health, and avoidance of toxic exposures are foundational interventions for all therapeutic regimens addressing mitochondrial dysfunction.

Addressing each patient's personalized ATMs and matrix dysfunctions is essential for favorable therapeutic outcomes.



See: “References: Mitochondropathy: Many Root Causes and Manifestations”

Long Description – **Mitochondropathy: Many Root Causes and Manifestations**

This graphic uses an "egg and chicken" metaphor to explain the two-way relationship between secondary mitochondrial dysfunction and chronic disease. It is divided into two panels:

Top Panel: "Mitochondropathy as Egg: Many Root Causes“

- This panel explains that secondary mitochondrial dysfunction can be caused by various lifestyle and environmental factors. These root causes include:
- Sedentary lifestyle or overexercise
- Oxidative stress
- Overeating
- Nutrient deficiencies
- Exposure to toxicants or medications
- These factors lead to **secondary mitochondrial dysfunction**, depicted as the “egg” in the metaphor.

Bottom Panel: "Mitochondropathy as Chicken: Many Manifestations"

- This panel illustrates that secondary mitochondrial dysfunction can also be a contributor to a wide range of health issues, referred to as the “chicken” in the metaphor. The potential disease manifestations include:
- Neurologic and cognitive symptoms
- Cardiovascular disease
- Musculoskeletal and gastrointestinal issues
- Endocrine and systemic dysfunction
- Liver and kidney disorders

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Sedentarism/Overexercise:

- Ostojic SM, Ratgeber L, Olah A, Betlehem J, Pongras A. What do over-trained athletes and patients with neurodegenerative diseases have in common? Mitochondrial dysfunction. *Exp Biol Med (Maywood)*. 2021;246(11):1241-1243. doi:10.1177/1535370221990619
- Raffin J, de Souto Barreto P, Le Traon AP, Vellas B, Aubertin-Leheudre M, Rolland Y. Sedentary behavior and the biological hallmarks of aging. *Ageing Res Rev*. 2023;83:101807. doi:10.1016/j.arr.2022.101807

Oxidative Stress:

- Kowalczyk P, Krych S, Kramkowski K, Jęczmyk A, Hrapkowicz T. Effect of oxidative stress on mitochondrial damage and repair in heart disease and ischemic events. *Int J Mol Sci*. 2024;25(22):12467. doi:10.3390/ijms252212467

Overeating:

- Engin A. The unrestrained overeating behavior and clinical perspective. *Adv Exp Med Biol*. 2024;1460:167-198. doi:10.1007/978-3-031-63657-8_6

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Nutrient Deficiencies:

- Bomer N, Pavez-Giani MG, Grote Beverborg N, Cleland JGF, van Veldhuisen DJ, van der Meer P. Micronutrient deficiencies in heart failure: mitochondrial dysfunction as a common pathophysiological mechanism? *J Intern Med*. 2022;291(6):713-731. doi:10.1111/joim.13456
- Izadi M, Sadri N, Abdi A, et al. Harnessing the fundamental roles of vitamins: the potent anti-oxidants in longevity. *Biogerontology*. 2025;26(2):58. doi:10.1007/s10522-025-10202-5

Toxicants/Medications:

- Duarte-Hospital C, Tête A, Brial F, et al. Mitochondrial dysfunction as a hallmark of environmental injury. *Cells*. 2021;11(1):110. doi:10.3390/cells11010110
- Ryu DY, Pang WK, Adegoke EO, Rahman MS, Park YJ, Pang MG. Bisphenol-A disturbs hormonal levels and testis mitochondrial activity, reducing male fertility. *Hum Reprod Open*. 2023;2023(4):hoad044. doi:10.1093/hropen/hoad044

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Neurologic/Cognitive:

- Monzio Compagnoni G, Di Fonzo A, Corti S, Comi GP, Bresolin N, Masliah E. The role of mitochondria in neurodegenerative diseases: the lesson from Alzheimer's disease and Parkinson's disease. *Mol Neurobiol*. 2020;57(7):2959-2980. doi:10.1007/s12035-020-01926-1
- Qiao L, Yang G, Wang P, Xu C. The potential role of mitochondria in the microbiota-gut-brain axis: implications for brain health. *Pharmacol Res*. 2024;209:107434. doi:10.1016/j.phrs.2024.107434

Cardiovascular:

- Rocca C, Soda T, De Francesco EM, et al. Mitochondrial dysfunction at the crossroad of cardiovascular diseases and cancer. *J Transl Med*. 2023;21(1):635. doi:10.1186/s12967-023-04498-5
- Wang X, Yu Q, Liao X, et al. Mitochondrial dysfunction in arrhythmia and cardiac hypertrophy. *Rev Cardiovasc Med*. 2023;24(12):364. doi:10.31083/j.rcm2412364

Musculoskeletal:

- Macchi C, Giachi A, Fichtner I, et al. Mitochondrial function in patients affected with fibromyalgia syndrome is impaired and correlates with disease severity. *Sci Rep*. 2024;14(1):30247. doi:10.1038/s41598-024-81298-x
- Qiu B, Xie X, Xi Y. Mitochondrial quality control: the real dawn of intervertebral disc degeneration? *J Transl Med*. 2024;22(1):1126. doi:10.1186/s12967-024-05943-9

References

MITOCHONDROPATHY: MANY ROOT CAUSES AND MANIFESTATIONS

Gastrointestinal:

- Xiao Y, Liu X, Xie K, et al. Mitochondrial dysfunction induced by HIF-1α under hypoxia contributes to the development of gastric mucosal lesions. *Clin Transl Med*. 2024;14(4):e1653. doi:10.1002/ctm2.1653
- Zhang YF, Fan MY, Bai QR, et al. Precision therapy for ulcerative colitis: insights from mitochondrial dysfunction interacting with the immune microenvironment. *Front Immunol*. 2024;15:1396221. doi:10.3389/fimmu.2024.1396221

Endocrine:

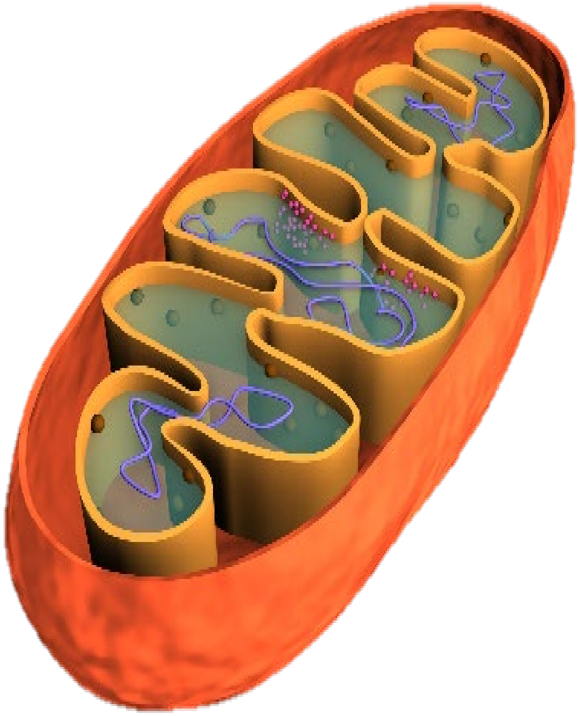
- Krako Jakovljevic N, Pavlovic K, Jotic A, et al. Targeting mitochondria in diabetes. *Int J Mol Sci*. 2021;22(12):6642. doi:10.3390/ijms22126642
- Siemers KM, Klein AK, Baack ML. Mitochondrial dysfunction in PCOS: insights into reproductive organ pathophysiology. *Int J Mol Sci*. 2023;24(17):13123. doi:10.3390/ijms241713123

Liver/Kidney:

- Diep TN, Liu H, Wang Y, Wang Y, Hoogewijs D, Yan LJ. Redox imbalance and mitochondrial abnormalities in kidney disease—volume II. *Biomolecules*. 2024;14(8):973. doi:10.3390/biom14080973
- Zhang R, Yan Z, Zhong H, et al. Gut microbial metabolites in MASLD: implications of mitochondrial dysfunction in the pathogenesis and treatment. *Hepatol Commun*. 2024;8(7):e0484. doi:10.1097/HC9.0000000000000484

Mitochondrial Functions:
More Than Just ATP and Energy Production

Mitochondria help maintain homeostasis.

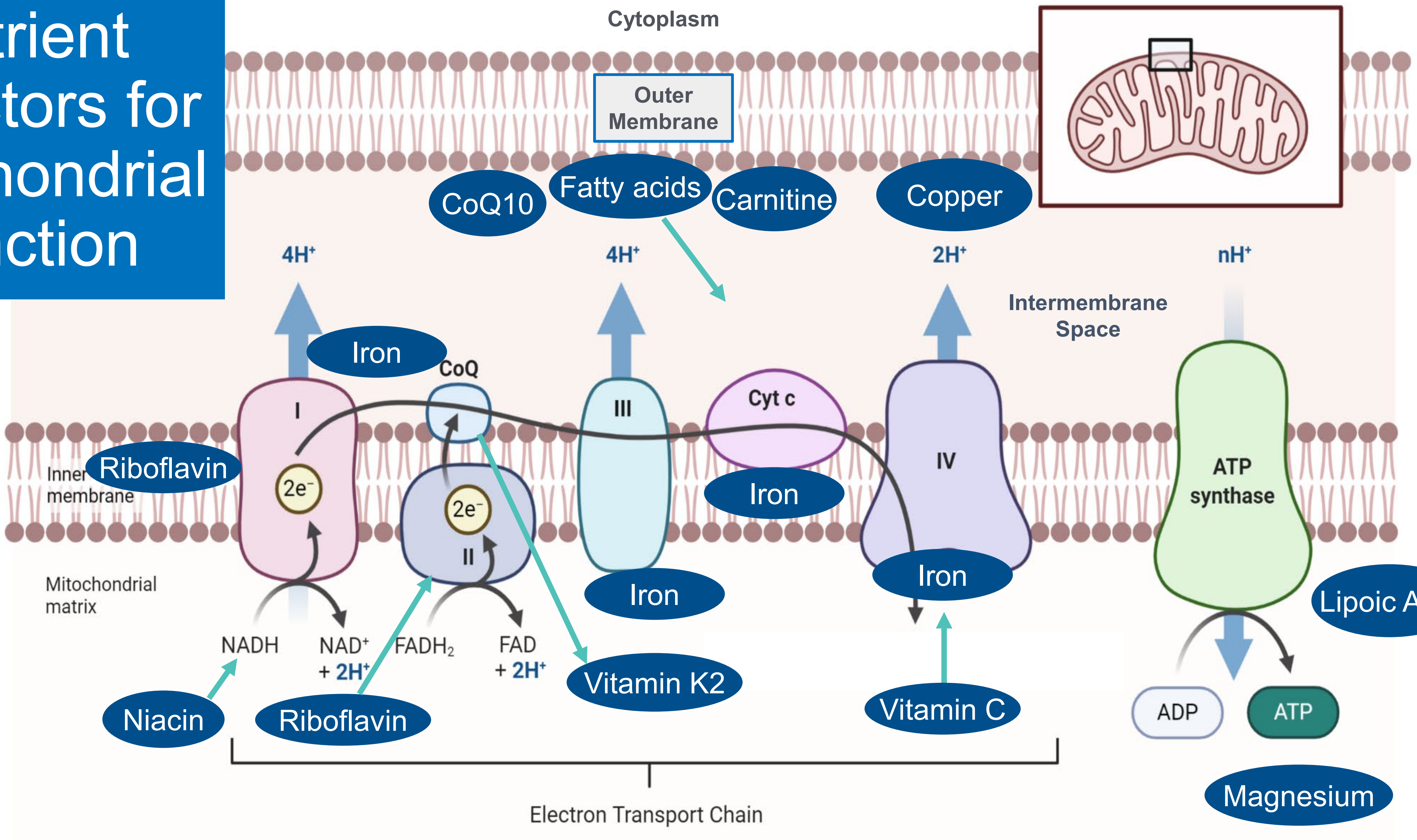


Bioenergetics

Biosynthesis

Signaling/Metabolism

Nutrient Cofactors for Mitochondrial Function



Wesselink E, Koekkoek WAC, Grefte S, Witkamp RF, van Zanten ARH. Feeding mitochondria: potential role of nutritional components to improve critical illness convalescence. *Clin Nutr.* 2019;38(3):982-995. doi:10.1016/j.clnu.2018.08.032

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Long Description

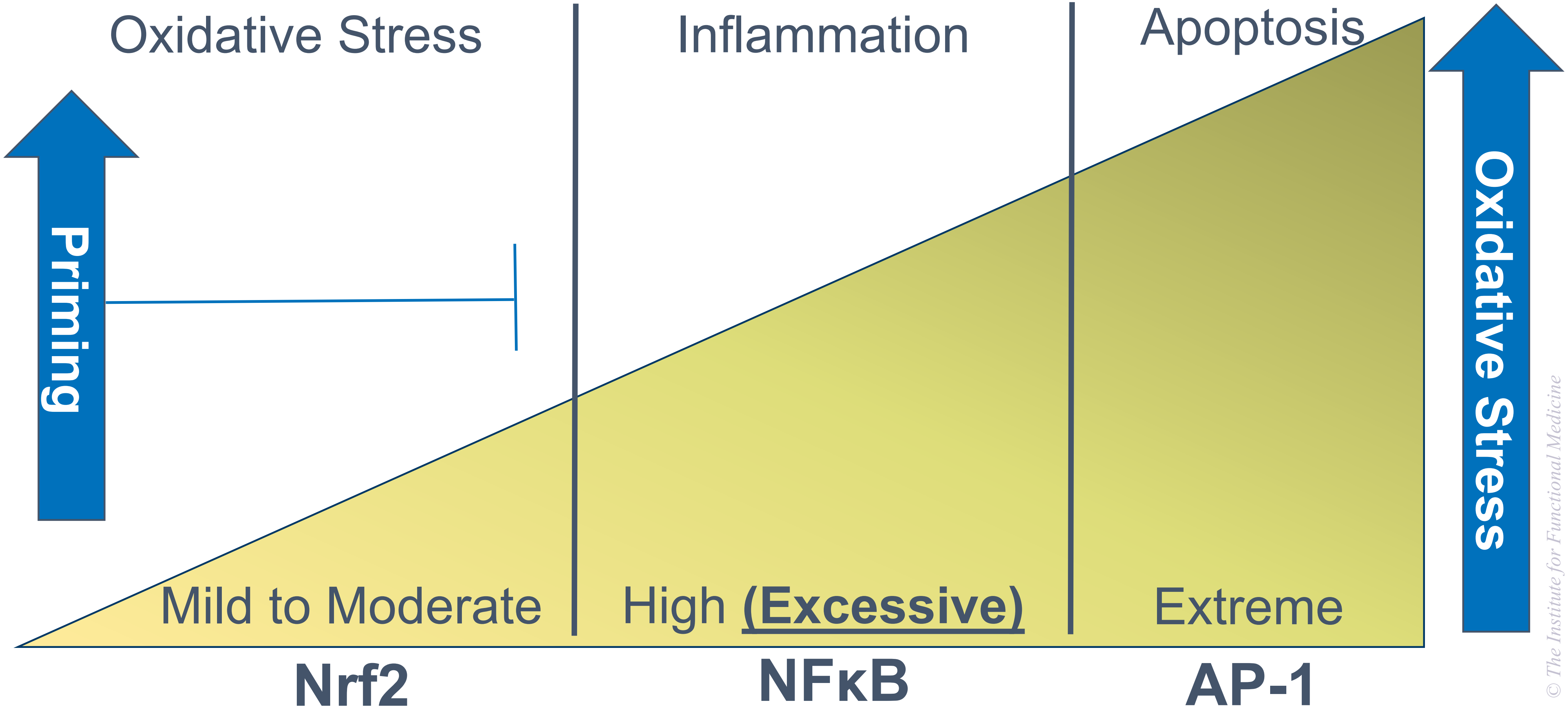
Nutrient Cofactors for Mitochondrial Function

This diagram shows the key vitamins, minerals, and other nutrients required for proper mitochondrial energy production through the electron transport chain.

- It depicts the inner mitochondrial membrane with protein complexes 1 through 4 and A T P synthase.
- The nutrients and nutrient cofactors needed are as follows:
  - Niacin and riboflavin are needed for N A D H and F A D H_2 production.
  - Riboflavin also supports Complex 2.
  - Iron is required at multiple points in Complexes 1, 2, 3, and 4, while Coenzyme Q 10 carries electrons between Complexes 1 or 2 and 3.
  - Vitamin K 2, carnitine, and fatty acids are linked to Complex 3 function.
  - Copper and iron are needed for Complex 4.
  - Vitamin C supports iron function and electron transfer. Lipoic acid and magnesium are needed for A T P synthase, which converts A D P to A T P.

The graphic emphasizes that these micronutrients are essential cofactors for mitochondrial oxidative phosphorylation and energy generation.

Differential Responses to Rising Oxidative Stress



Adapted from: Stefanson AL, Bakovic M. Dietary regulation of Keap1/Nrf2/ARE pathway: focus on plant-derived compounds and trace minerals. *Nutrients*. 2014;6(9):3777-3801. doi:10.3390/nu6093777

Long Description for
Differential Responses to Rising Oxidative Stress

- This graphic illustrates how the body responds differently to increasing levels of oxidative stress.
- At mild to moderate oxidative stress, the N r f 2 pathway is activated, which supports protective and adaptive responses, a stage referred to as “priming.”
- When oxidative stress becomes high or excessive, the N F κ B pathway is triggered, leading to inflammation.
- At extreme oxidative stress levels, the AP-1 pathway is activated, resulting in apoptosis, or programmed cell death.
- The diagram shows oxidative stress increasing from left to right, with priming occurring at lower levels, inflammation at higher levels, and apoptosis at the highest levels.

Qualitative Data: Single Lines of Evidence

Symptoms

Headaches
Migraines
Neuropathic pain
Chronic pain
Memory loss
Vision loss
Hearing loss
Fainting
Cognitive deficit
Neuropsych sx
Fatigue
Weakness
Exercise intolerance

Syndromes

Brain/Neuro:
Depression
Dementia
Parkinson’s
Autism
Dysautonomia
Neurodegenerative

Systemic:
Chronic fatigue
Fibromyalgia
Sleep apnea

Endocrine:
Diabetes/Met S
Hypothyroidism

Organs:
Cardiomyopathy
CHF
Dysrhythmias
NAFLD/NASH
Liver failure
Exocrine pancreas

GI:
IBS
GERD
Dysmotility
Chronic diarrhea
Chronic constipation
Cyclic vomiting

Signs

Wasting
Sarcopenia
Ptosis
Ophthalmoplegia
Hypotonia
Absent reflexes

See: “References: Clinical Findings of Mitochondropathy”

Long Description for
Qualitative Data: Single Lines of Evidence

The graphic outlines categories of qualitative data that serve as single lines of evidence for identifying mitochondrial dysfunction. It is divided into three main columns: Symptoms, Sydromes, Signs

1. Symptoms – Patient-reported experiences that may suggest mitochondrial issues:

- Headaches
- Migraines
- Neuropathic pain
- Chronic pain
- Memory loss
- Vision loss
- Hearing loss
- Fainting
- Cognitive deficit
- Neuropsychiatric symptoms
- Fatigue
- Weakness
- Exercise intolerance

2. Syndromes – Recognized clinical patterns, grouped by system:

- **Brain/Neuro:** Depression, dementia, Parkinson's disease, autism, dysautonomia, neurodegenerative disorders
- **Systemic:** Chronic fatigue, fibromyalgia, sleep apnea
- **Endocrine:** Diabetes/metabolic syndrome, hypothyroidism
- **Organs:** Cardiomyopathy, congestive heart failure (CHF), dysrhythmias, non-alcoholic fatty liver disease/non-alcoholic steatohepatitis (NAFLD/NASH), liver failure, exocrine pancreatic insufficiency
- **Gastrointestinal (GI):** Irritable bowel syndrome (IBS), gastroesophageal reflux disease (GERD), dysmotility, chronic diarrhea, chronic constipation, cyclic vomiting

3. Signs – Clinically observed findings:

- Wasting
- Sarcopenia
- Ptosis
- Ophthalmoplegia
- Hypotonia
- Absent reflexes

Overall meaning: This chart presents a framework for recognizing possible mitochondrial dysfunction using qualitative clinical information, which includes patient symptoms, known syndromes, and observable signs.

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (CBC & Iron Status)	Biomarker of Susceptibility	Biomarker of Effect	Comment
Hemoglobin (Low)	✓		May affect oxygen availability for oxidative phosphorylation
WBC (Low)		✓	Line of evidence: high oxidative stress states
Neutrophils (Low)		✓	Line of evidence: high oxidative stress states
Platelets (Low)		✓	Line of evidence: high oxidative stress states
Ferritin (Low)	✓		Iron (Fe) is a cofactor in the mitochondrial electron transport chain

See: "References: Laboratory Tests"

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (Basic Chemistries)	Biomarker of Susceptibility	Biomarker of Effect	Comment
Carbon Dioxide (CO_2) (High)	✓	✓	Line of evidence for acidosis; complex associations with mito function
Glucose (HbA1c) (High)		✓	Mito dysfunction impairs insulin signaling and mediates insulin resistance
Albumin (Low)	✓		Albumin is the most abundant antioxidant in the blood
Total Protein (Low)		✓	Line of evidence for impaired mitochondrial biosynthetic function
Creatinine (High)		✓	Line of evidence for kidney disease mediated by mitochondrial dysfunction

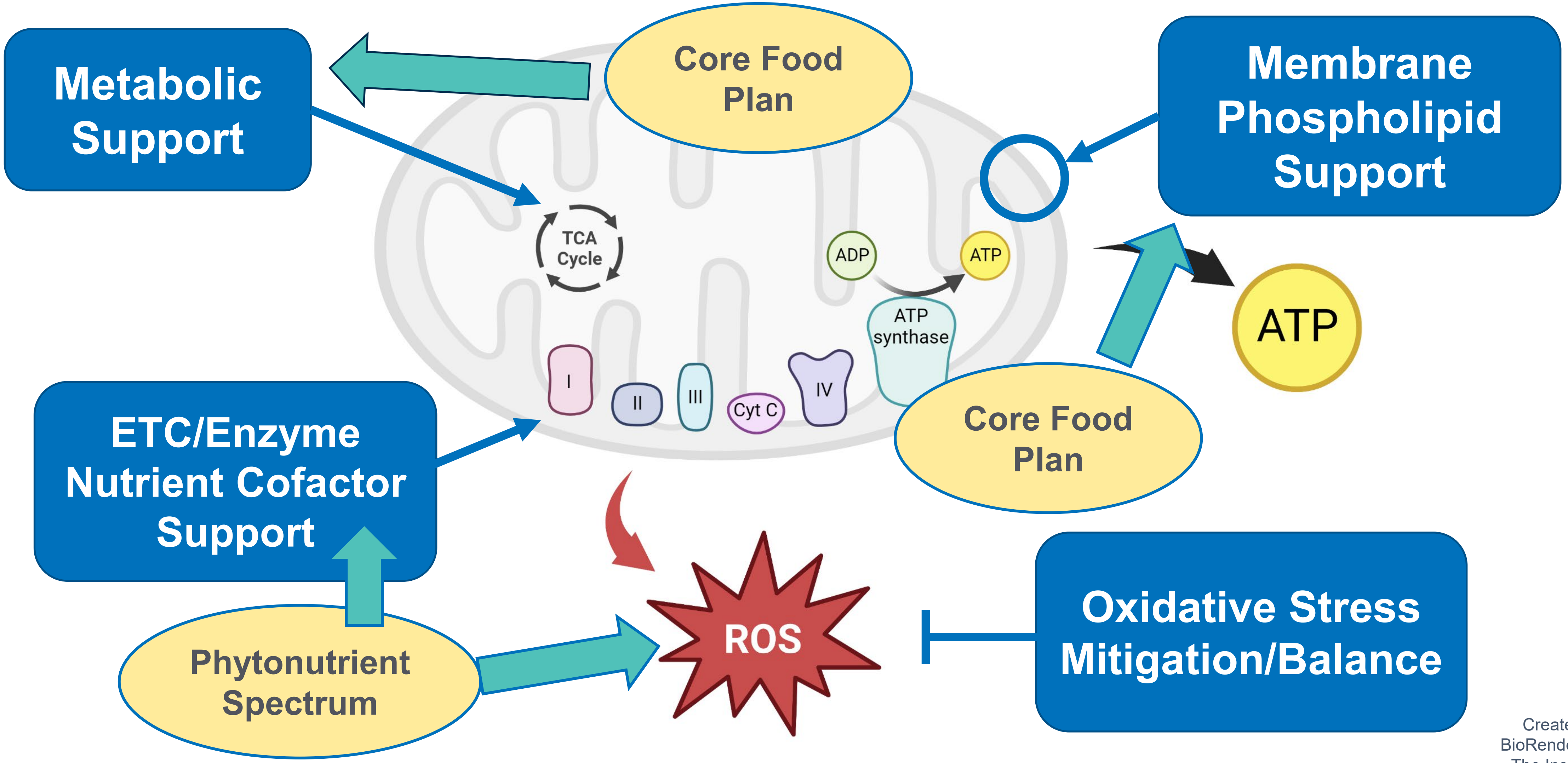
See: "References: Laboratory Tests"

Laboratory Tests
Single Lines of Evidence

Laboratory Analyte (Basic Chemistries)	Biomarker of Susceptibility	Biomarker of Effect	Comment
ALT and AST (High)		✓	Enzymes catalyzing alanine and aspartate entry into TCA cycle; damage if in blood
GGT (High)		✓	Indicator of low glutathione and increased oxidative stress
RBC Magnesium (Low)	✓		Cofactor in mitochondrial electron transport chain
B Vitamins (Low)	✓		Cofactor in mitochondrial electron transport chain
RBC Copper (Low)	✓		Cofactor in mitochondrial electron transport chain

See: “References: Laboratory Tests”

Multimodal Therapeutic Regimen to Address All Aspects of Mitochondrial Dysfunction



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Long Description for
Multimodal Therapeutic Regimen to Address All Aspects of Mitochondrial Dysfunction (Core Food Plan included)

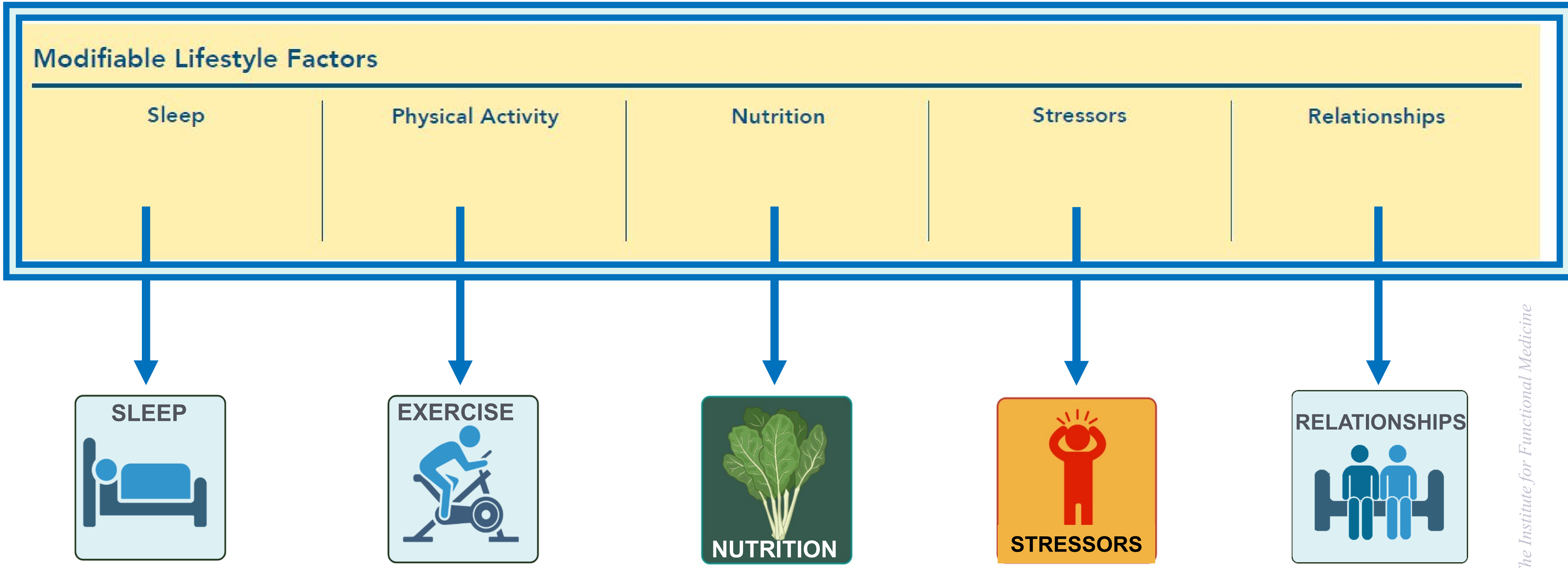
The diagram presents a multimodal therapeutic strategy for addressing all aspects of mitochondrial dysfunction, with the **Core Food Plan** integrated into multiple support areas.

Key intervention areas are:

- **Metabolic Support** – assisted by the Core Food Plan to optimize energy metabolism.
- **Membrane Phospholipid Support** – maintains mitochondrial membrane health, supporting ATP production.
- **ETC/Enzyme Nutrient Cofactor Support** – aided by the phytonutrient spectrum to enhance electron transport chain function and enzyme activity.
- **Oxidative Stress Mitigation/Balance** – reduces excess reactive oxygen species (ROS) to protect mitochondrial structures and function.

- The Core Food Plan is shown as central to both metabolic support and overall mitochondrial function, with its role in promoting ATP generation and balancing oxidative stress through nutrient-rich dietary strategies.

Mitochondria and Lifestyle Factors



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Second-Tier Support for Mitochondrial Function: Nutraceuticals That Reduce Oxidative Stress

Phytochemical	Dose
Curcumin	500-1,500 mg QD
Quercetin	250-500 mg TID
Green Tea Polyphenols	50-100 mg QD

- See: "References: Second-tier Support for Mitochondrial Function"

Second-Tier Support for Mitochondrial Function: Nutraceuticals That Reduce Oxidative Stress

Nutrient	Dose
Coenzyme Q10	50-200 mg
N-Acetylcysteine	500-3,000 mg

- See: "References: Second-tier Support for Mitochondrial Function"

References

SECOND-TIER SUPPORT FOR MITOCHONDRIAL FUNCTION

Curcumin to reduce oxidative stress (decreased malondialdehyde) (Strong Evidence)

- Qiu L, Gao C, Wang H, et al. Effects of dietary polyphenol curcumin supplementation on metabolic, inflammatory, and oxidative stress indices in patients with metabolic syndrome: a systematic review and meta-analysis of randomized controlled trials. *Front Endocrinol (Lausanne)*. 2023;14:1216708. Published 2023 Jul 14. doi:10.3389/fendo.2023.1216708

Curcumin to reduce inflammation (decreased CRP) (Strong Evidence)

- Dehzad MJ, Ghalandari H, Nouri M, Askarpour M. Antioxidant and anti-inflammatory effects of curcumin/turmeric supplementation in adults: A GRADE-assessed systematic review and dose-response meta-analysis of randomized controlled trials. *Cytokine*. 2023 Apr;164:156144. doi: 10.1016/j.cyto.2023.156144. Epub 2023 Feb 15.

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References

SECOND-TIER SUPPORT FOR MITOCHONDRIAL FUNCTION

Quercetin to reduce inflammation (CRP) (Moderate, Conflicting Evidence)

- Mohammadi-Sartang M, Mazloom Z, Sherafatmanesh S, Ghorbani M, Firoozi D. Effects of supplementation with quercetin on plasma C-reactive protein concentrations: a systematic review and meta-analysis of randomized controlled trials. *Eur J Clin Nutr.* 2017;71(9):1033-1039. doi:10.1038/ejcn.2017.55
- Ou Q, Zheng Z, Zhao Y, Lin W. Impact of quercetin on systemic levels of inflammation: a meta-analysis of randomised controlled human trials. *Int J Food Sci Nutr.* 2020;71(2):152-163. doi:10.1080/09637486.2019.1627515

Green tea polyphenols to improve oxidative stress (increased total antioxidant capacity and decreased malondialdehyde) (Moderate Evidence)

- Rasaei N, Asbaghi O, Samadi M, et al. Effect of Green Tea Supplementation on Antioxidant Status in Adults: A Systematic Review and Meta-Analysis of Randomized Clinical Trials. *Antioxidants (Basel).* 2021;10(11):1731. Published 2021 Oct 29. doi:10.3390/antiox10111731
- Martinez-Negrin G, Acton JP, Cocksedge SP, Bailey SJ, Clifford T. The effect of dietary (poly)phenols on exercise-induced physiological adaptations: A systematic review and meta-analysis of human intervention trials. *Crit Rev Food Sci Nutr.* 2022;62(11):2872-2887. doi:10.1080/10408398.2020.1860898

References

SECOND-TIER SUPPORT FOR MITOCHONDRIAL FUNCTION

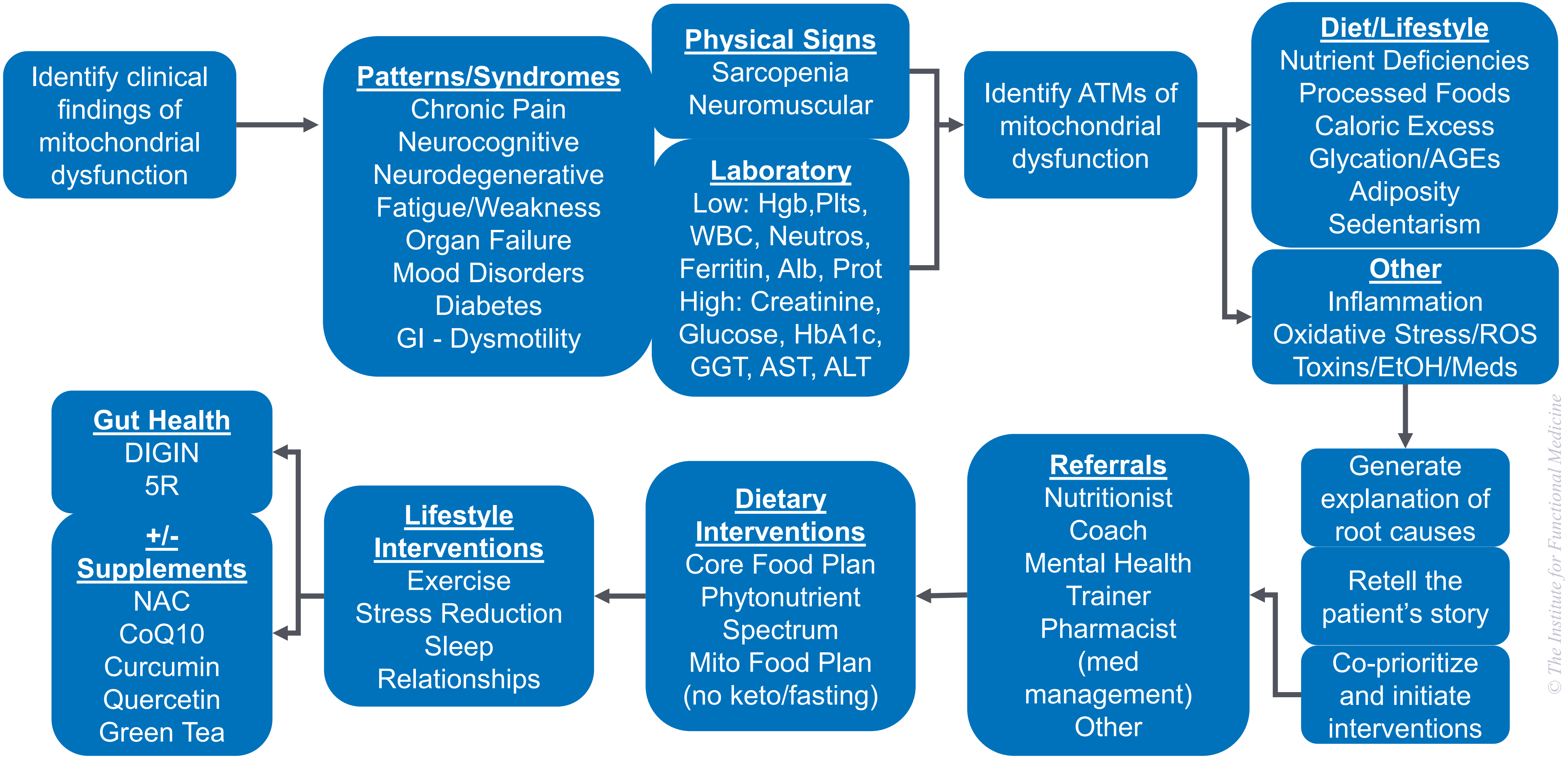
Coenzyme Q10 to improve oxidative stress (reduced malondialdehyde, increased total antioxidant capacity) (Strong Evidence)

- Dabbaghi Varnousfaderani S, Musazadeh V, Ghalichi F, et al. Alleviating effects of coenzyme Q10 supplements on biomarkers of inflammation and oxidative stress: results from an umbrella meta-analysis. *Front Pharmacol.* 2023;14:1191290. Published 2023 Aug 8. doi:10.3389/fphar.2023.1191290
- Akbari A, Mobini GR, Agah S, et al. Coenzyme Q10 supplementation and oxidative stress parameters: a systematic review and meta-analysis of clinical trials. *Eur J Clin Pharmacol*. 2020;76(11):1483-1499. doi:10.1007/s00228-020-02919-8

N-Acetylcysteine to reduce inflammation (reduced CRP) (Strong Evidence)

- Askari M, Faryabi R, Mozaffari H, Darooghegi Mofrad M. The effects of N-Acetylcysteine on serum level of inflammatory biomarkers in adults. Findings from a systematic review and meta-analysis of randomized clinical trials. *Cytokine*. 2020;135:155239. doi:10.1016/j.cyto.2020.155239

Flow Path: Mitochondrial Dysfunction



Long description - **Flow Path: Mitochondrial Dysfunction**

This flow chart outlines the clinical process for evaluating and addressing mitochondrial dysfunction.

1. **Identify clinical findings** – This step looks for signs of mitochondrial dysfunction.
2. **Patterns/Syndromes** – Possible patterns include chronic pain, neurocognitive issues, neurodegenerative conditions, fatigue or weakness, organ failure, mood disorders, diabetes, and gastrointestinal dysmotility.
3. **Physical Signs** – Examples include sarcopenia and neuromuscular problems.
4. **Laboratory Findings** – Low values may include hemoglobin, platelets, white blood cells, neutrophils, ferritin, albumin, and protein; high values may include creatinine, glucose, HbA1c, GGT, AST, and ALT.
5. **Identify ATMs (antecedents, triggers, mediators)** – Factors may include:
    - **Diet/Lifestyle:** nutrient deficiencies, processed foods, caloric excess, glycation/AGEs, adiposity, and sedentary behavior.
    - **Other:** inflammation, oxidative stress/ROS, toxins, alcohol, and medications.
6. **Generate explanation of root causes** – Clinician retells the patient’s story and develops a co-prioritized plan to start interventions.
7. **Referrals** – May include a nutritionist, health coach, mental health professional, trainer, or pharmacist for medication management.
8. **Dietary Interventions** – Core Food Plan, phytonutrient spectrum, and Mito Food Plan (with or without keto/fasting).
9. **Lifestyle Interventions** – Exercise, stress reduction, and adequate sleep.
10. **Gut Health** – DIGIN protocol, 5R approach, and supplements such as NAC, CoQ10, curcumin, quercetin, and green tea.

The process integrates assessment, lifestyle and dietary changes, targeted supplementation, and specialist referrals to address mitochondrial dysfunction comprehensively.

Bioenergetics and Mitochondropathy

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE